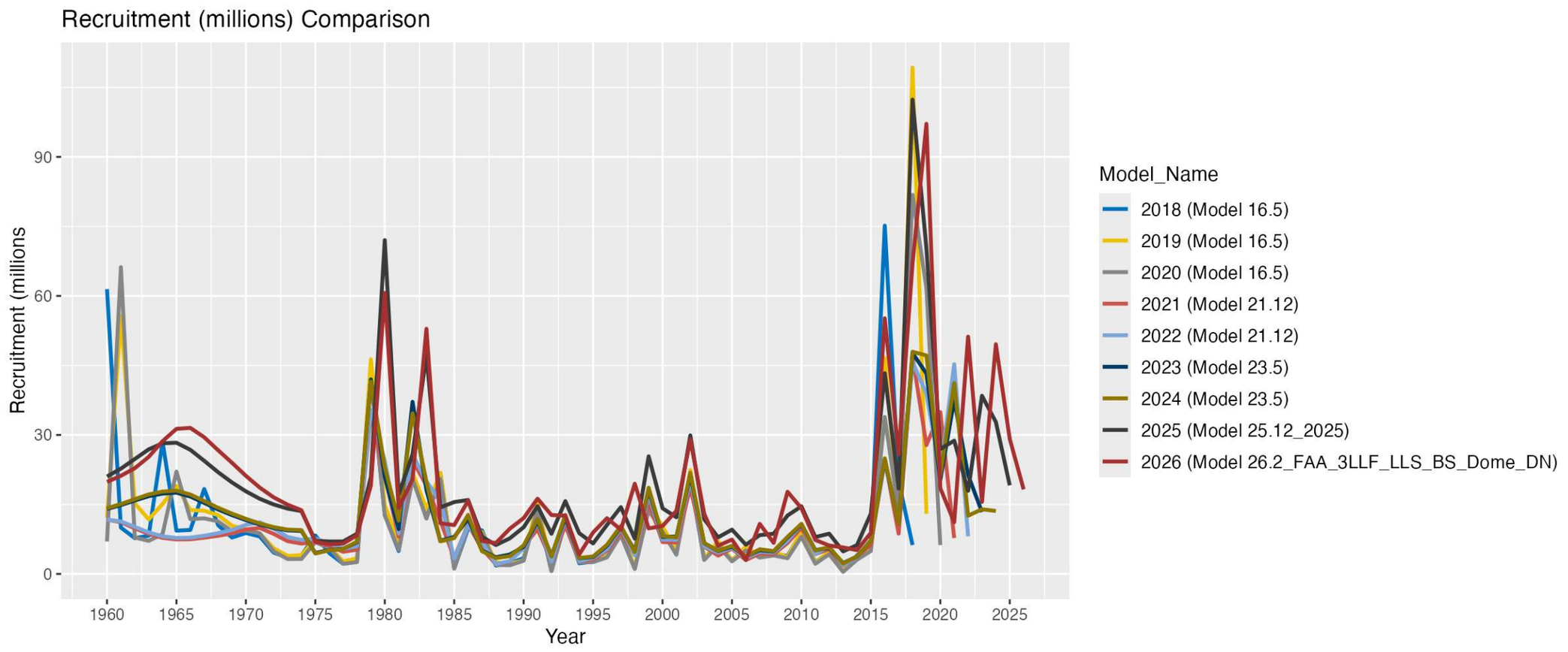
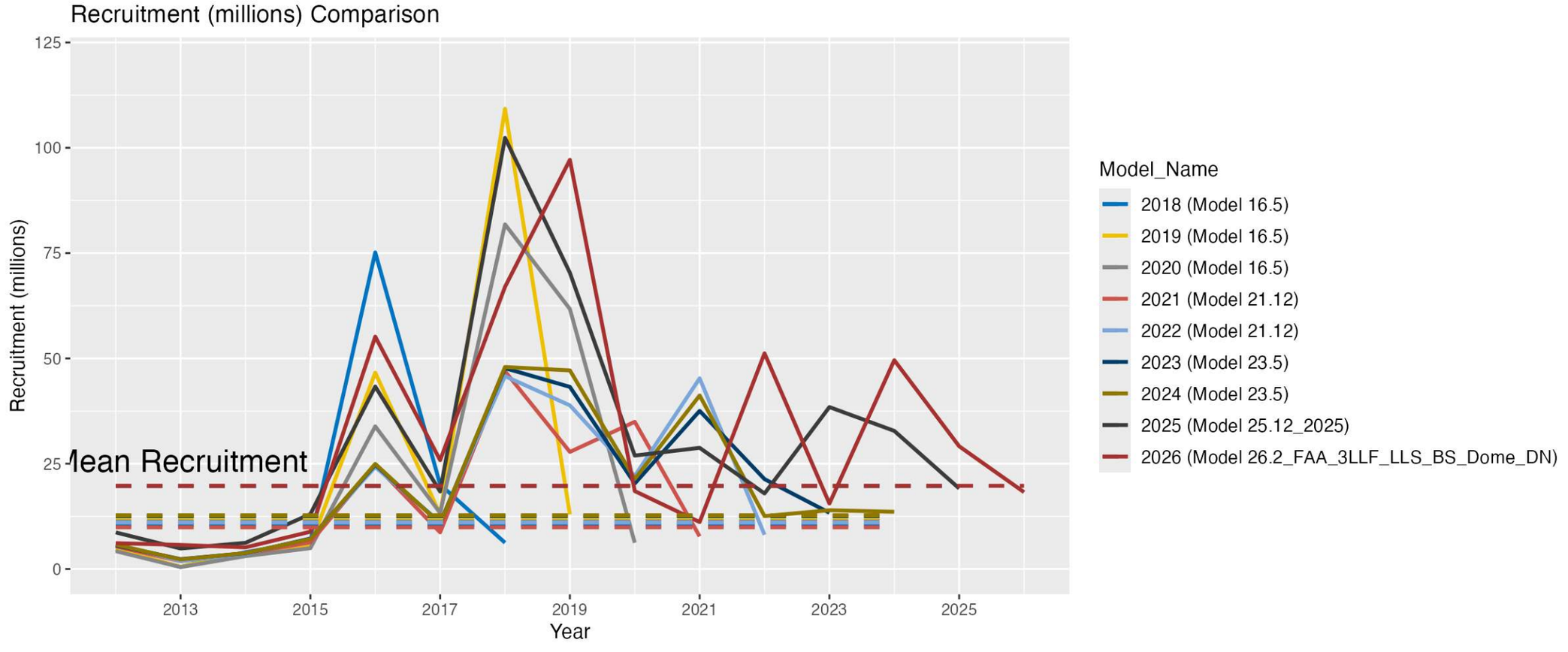
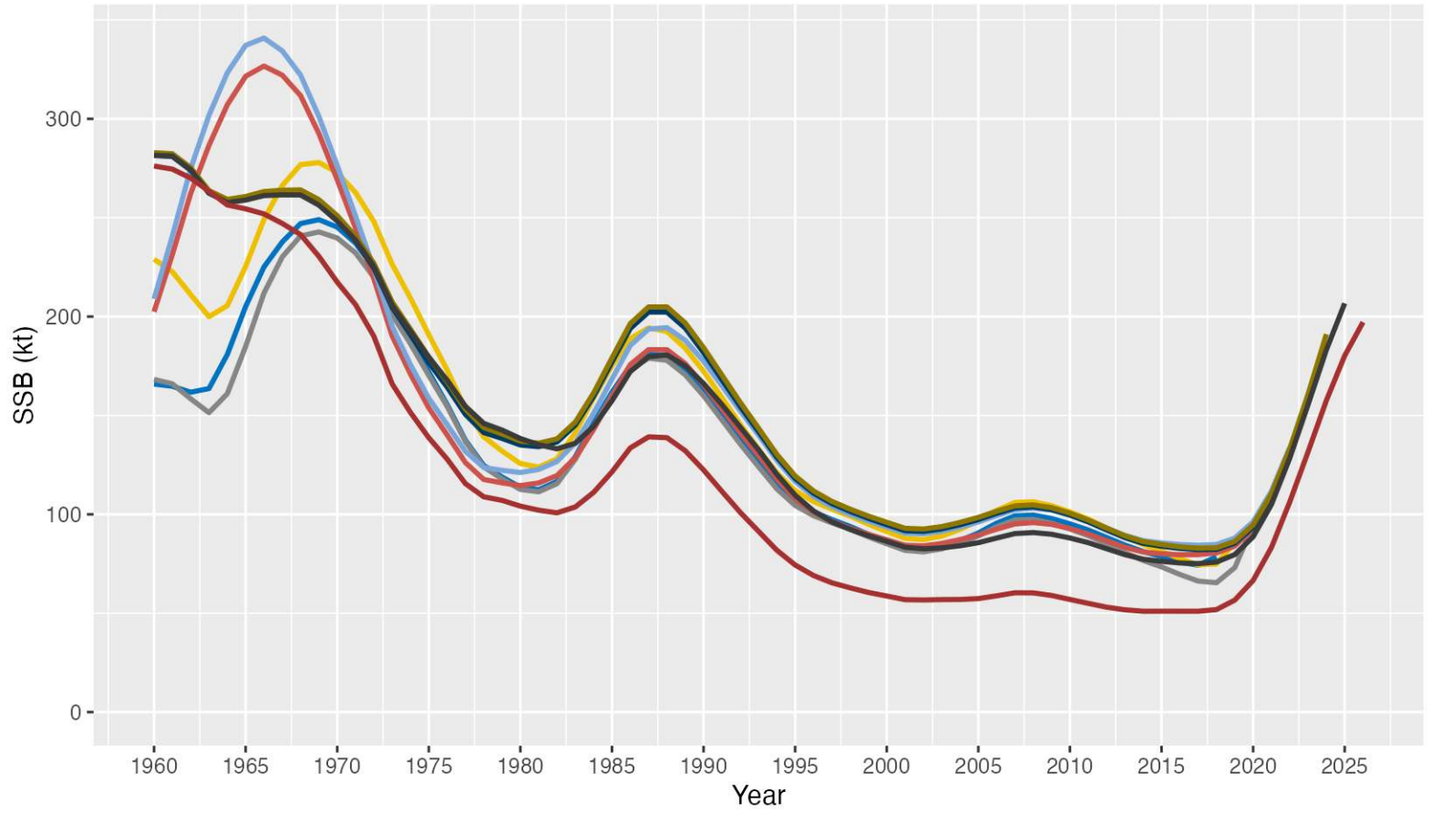
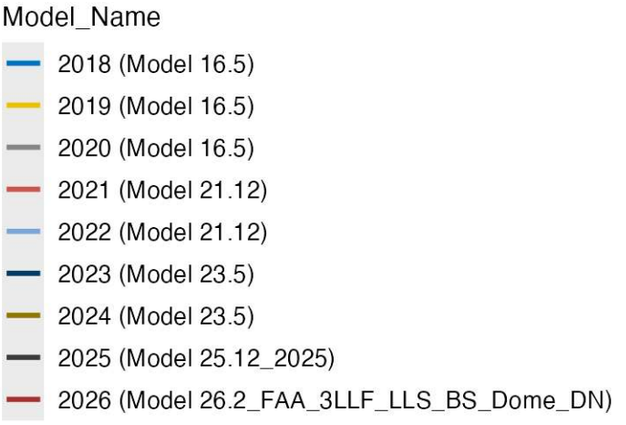

Recruitment (millions) Comparison
Recruitment (millions
0
30
60
90
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_DN)

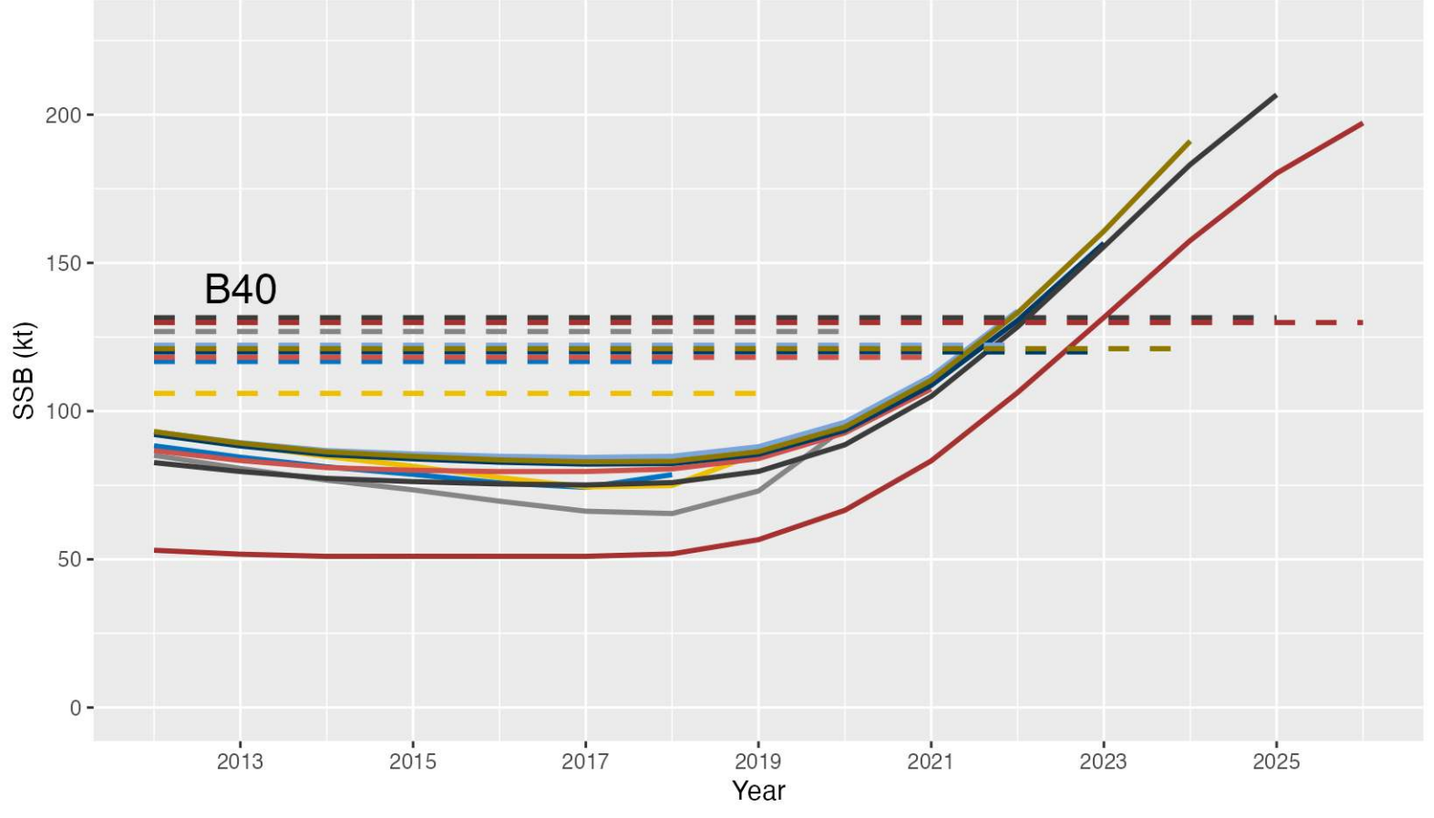

Recruitment (millions) Comparison
Recruitment (millions)
0
25
50
75
100
125
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_DN)

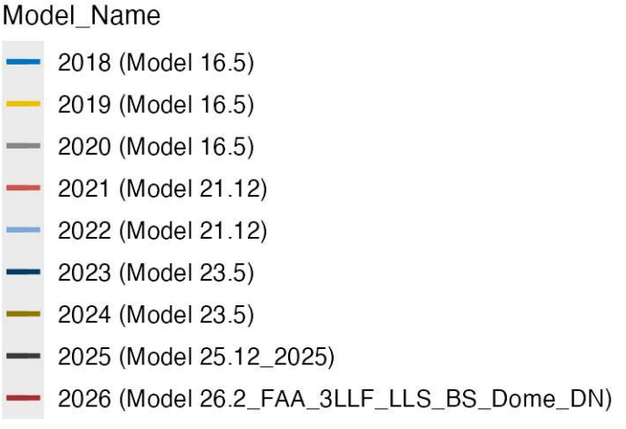
SSB (kt) Comparison





SSB (kt) Comparison





colour — First Iter — Last Iter

Fixed Gear AI Fishery | Fixed Gear BS Fishery | Fixed Gear GOA Fishery | Trawl Fishery

Fishery Ages

Fishery Lengths

Mean Composition Bin

25
20
15
10
5

65
60
55

1990 2000 2010 2020

Year

colour — First Iter — Last Iter

Japan LL Survey | NOAA LL AI Survey | NOAA LL BS Survey | NOAA LL GOA Survey

Survey Ages

Survey Lengths

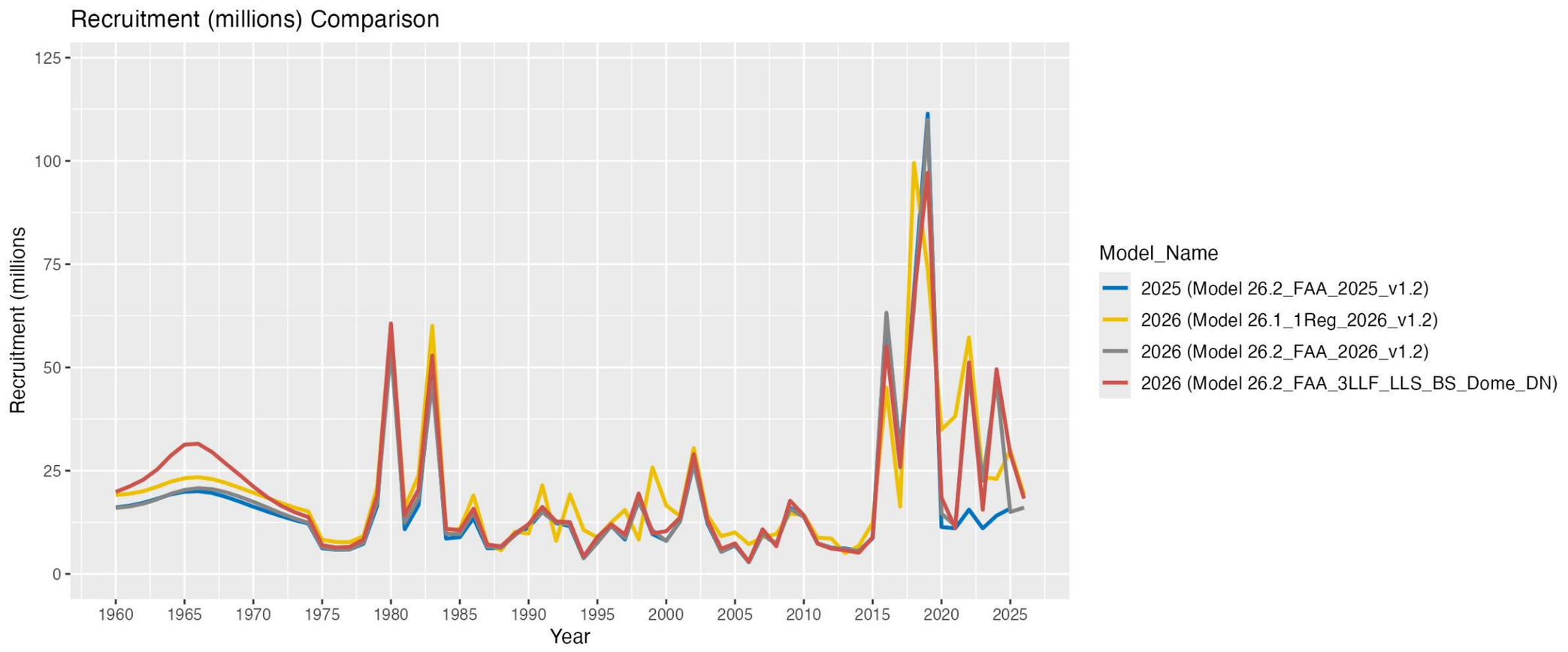
Mean Composition Bin

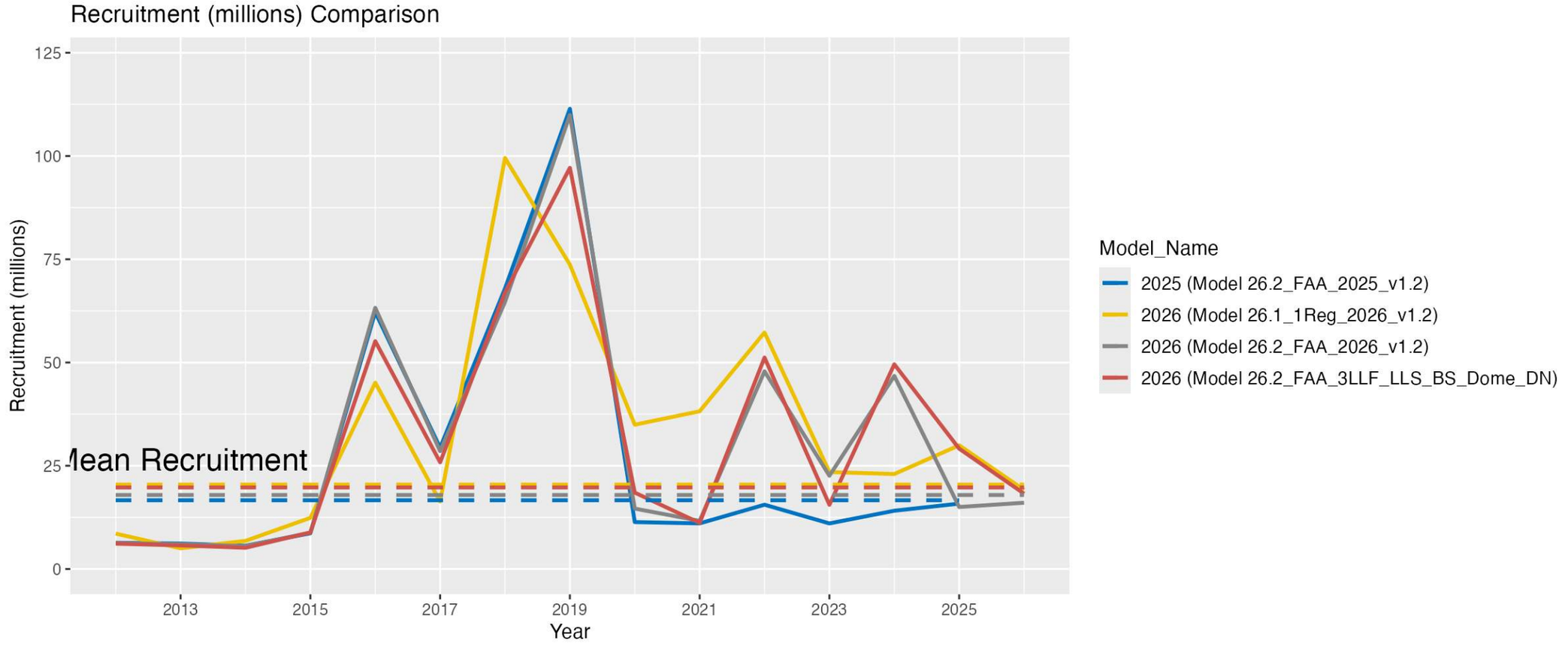
10.0
7.5
5.0

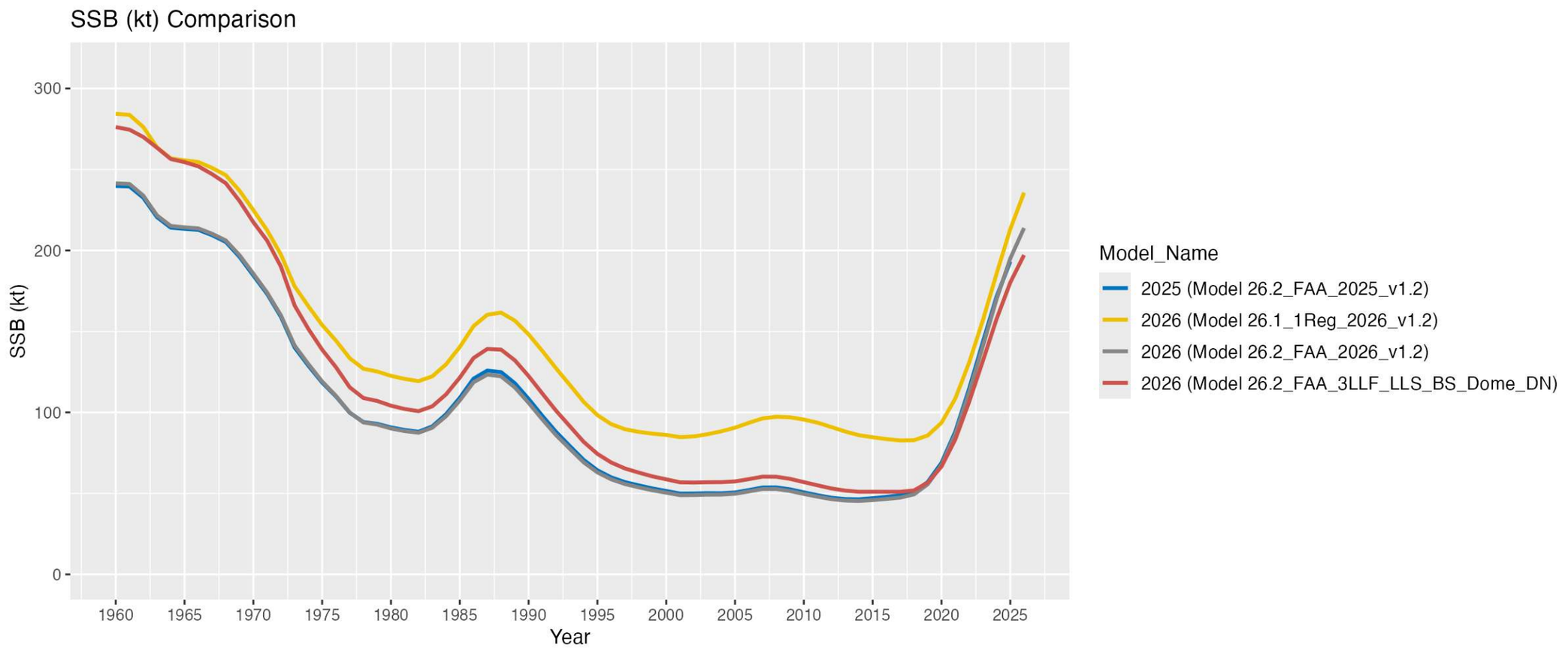
67.5
65.0
62.5

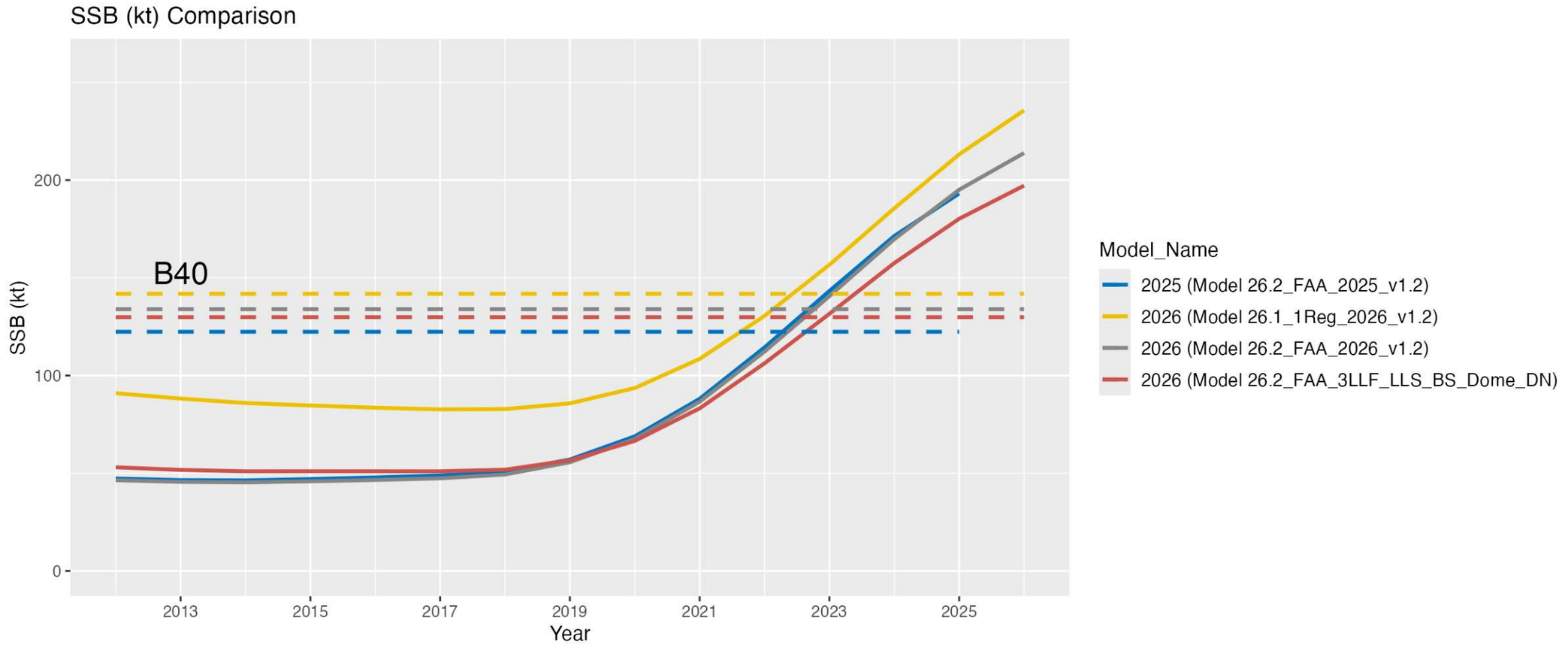
1980 1990 2000 2010 2020

Year

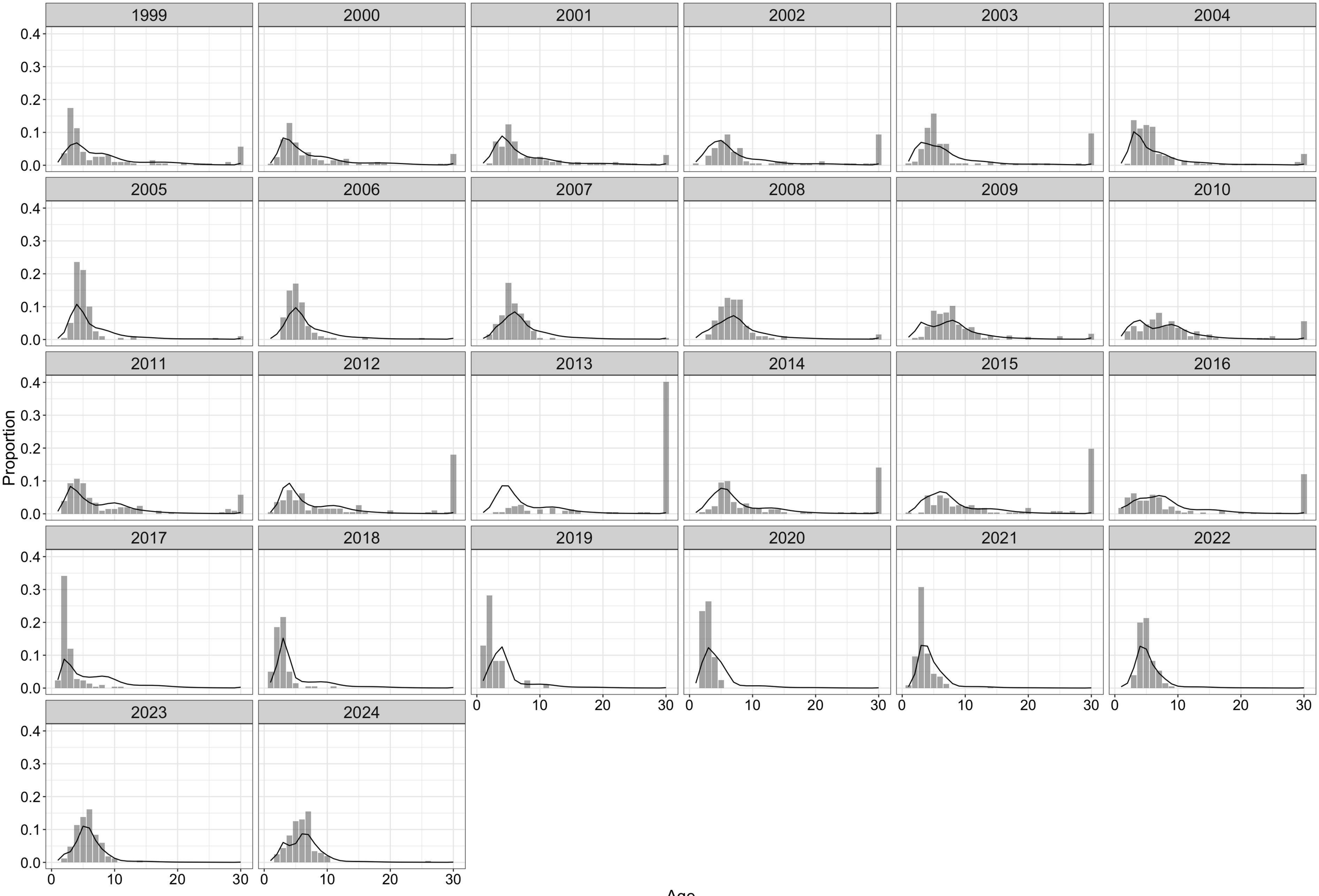

Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_DN)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_DN)

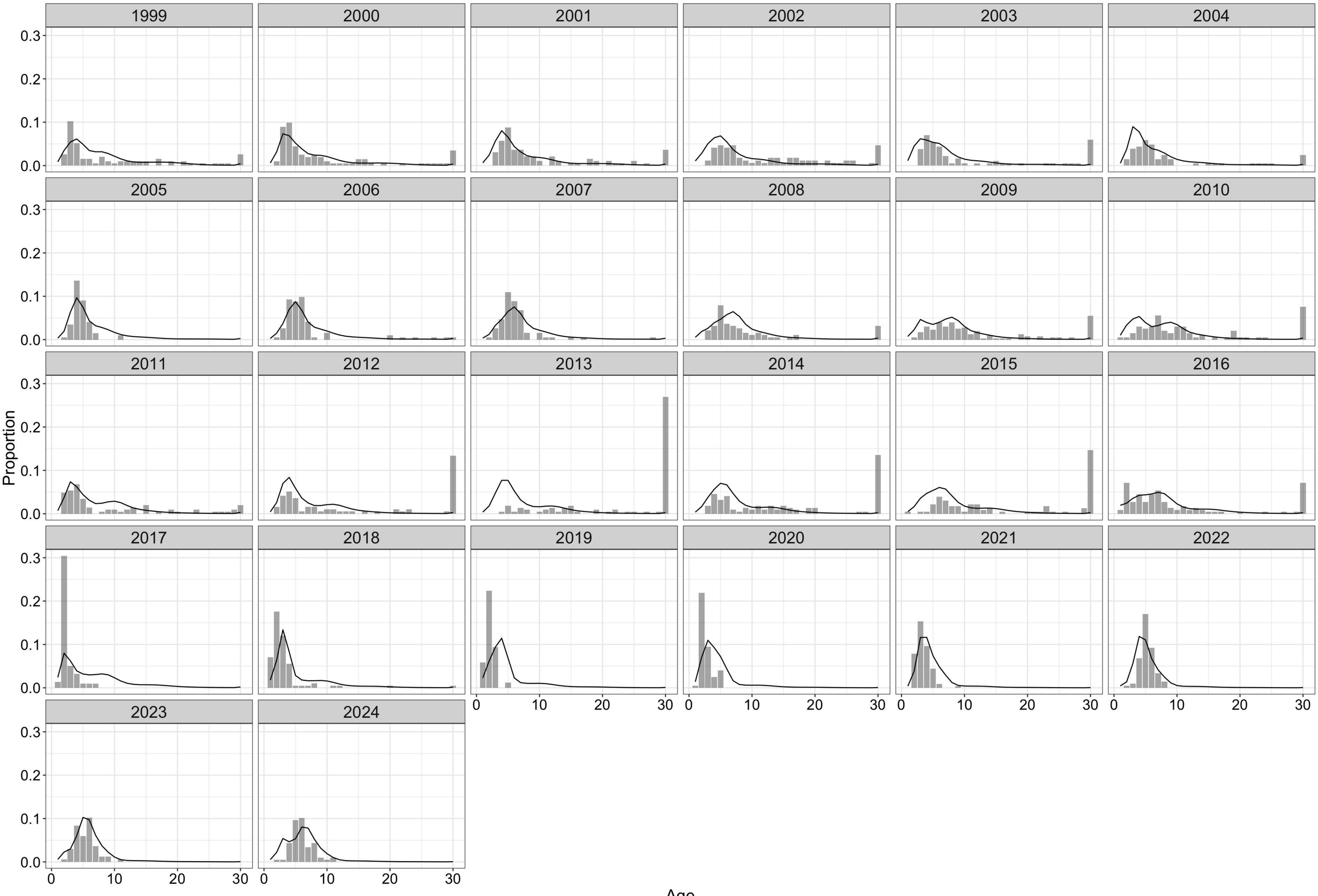

SSB (kt) Comparison
300
200
100
0
SSB (kt)
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_DN)


SSB (kt) Comparison
200
100
0
SSB (kt)
B40
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_DN)

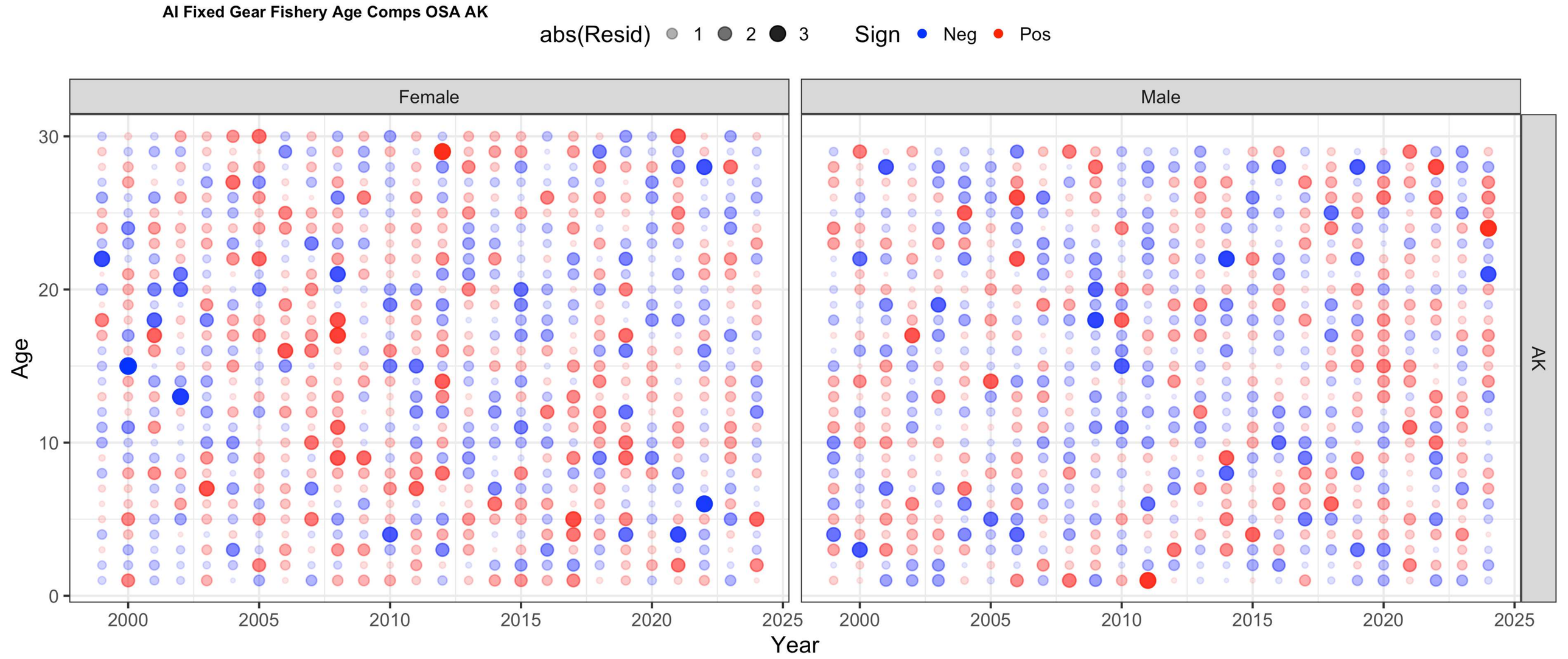
AI Fixed Gear Fishery Age Comps Female AK

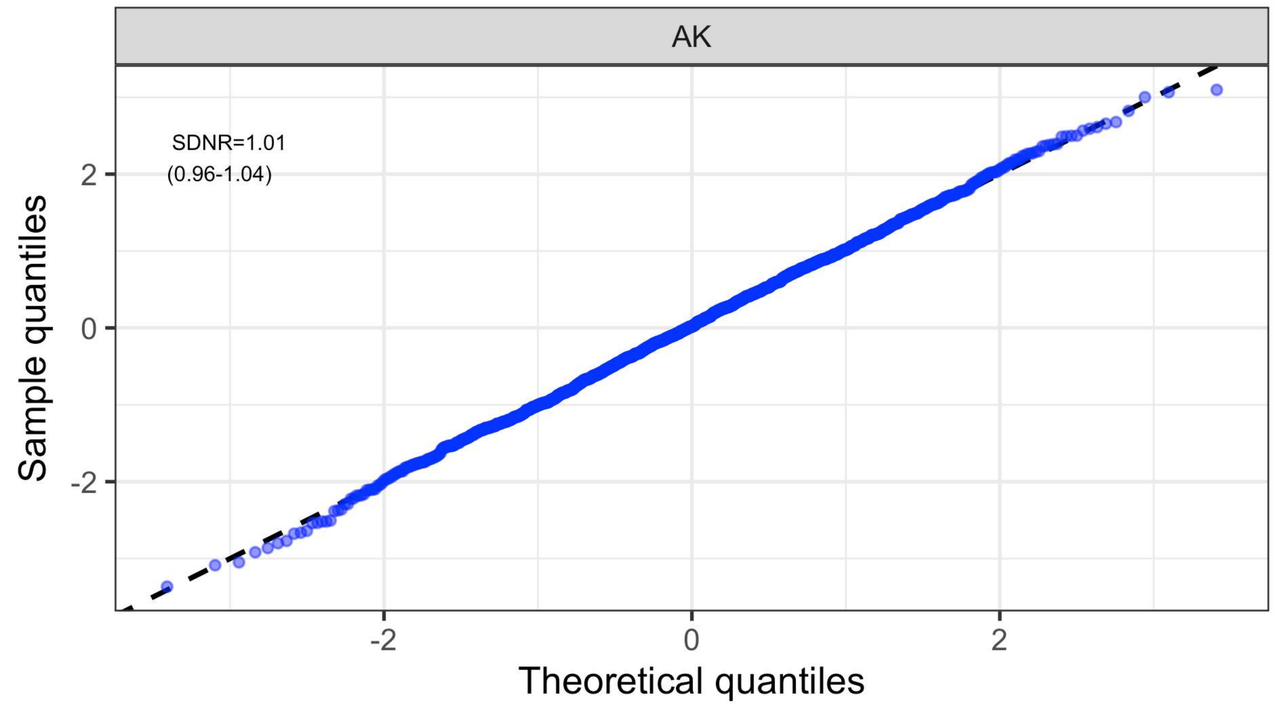


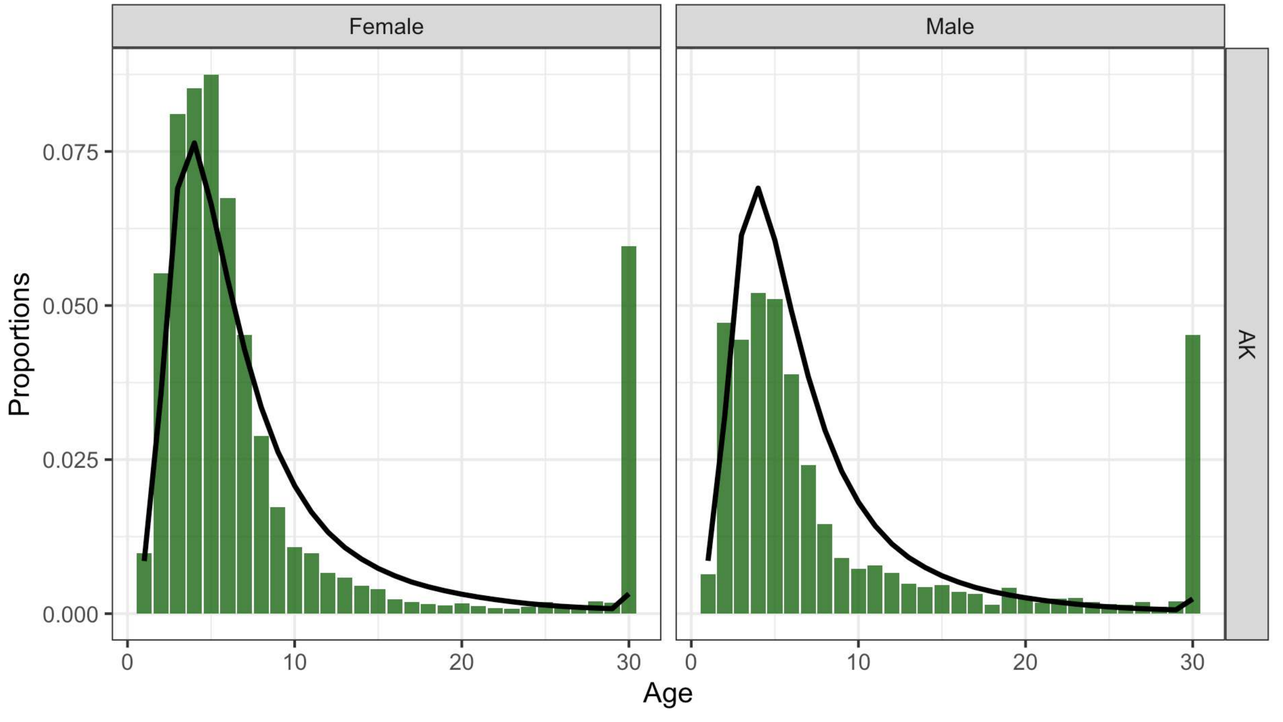
AI Fixed Gear Fishery Age Comps Male AK

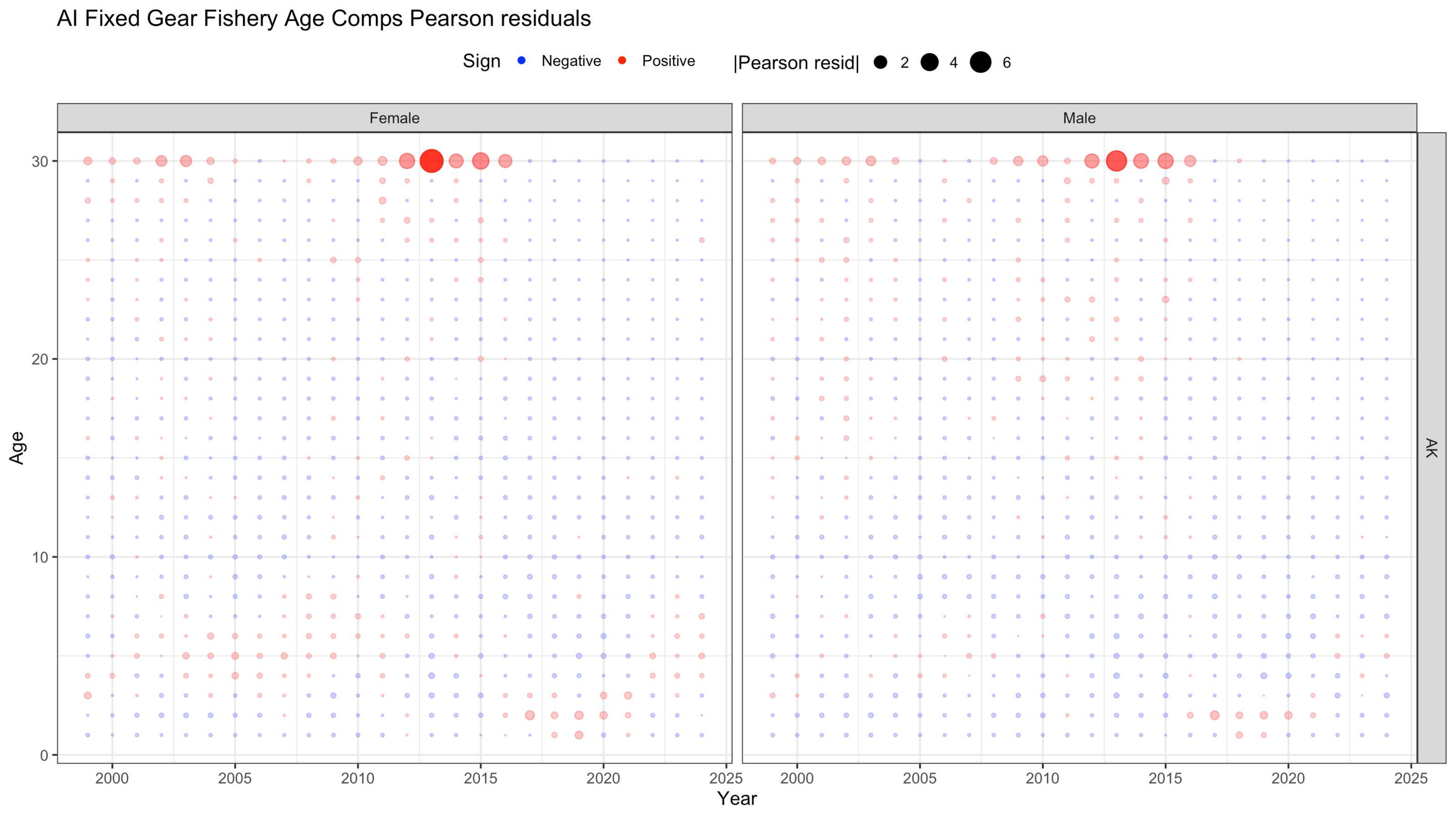


AI Fixed Gear Fishery Age Comps OSA AK

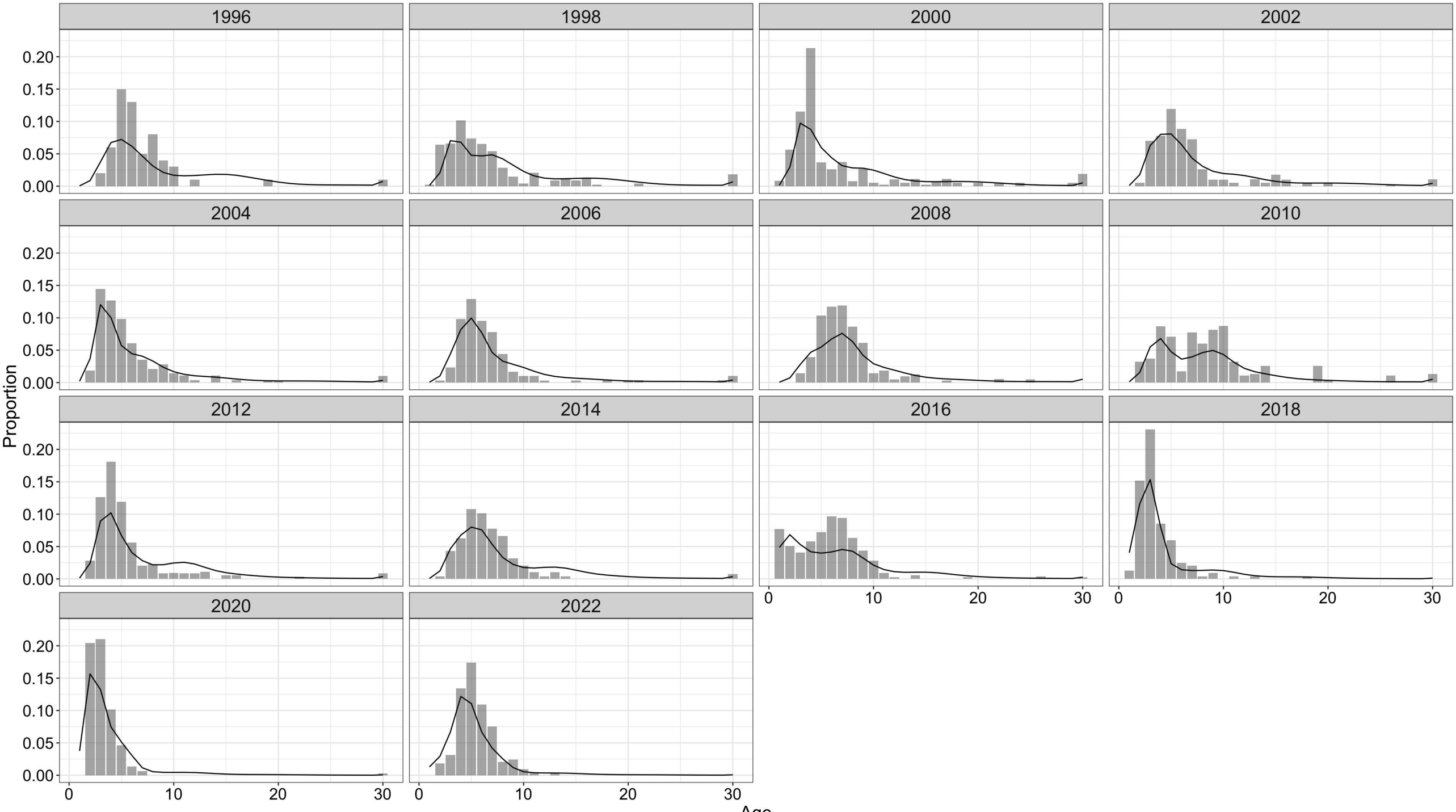







AI Fixed Gear Fishery Age Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
2
4
6
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

AI LL Survey Age Comps Female AK

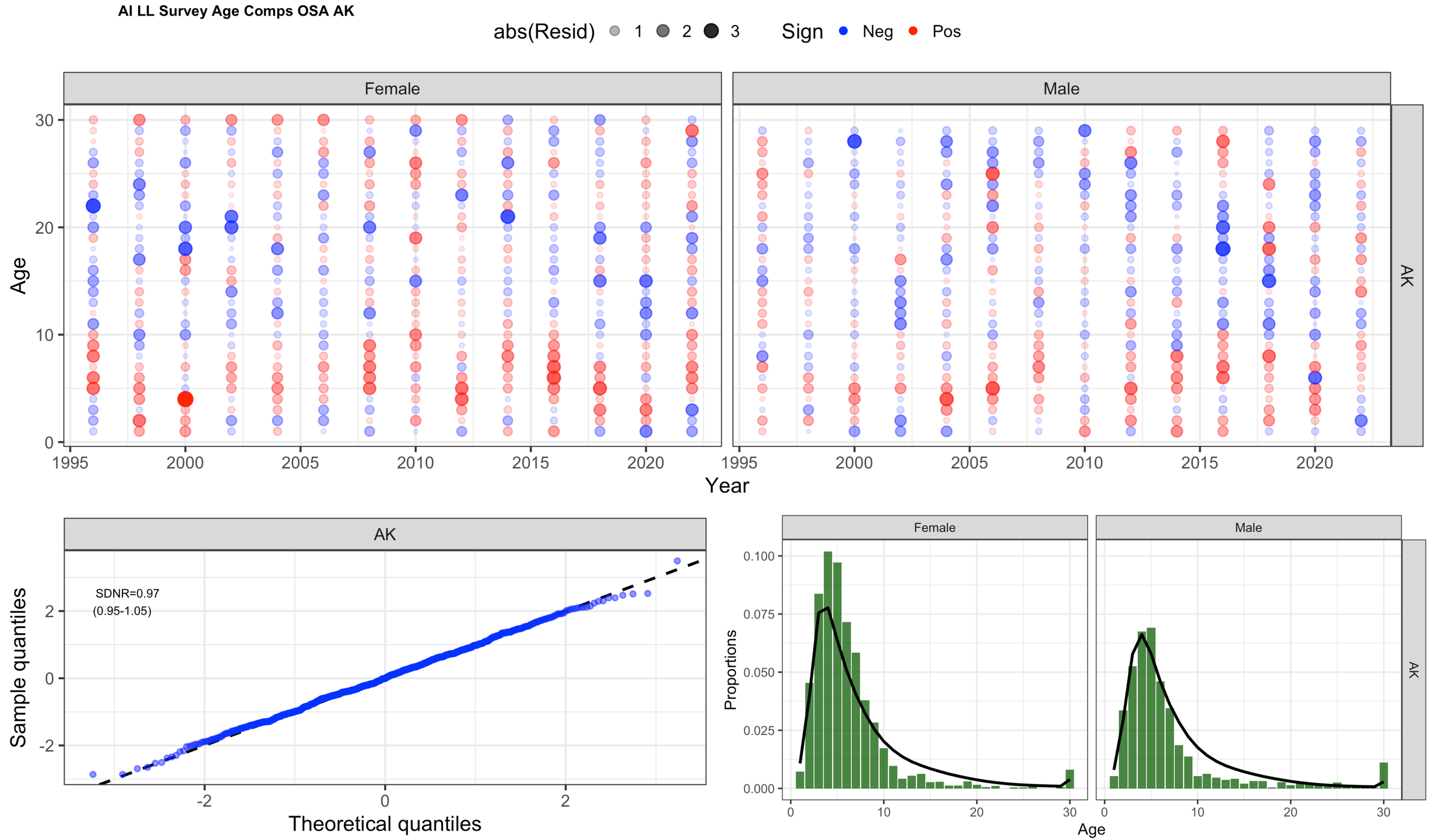


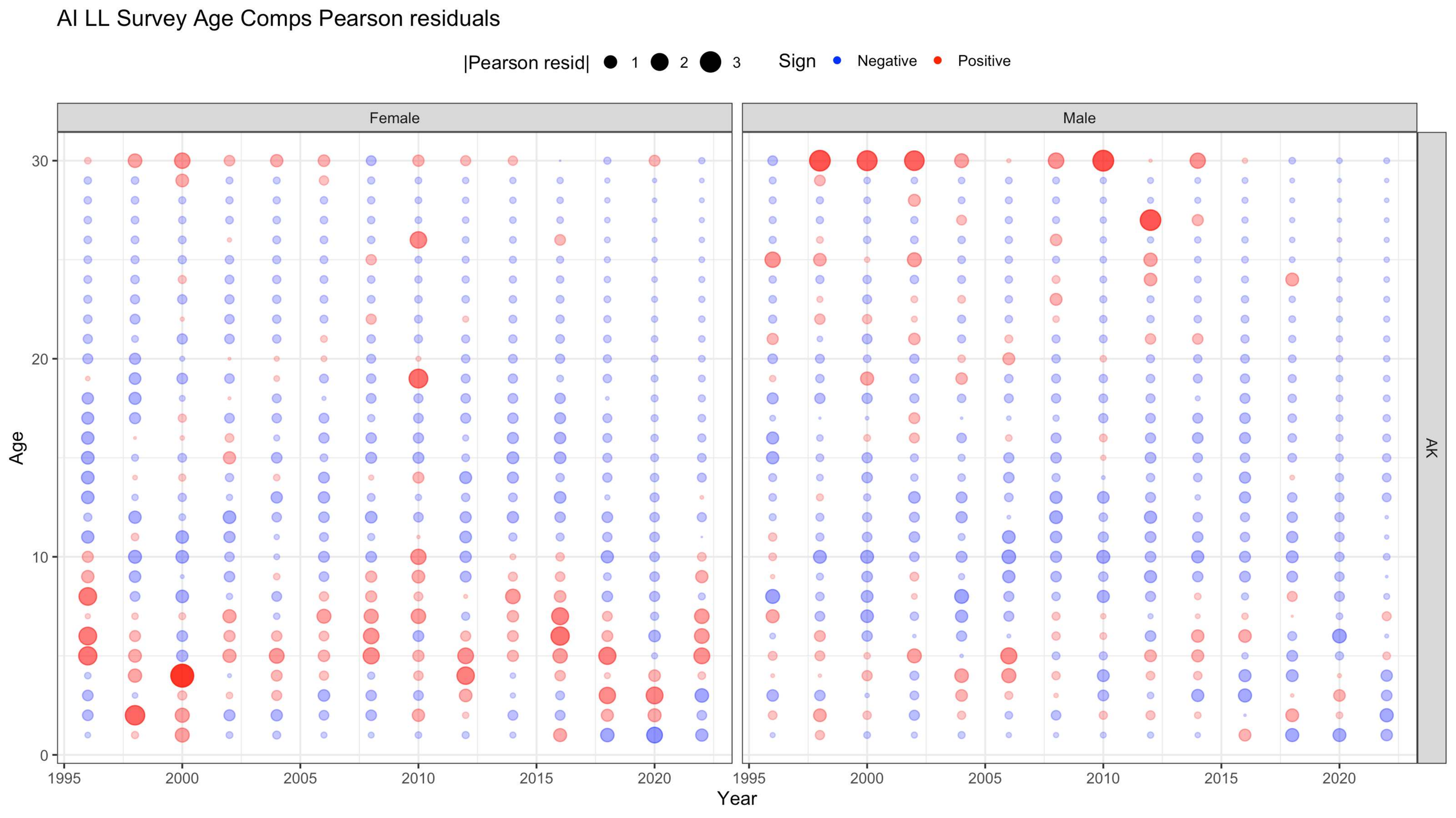
AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK




AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

Model 26.2_FAA_3LLF_LLS_BS_Dome_DN Population Population 1

Female | Male | AK

Spawning Weight-at-Age (kg)

0.100
0.075
0.050
0.025
0.000

0 10 20 30

Age

Model 26.2_FAA_3LLF_LLS_BS_Dome_DN Population Population 1

Female | Male | AK

Maturity at Age

1.00
0.75
0.50
0.25
0.00

0 10 20 30

Age

Model 26.2_FAA_3LLF_LLS_BS_Dome_DN Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

2

0

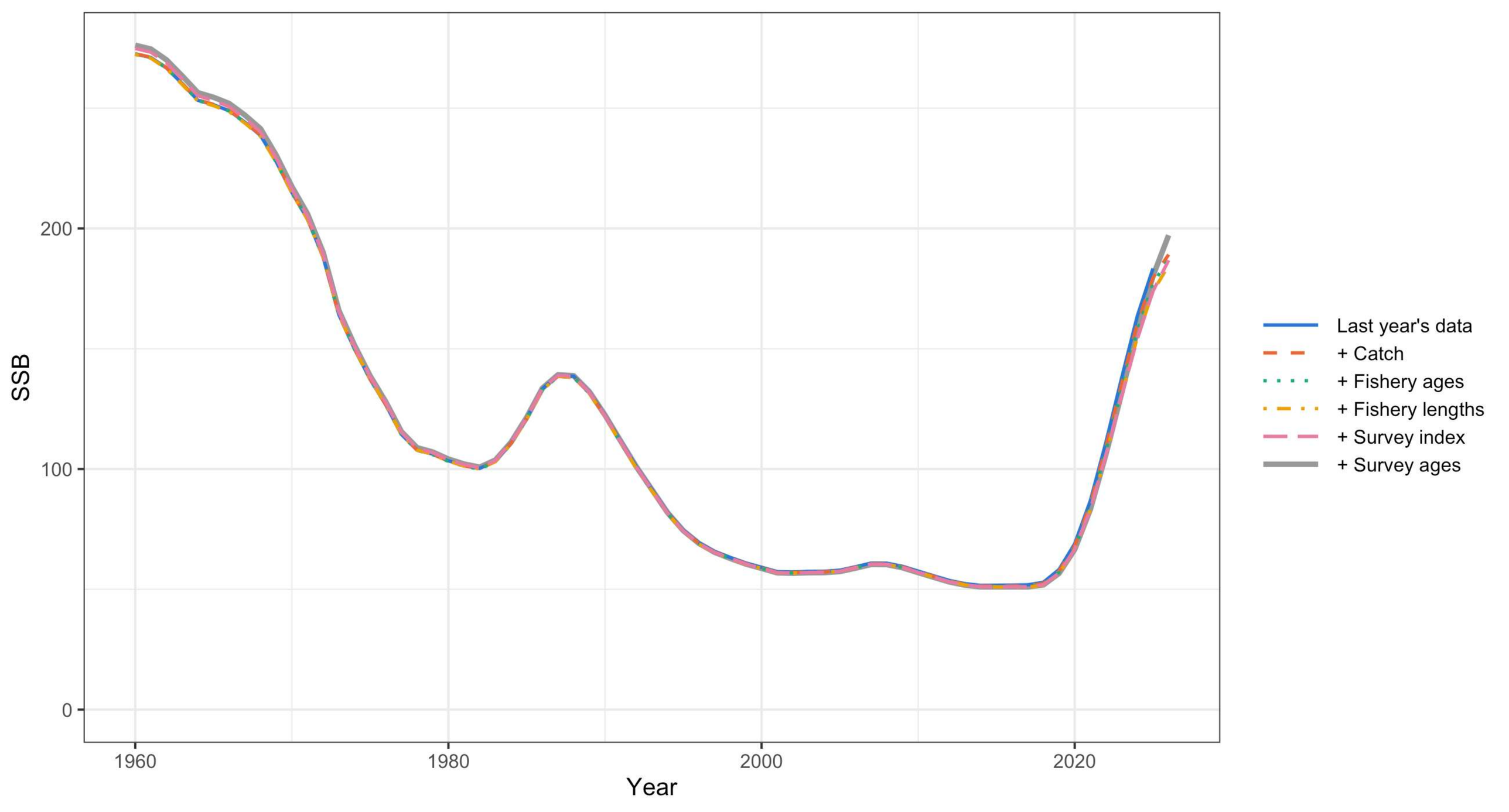
0

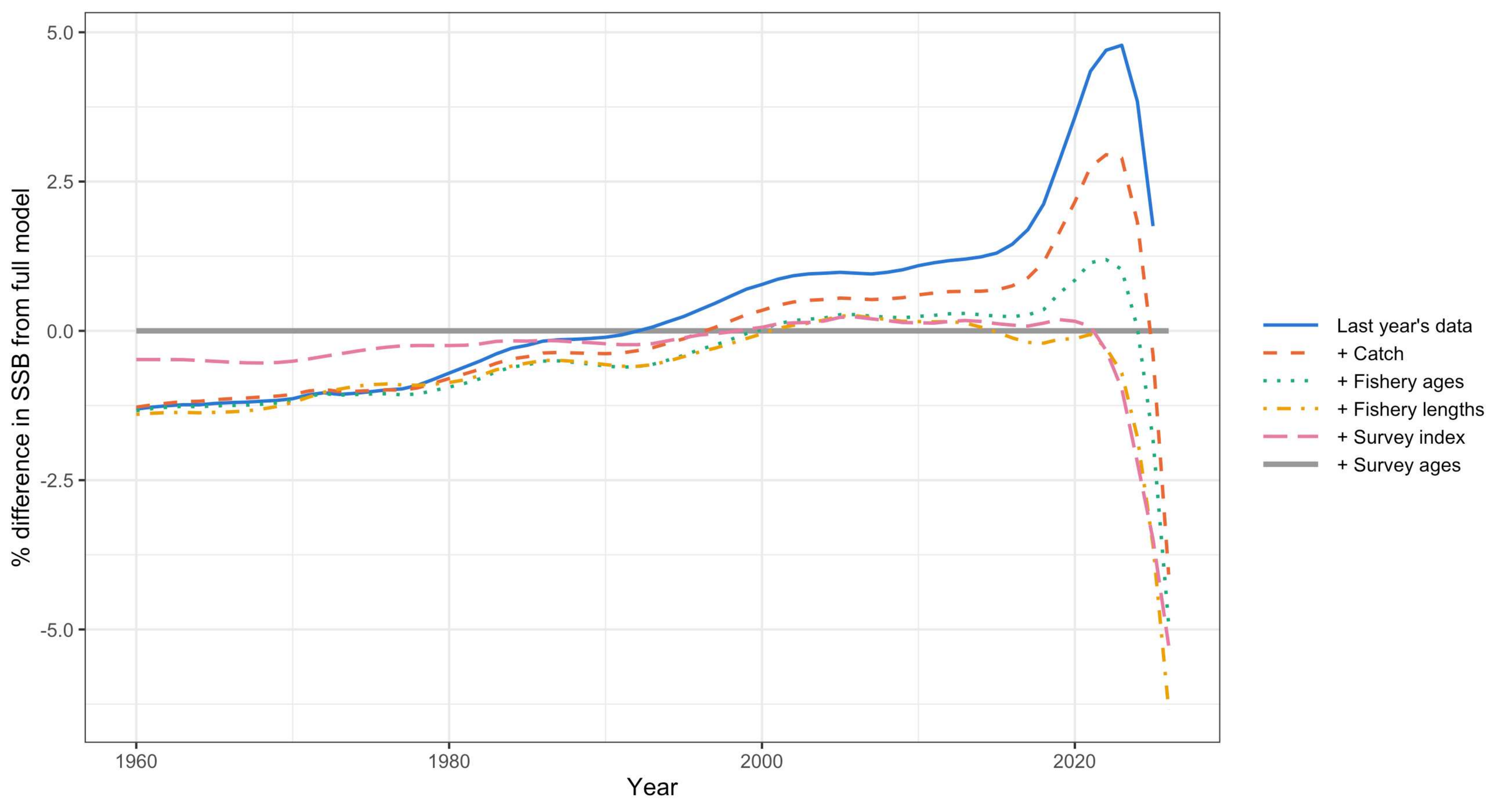
10

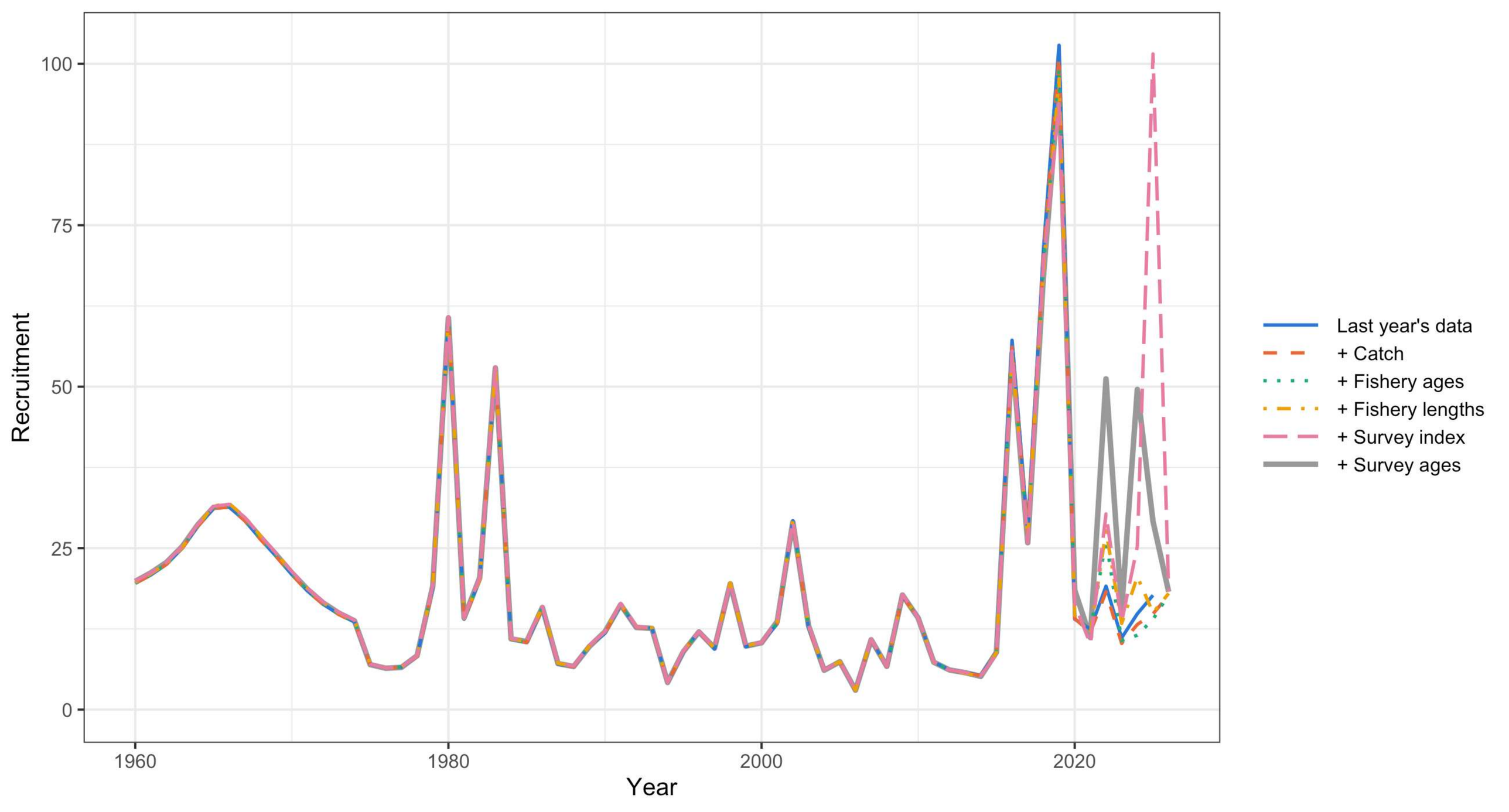
20

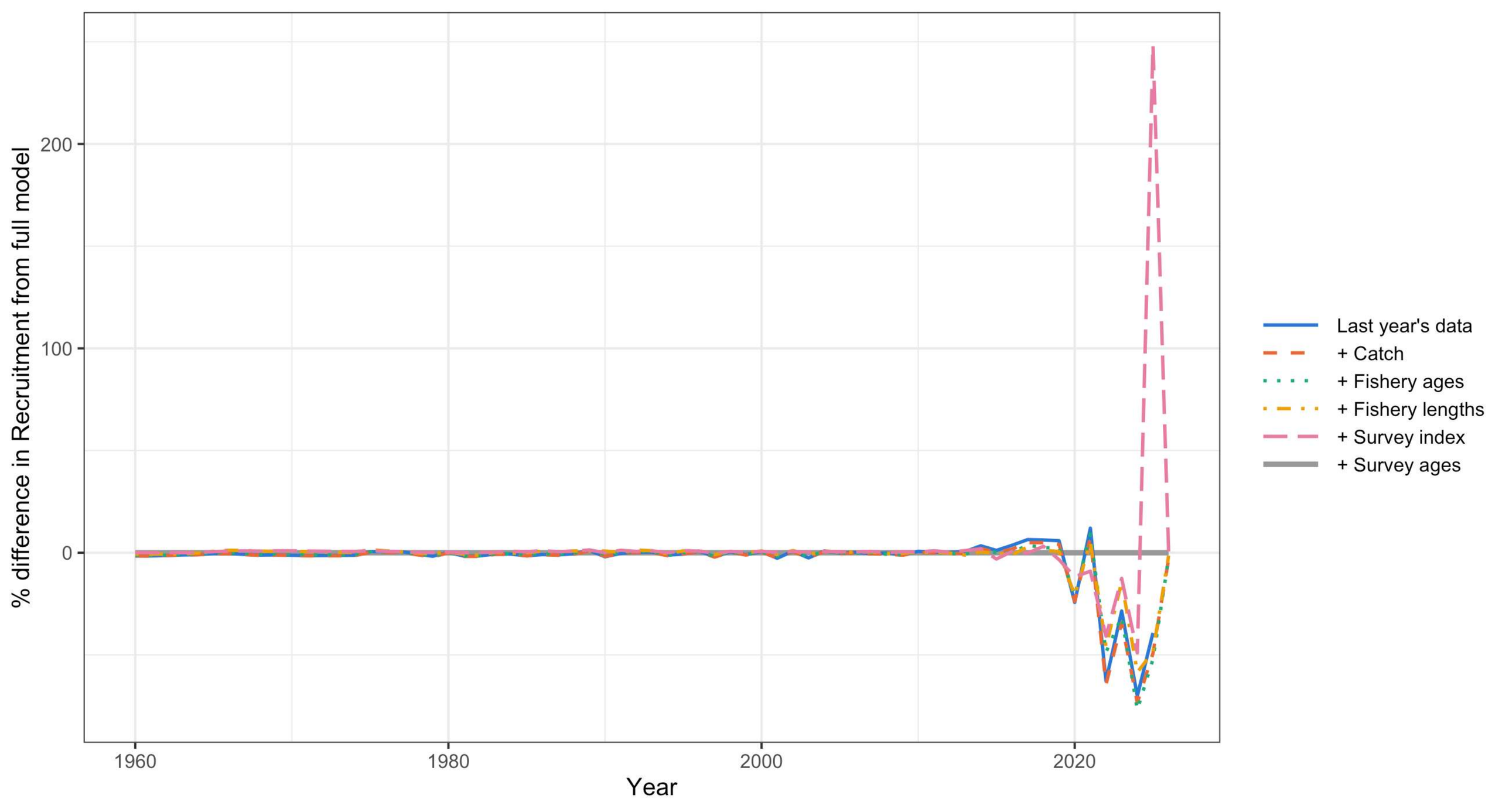
30

Age

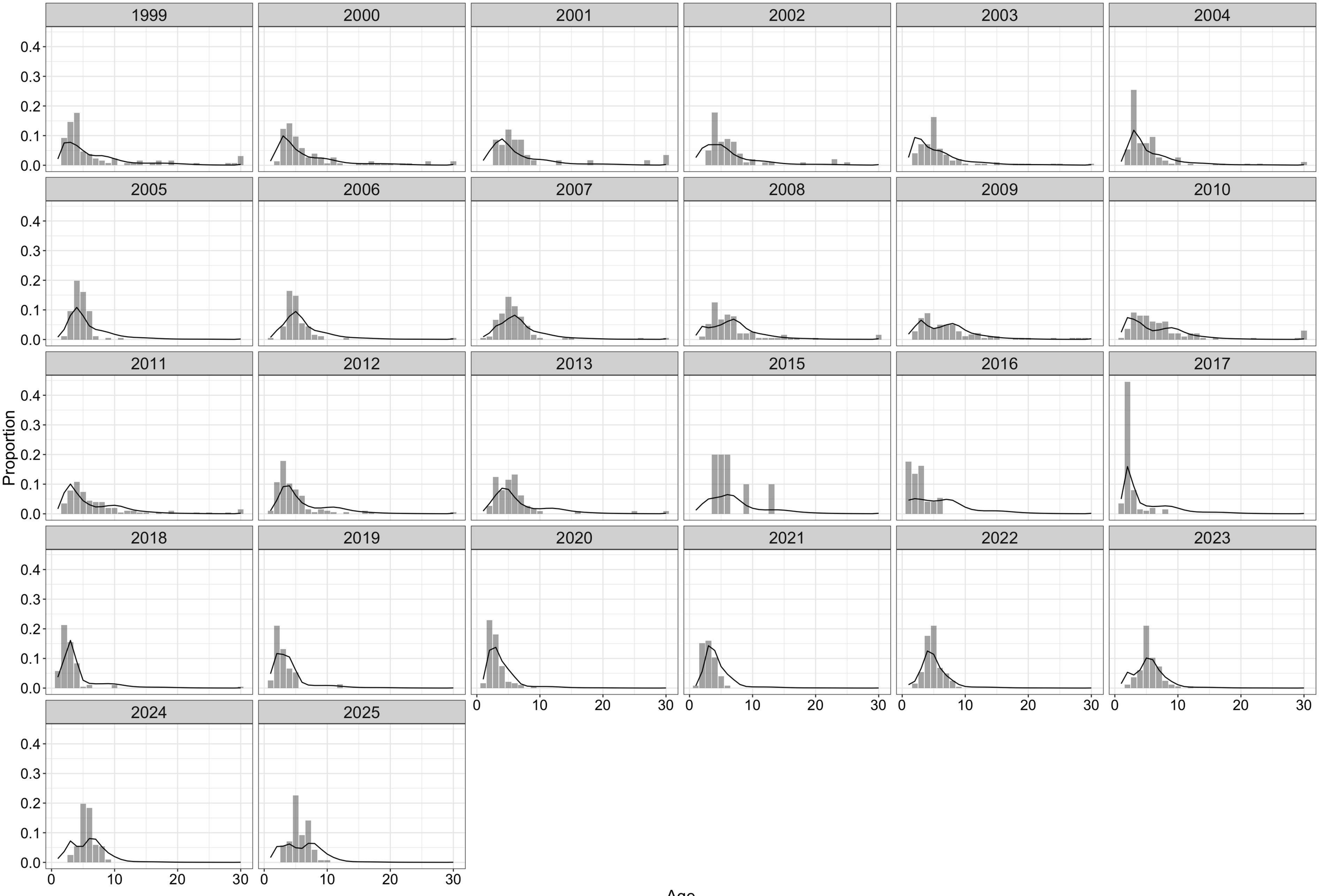

SSB
200
100
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


5.0
2.5
0.0
-2.5
-5.0
% difference in SSB from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

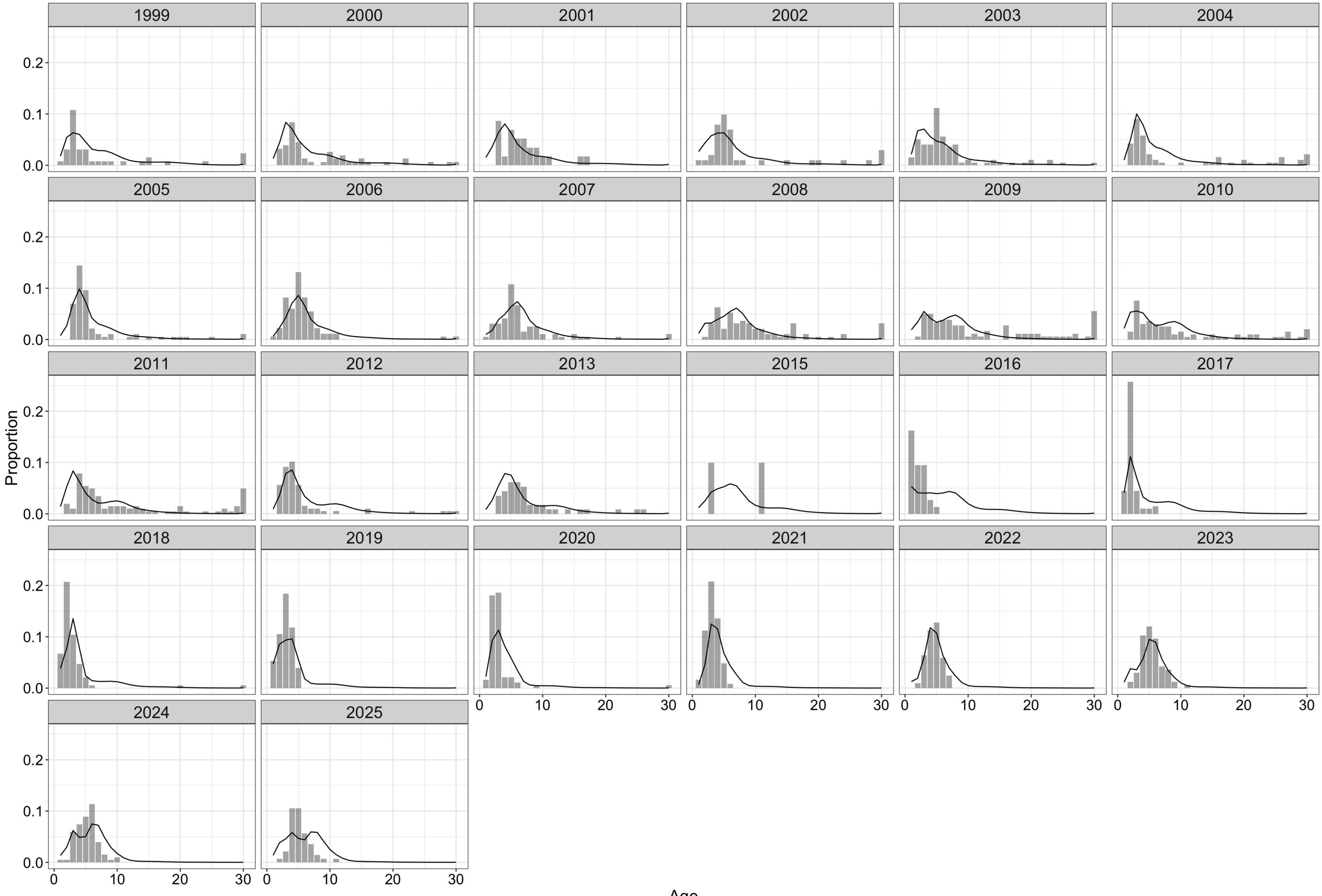

Recruitment
100
75
50
25
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


% difference in Recruitment from full model
200
100
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

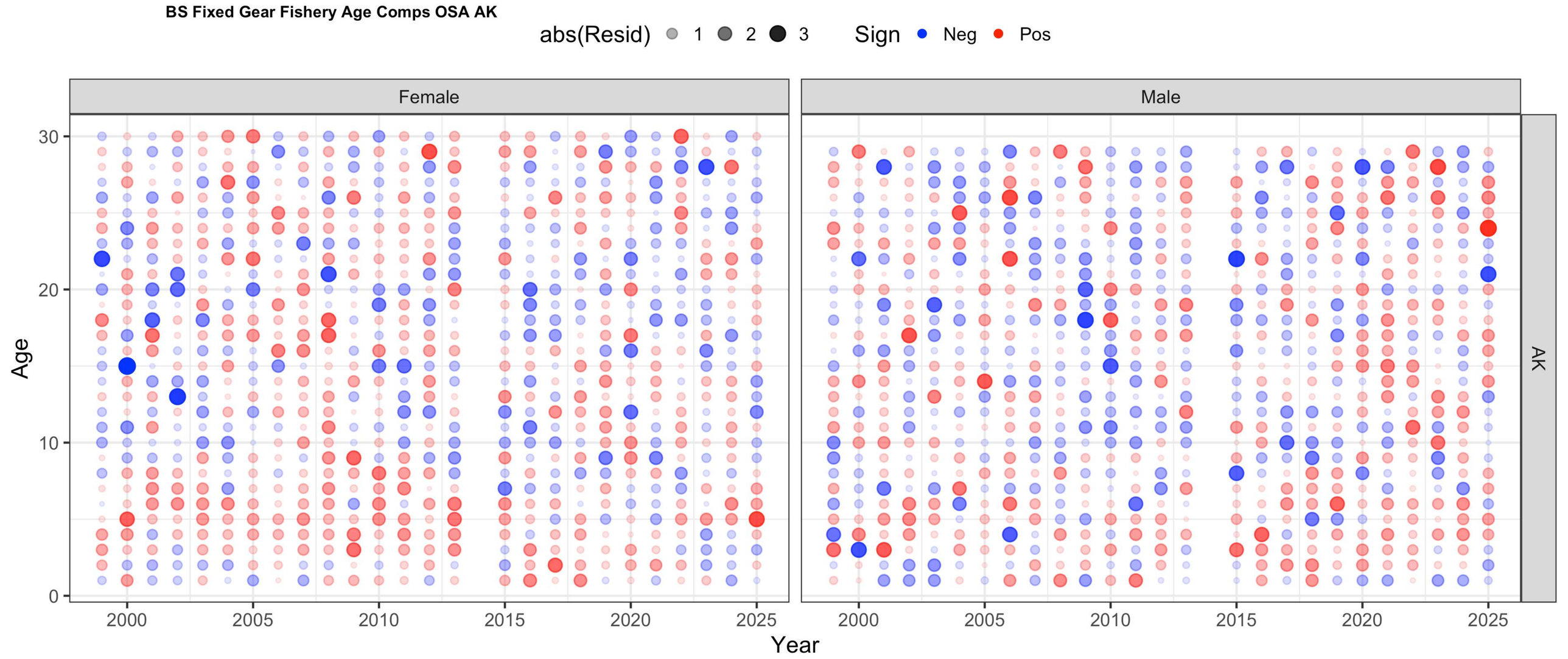
BS Fixed Gear Fishery Age Comps Female AK

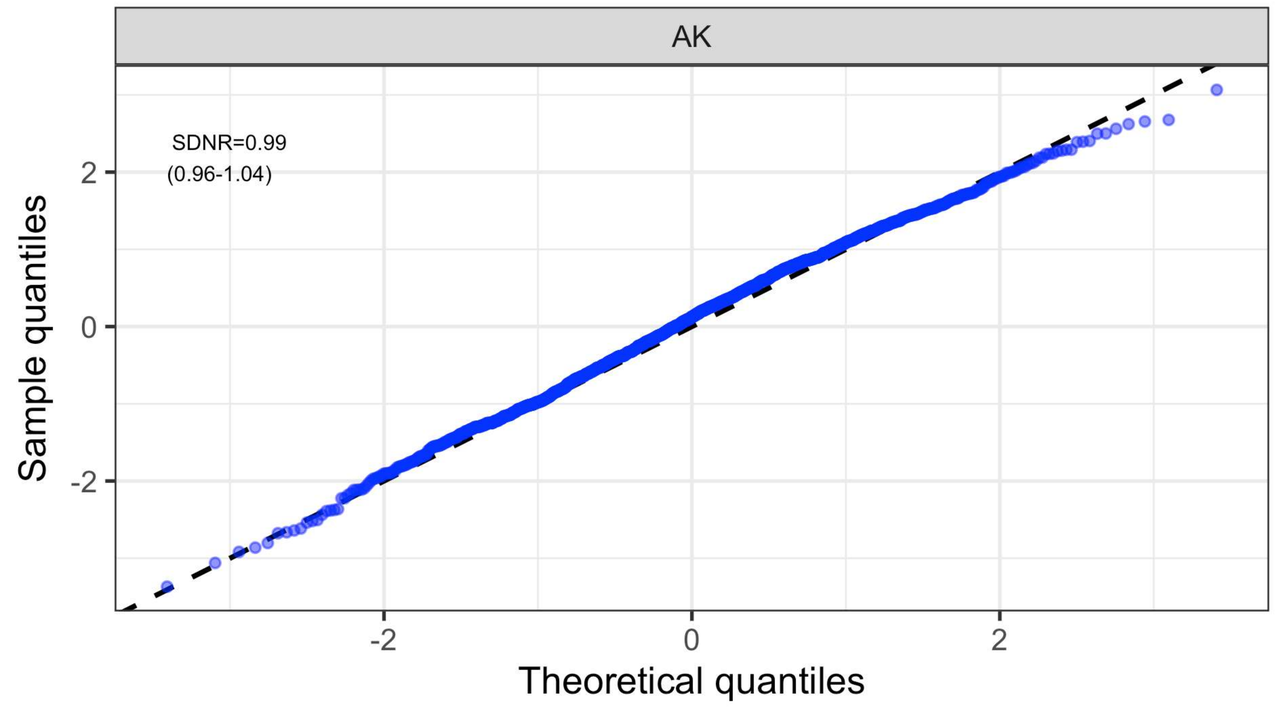


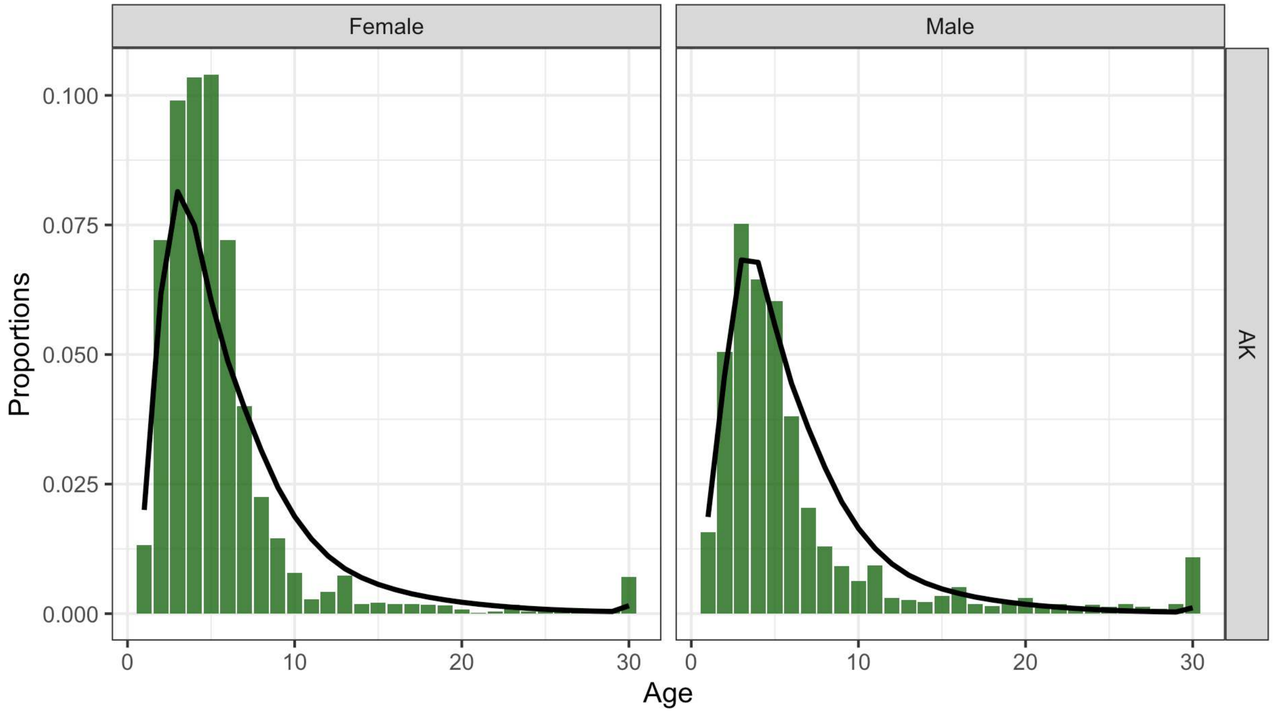
BS Fixed Gear Fishery Age Comps Male AK

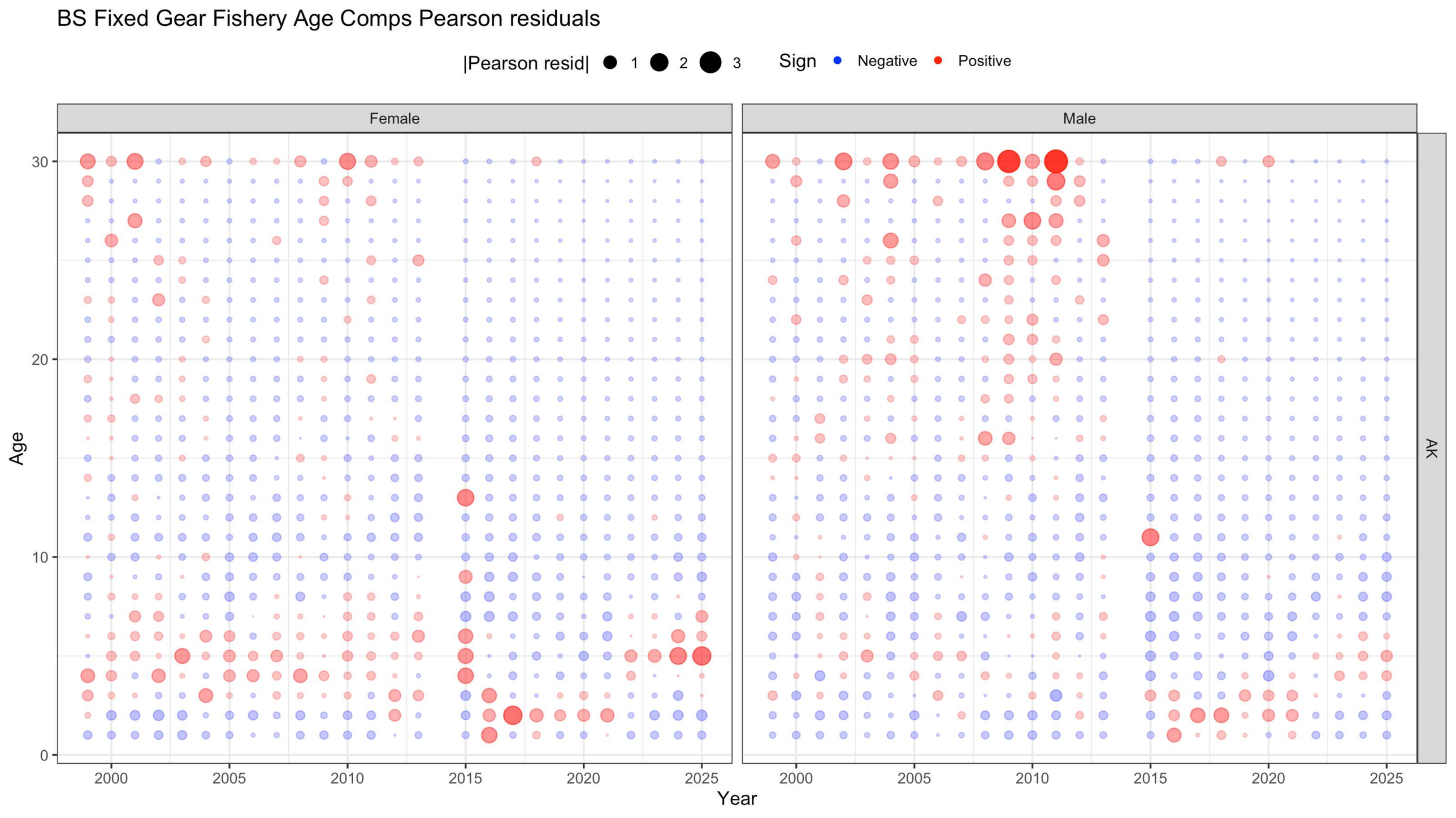


BS Fixed Gear Fishery Age Comps OSA AK








BS Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

BS LL Survey Age Comps Female AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
Age

BS LL Survey Age Comps Male AK

1997 1999 2001 2003

2005 2007 2009 2011

2013 2015 2017 2019

2021 2023

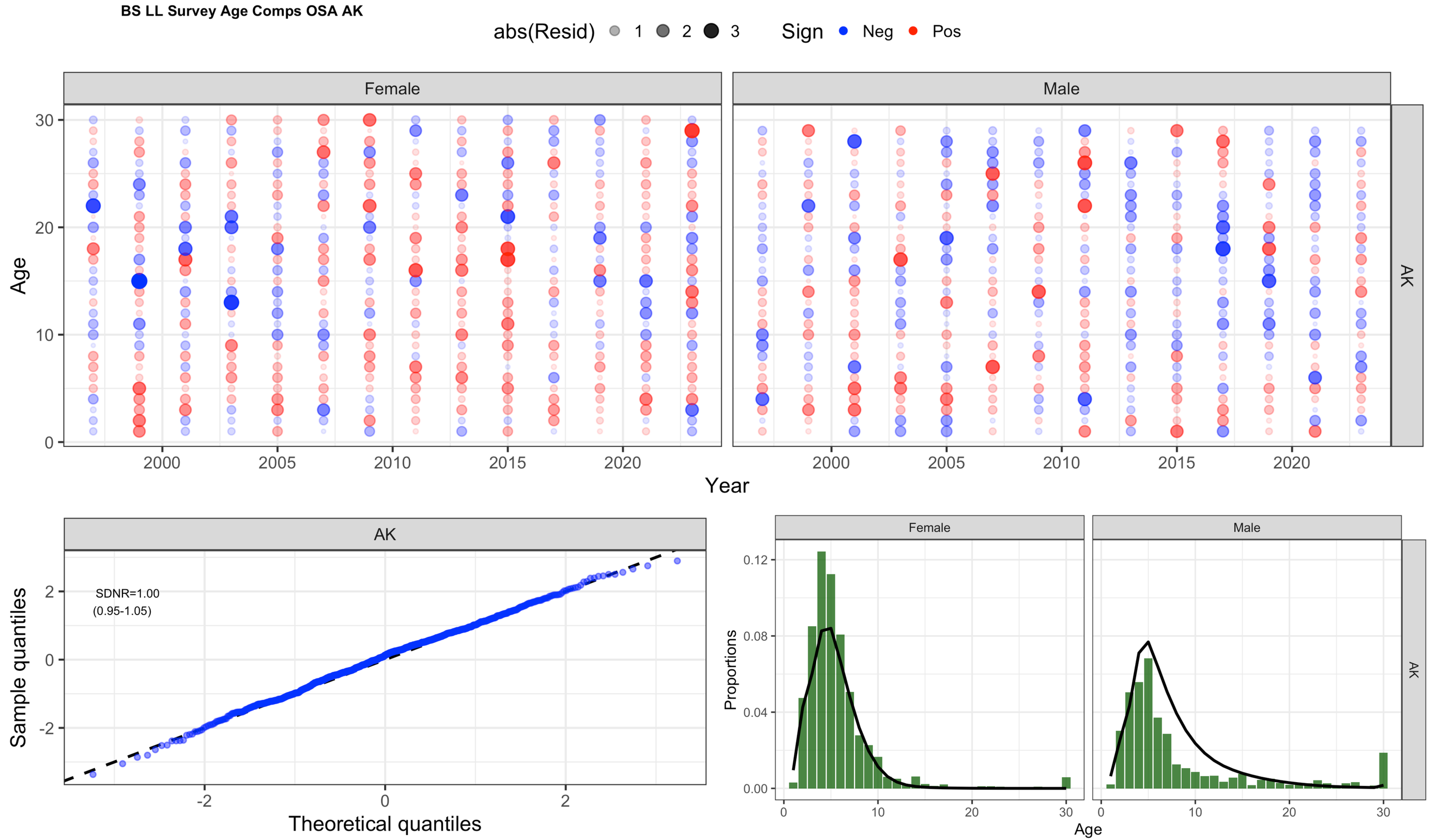
Proportion

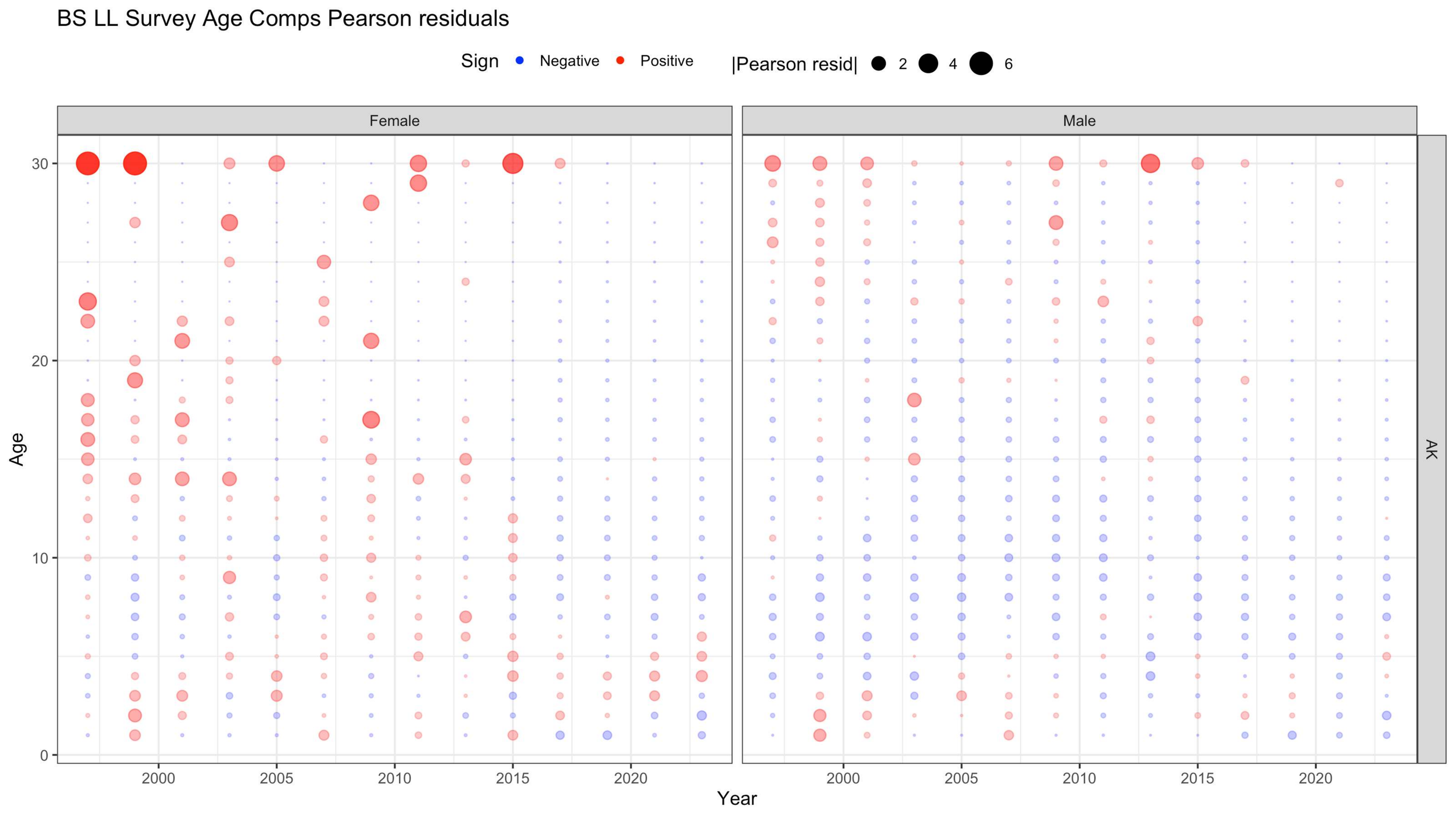
0.00 0.05 0.10 0.15

0 10 20 30

Age

BS LL Survey Age Comps OSA AK




BS LL Survey Age Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
2
4
6
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

Model — 26.2_FAA_3LLF_LLS_BS_Dome_DN

AK

BS LLF

AI LLF

GOA LLF

TF

Catch

Year

AK

26.2_FAA_3LLF_LLS_BS_Dome_DN

JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
GOA LLF Catch
AI LLF Catch
BS LLF Catch
GOA LLF Ages
AI LLF Ages
BS LLF Ages

1960
1980
2000
2020

Year

GOA Fixed Gear Fishery Age Comps Female AK

1999 2000 2001 2002 2003 2004
2005 2006 2007 2008 2009 2010
2011 2012 2013 2014 2015 2016
2017 2018 2019 2020 2021 2022
2023 2024 2025

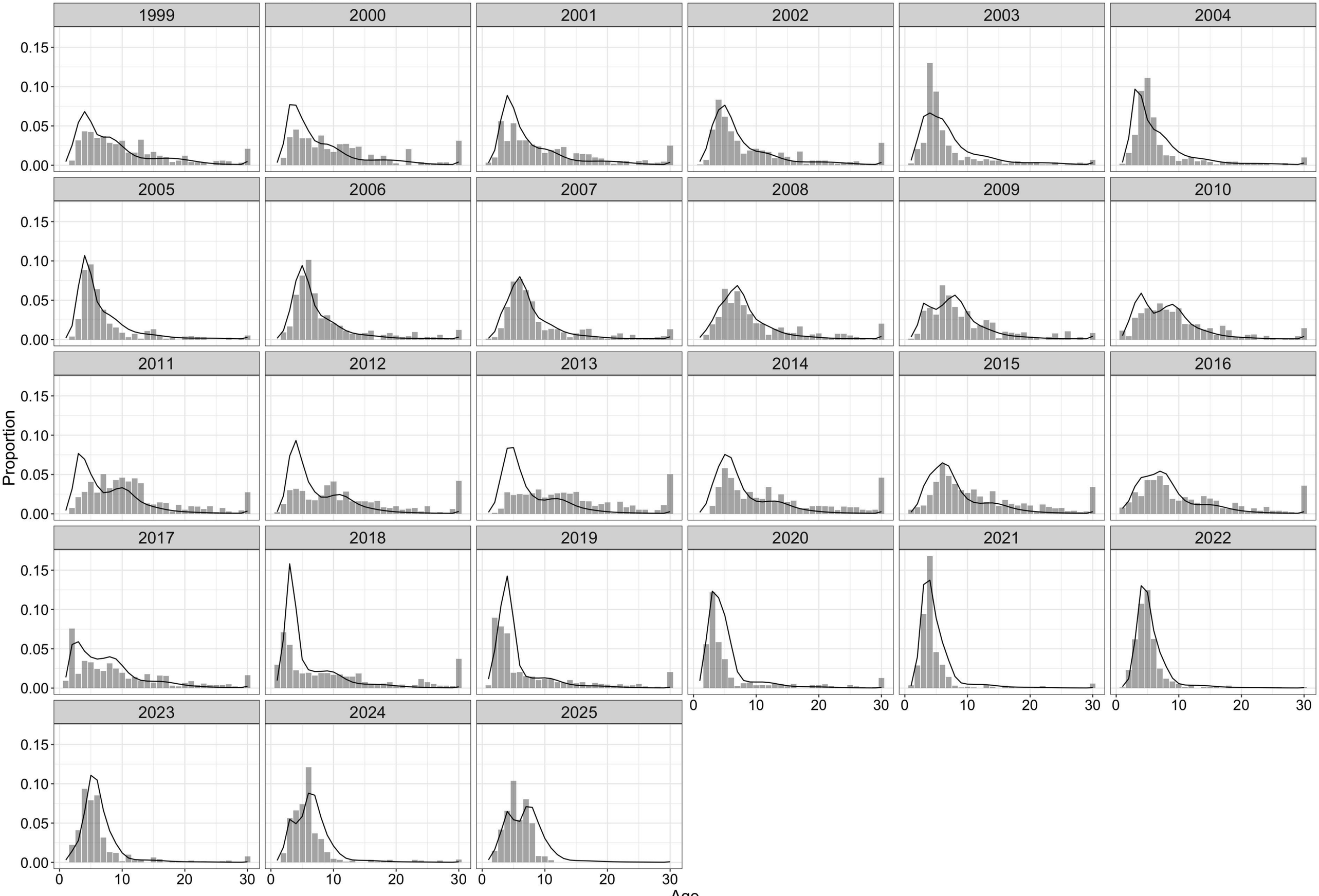
Proportion

0.0 0.1 0.2

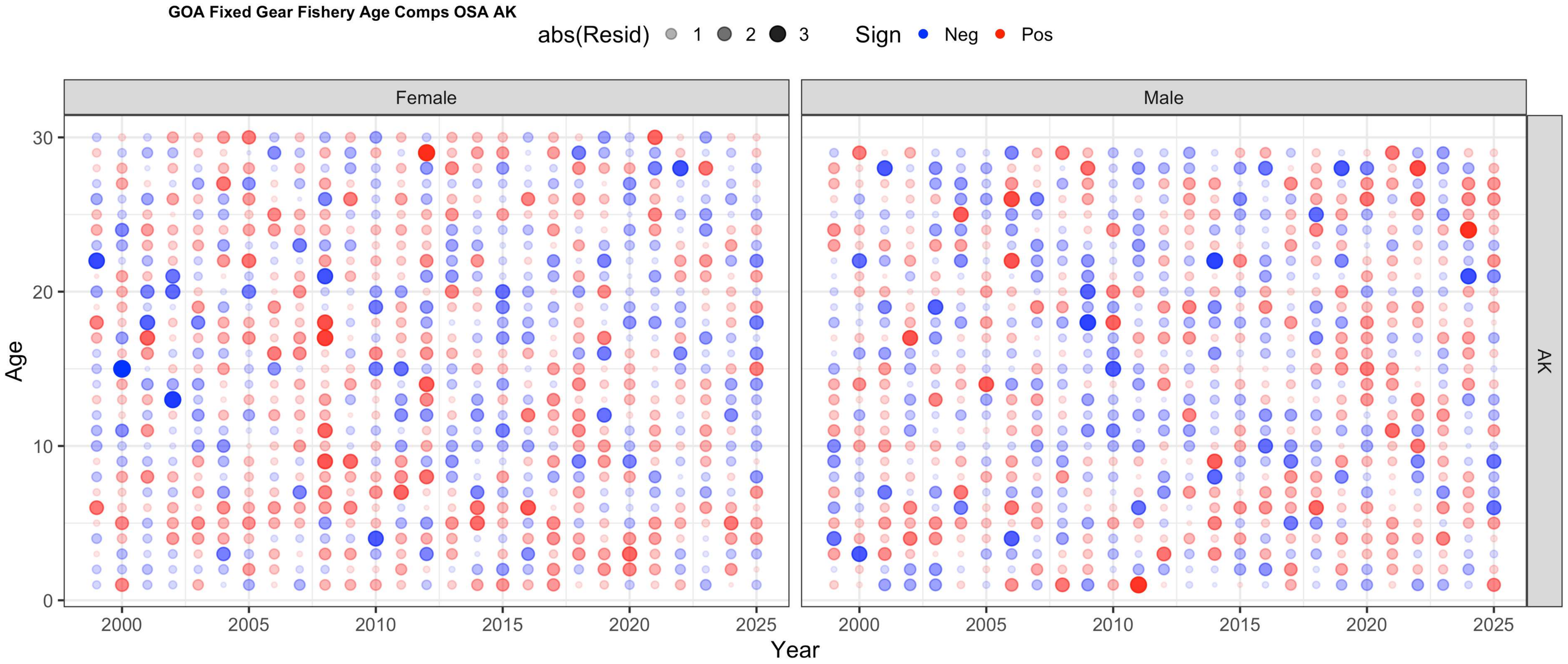
0 10 20 30

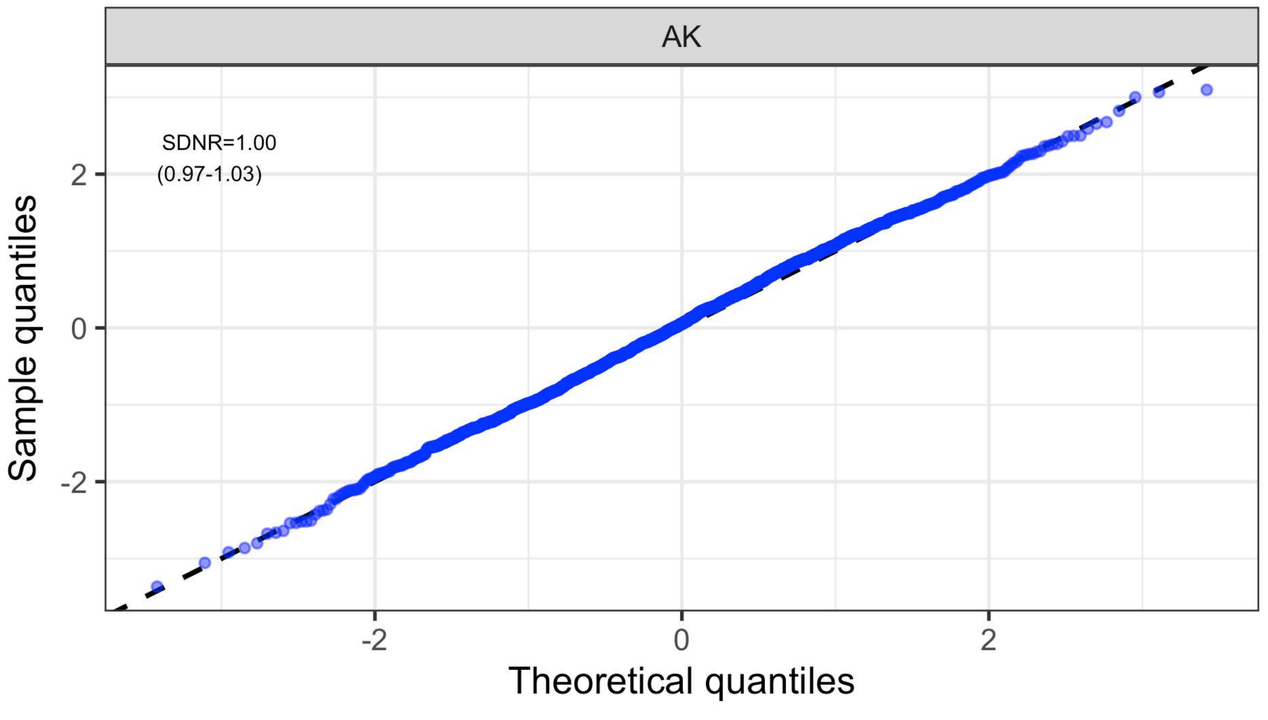
Age

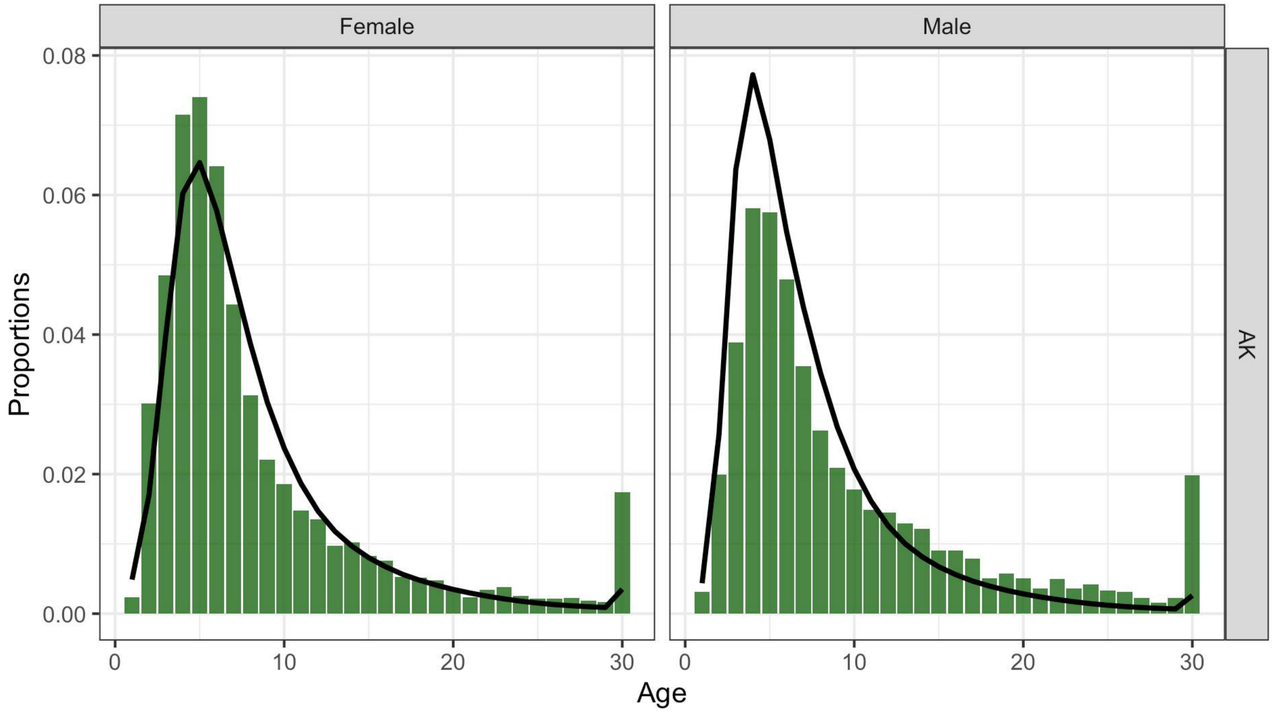
GOA Fixed Gear Fishery Age Comps Male AK

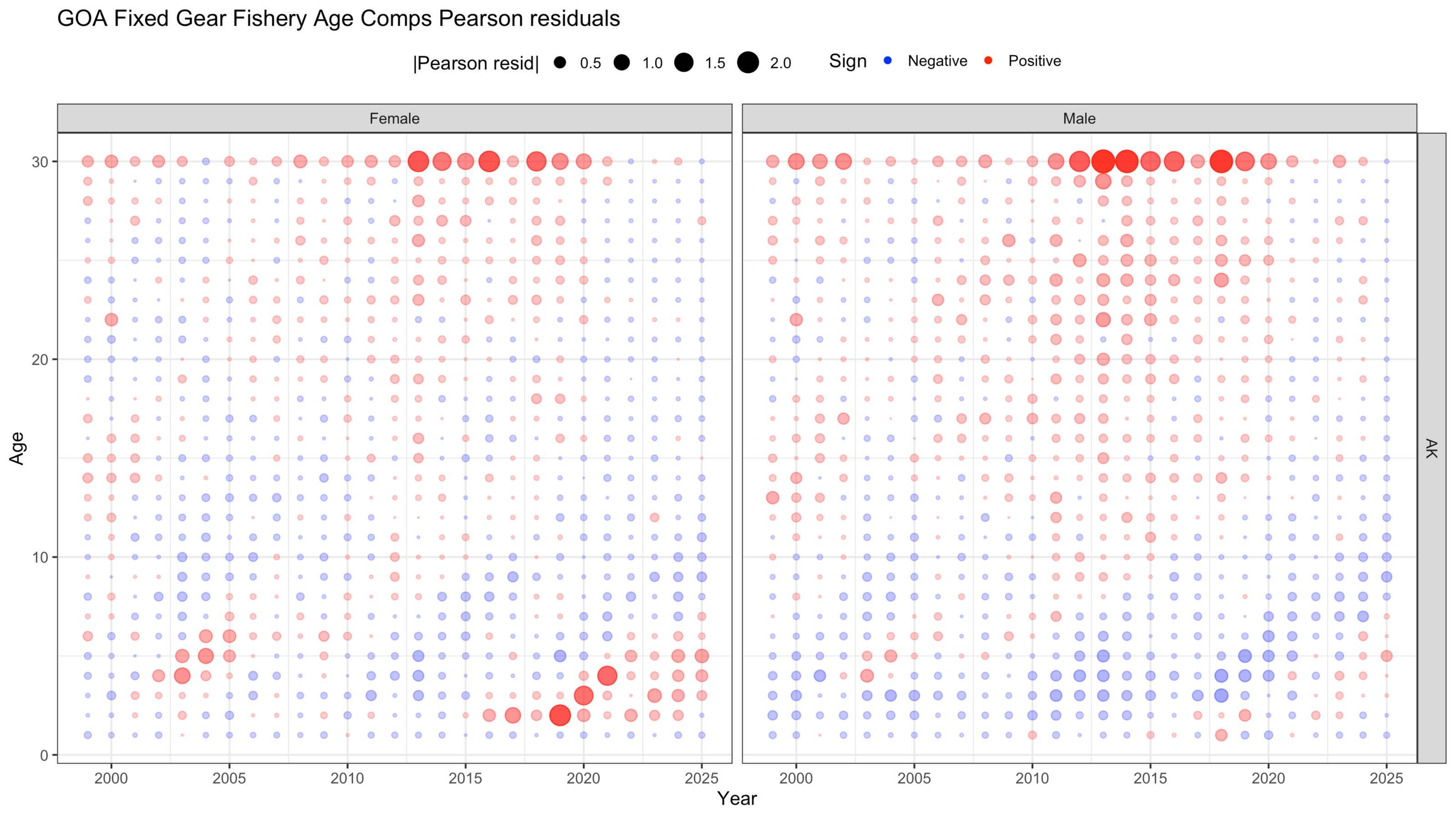


GOA Fixed Gear Fishery Age Comps OSA AK








GOA Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

GOA LL Survey Age Comps Female AK

1996 1997 1998 1999 2000 2001

2002 2003 2004 2005 2006 2007

2008 2009 2010 2011 2012 2013

2014 2015 2016 2017 2018 2019

2020 2021 2022 2023 2025

Proportion

0.0 0.1 0.2

0 10 20 30

Age

GOA LL Survey Age Comps Male AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

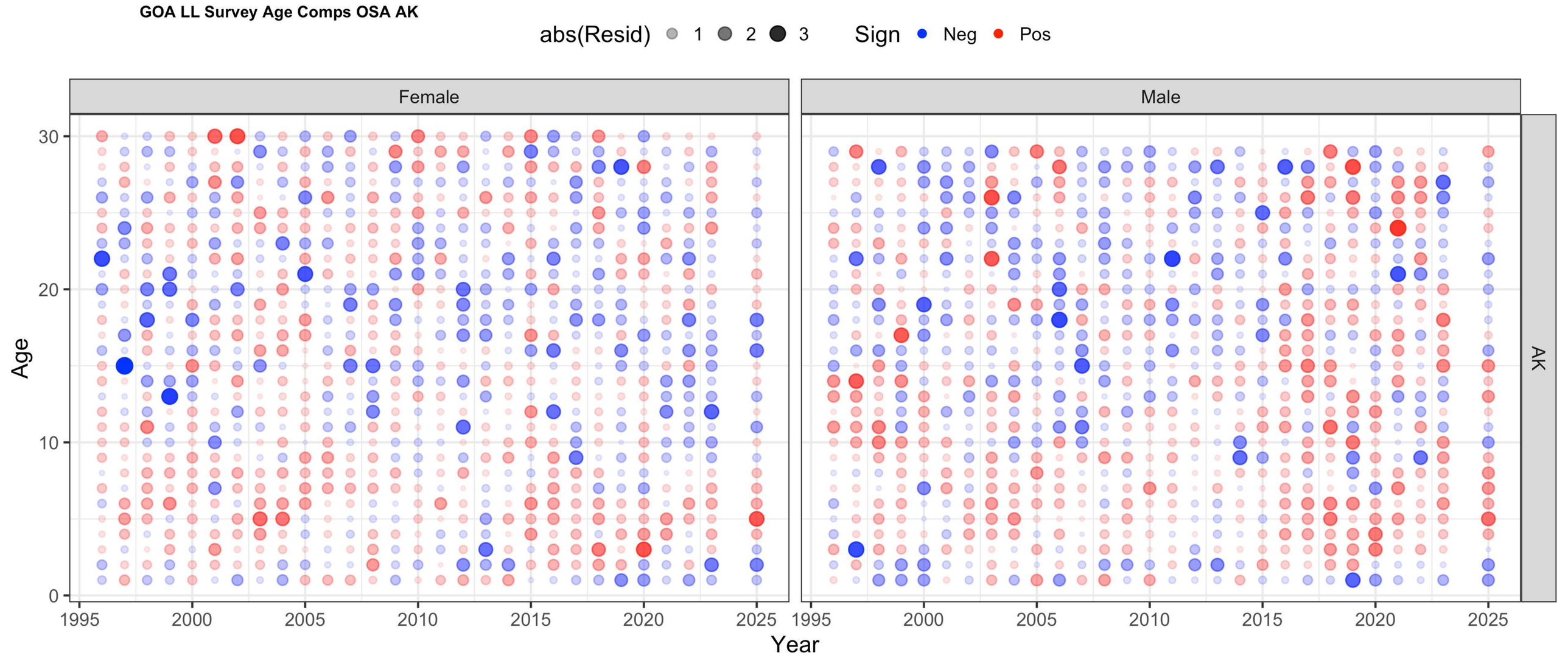
Proportion

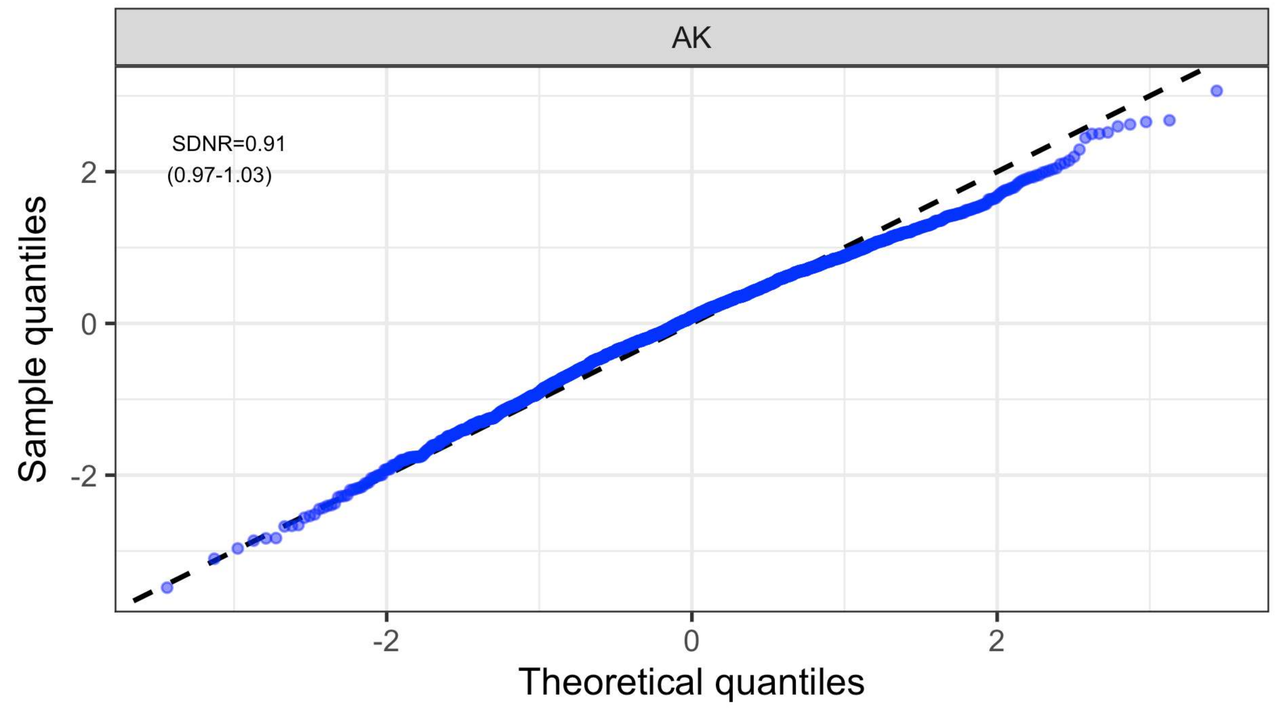
0.00 0.05 0.10 0.15

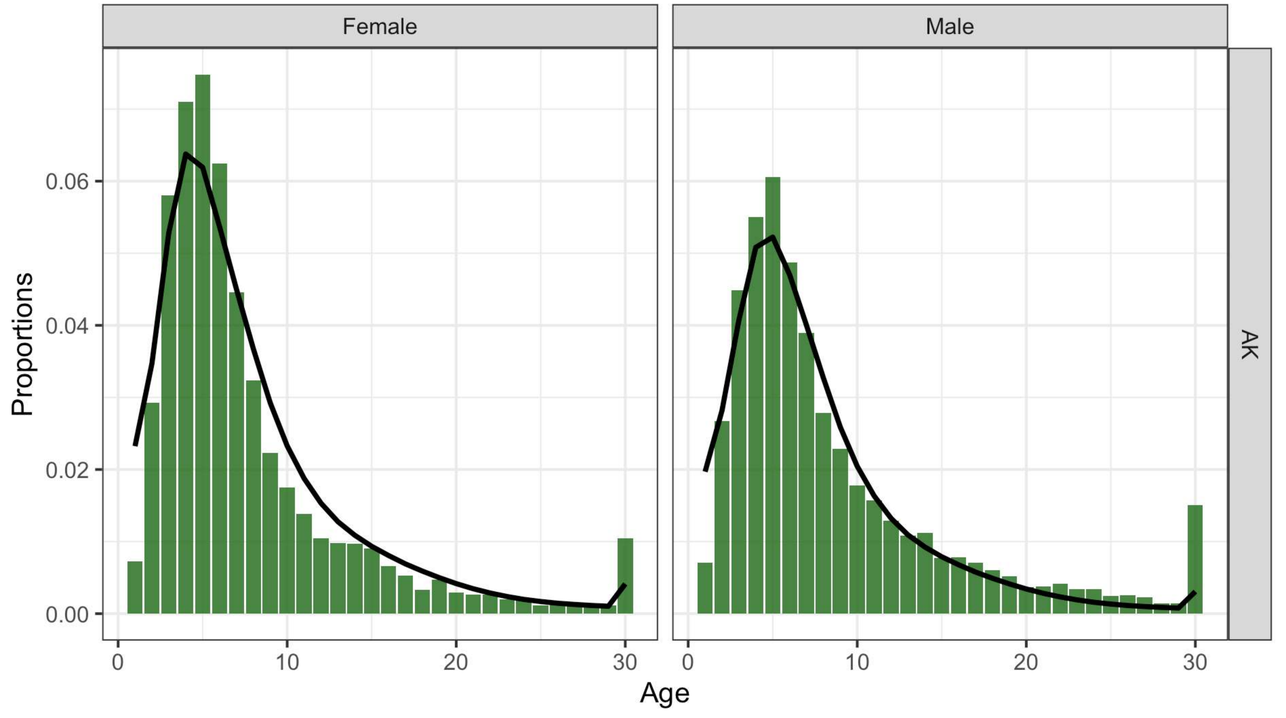
0 10 20 30

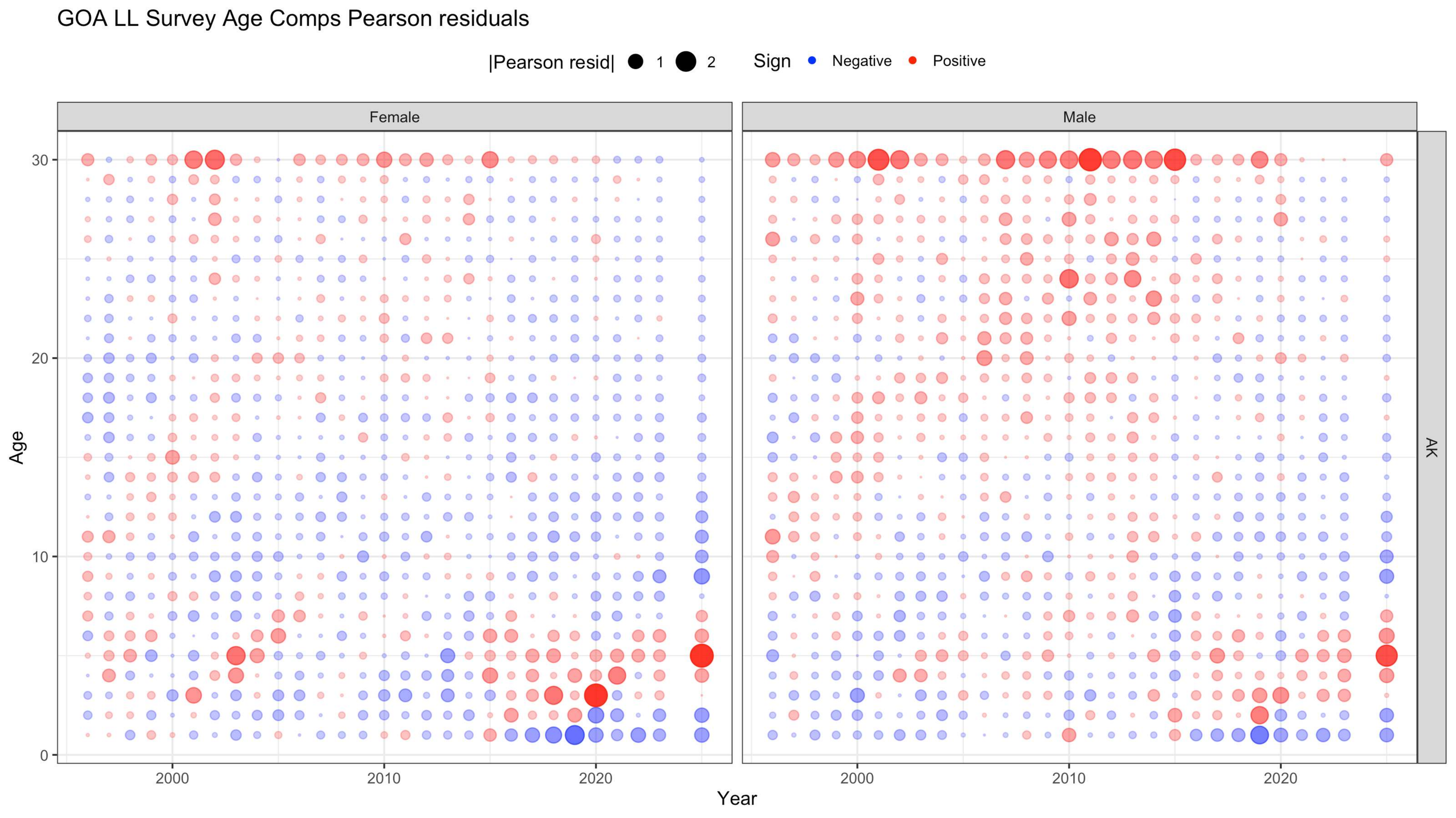
Age

GOA LL Survey Age Comps OSA AK








GOA LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

Model — 26.2_FAA_3LLF_LLS_BS_Dome_DN

BS LLS

AI LLS

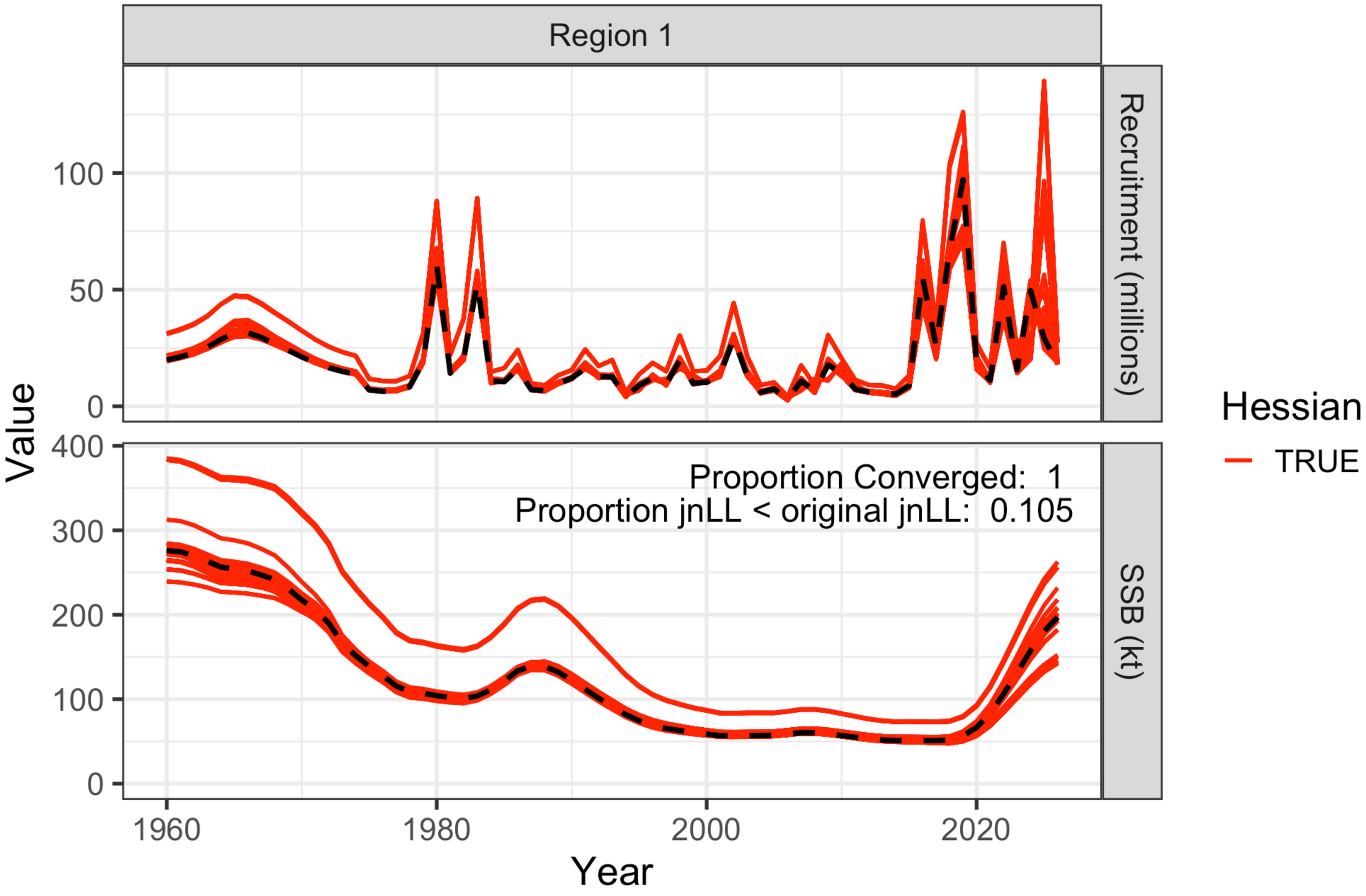
GOA LLS

JPN LLS

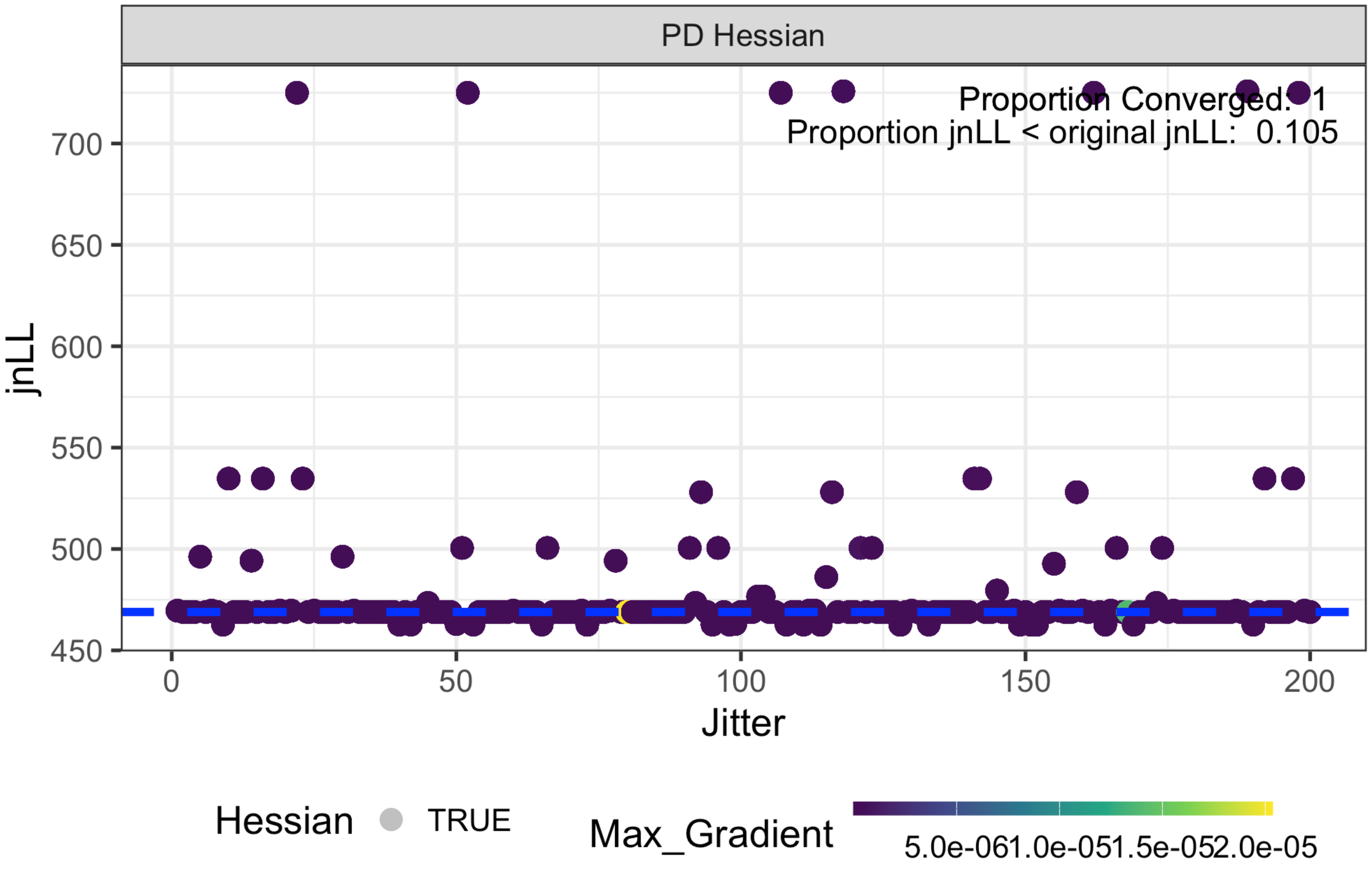
RPNs (1000s)

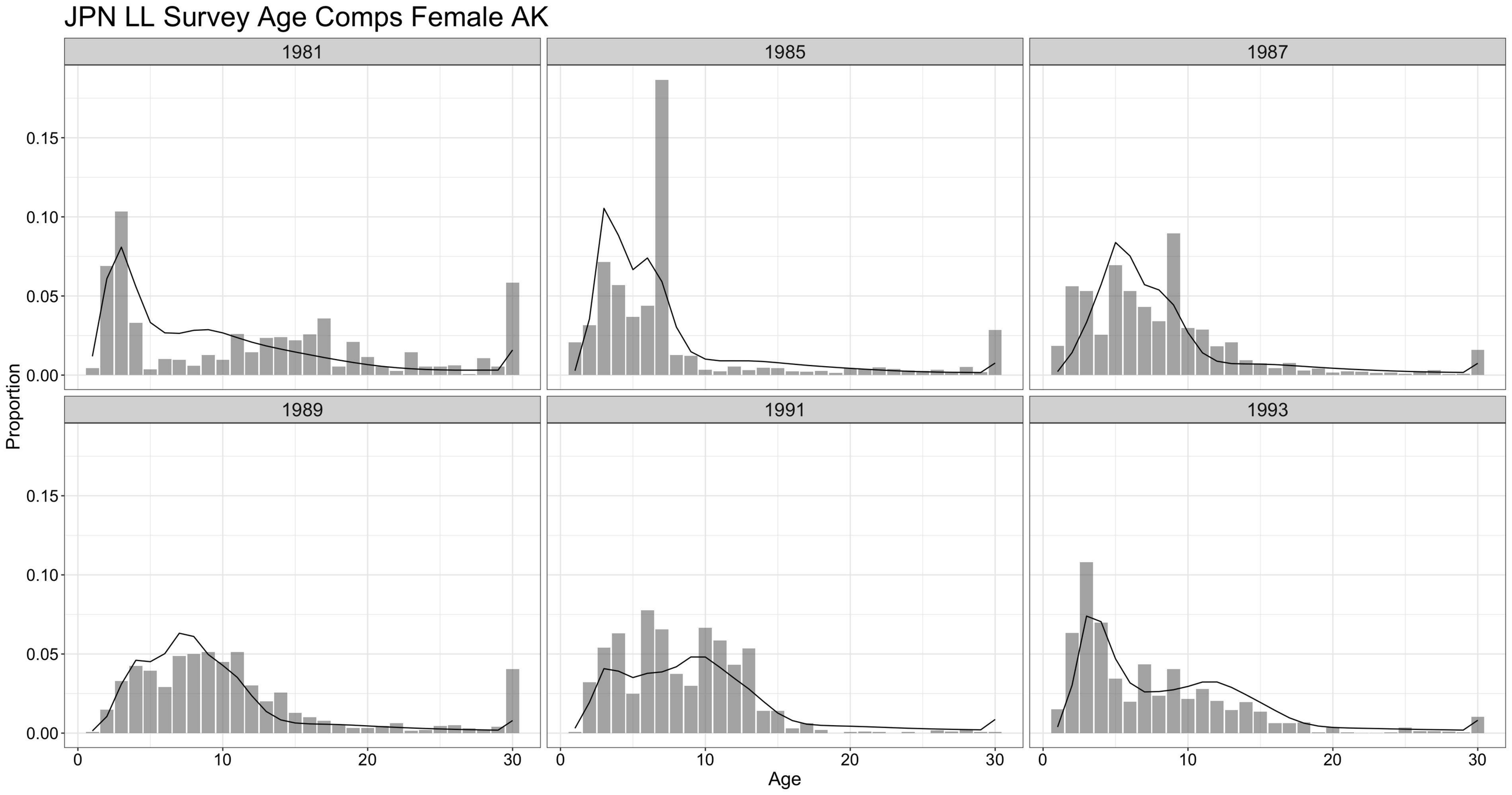
Year

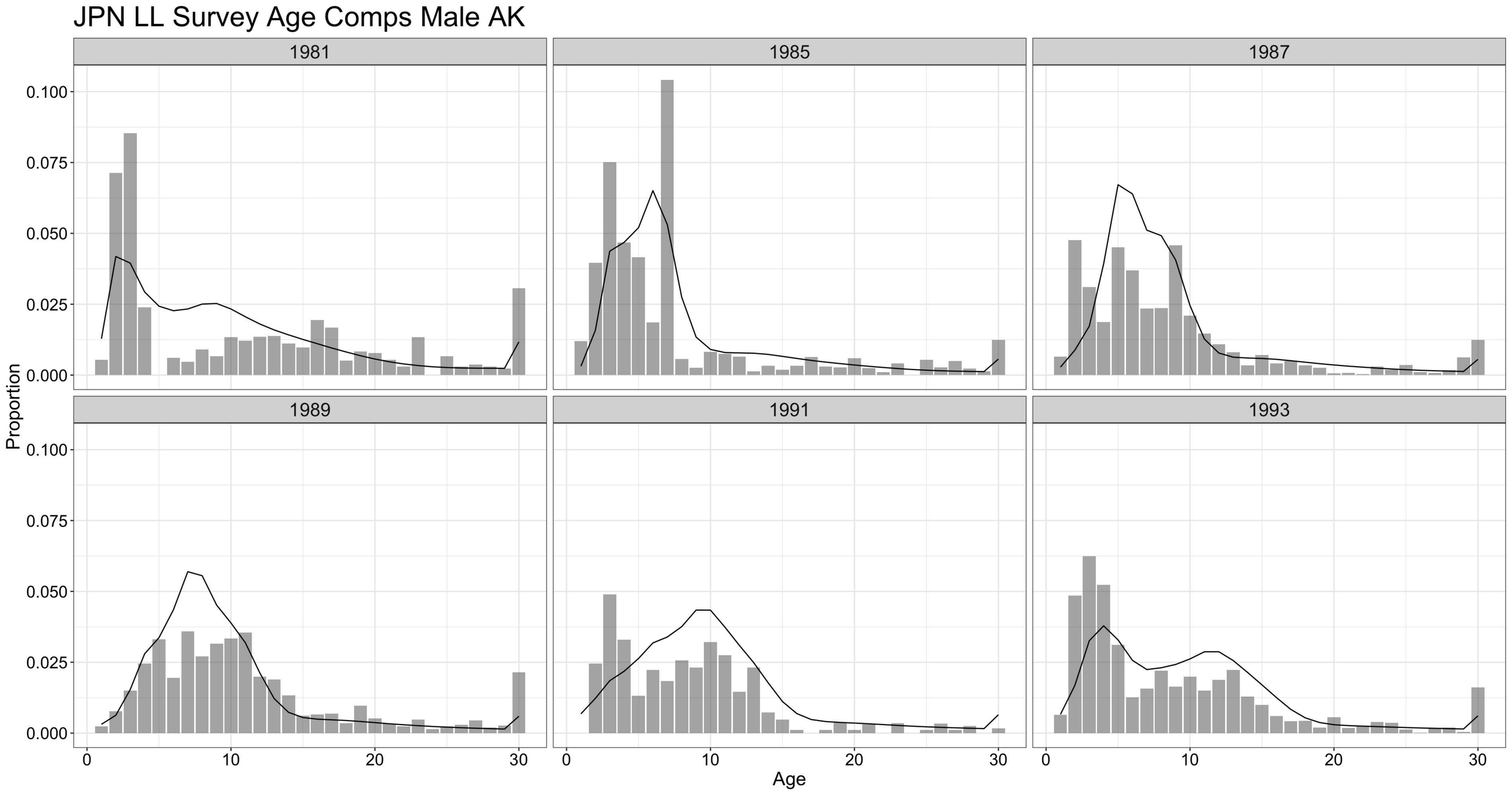
Jitter SSB and Recruitment



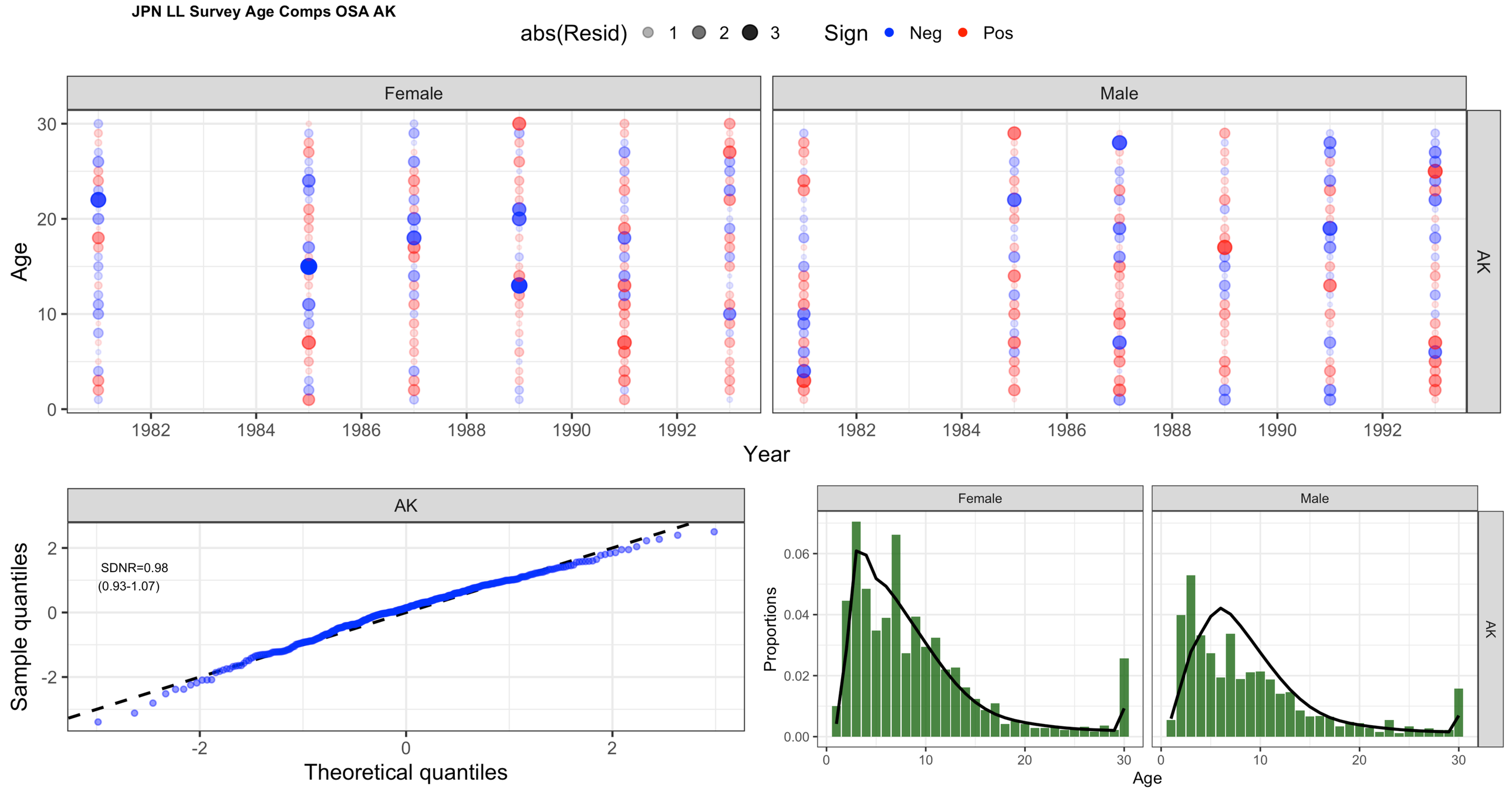
Jitter nLL and Convergence

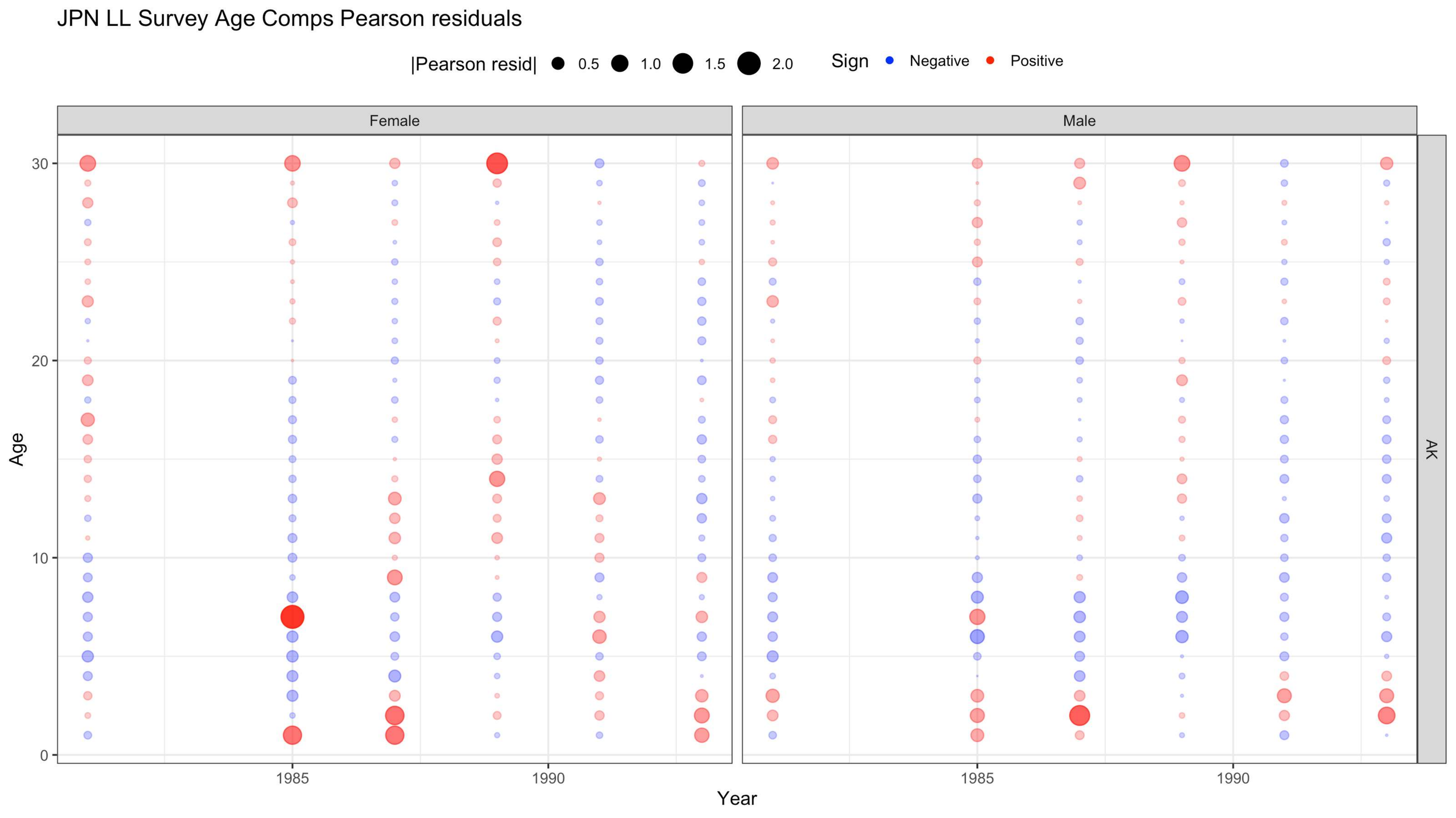



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

JPN LL Survey Age Comps OSA AK




JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

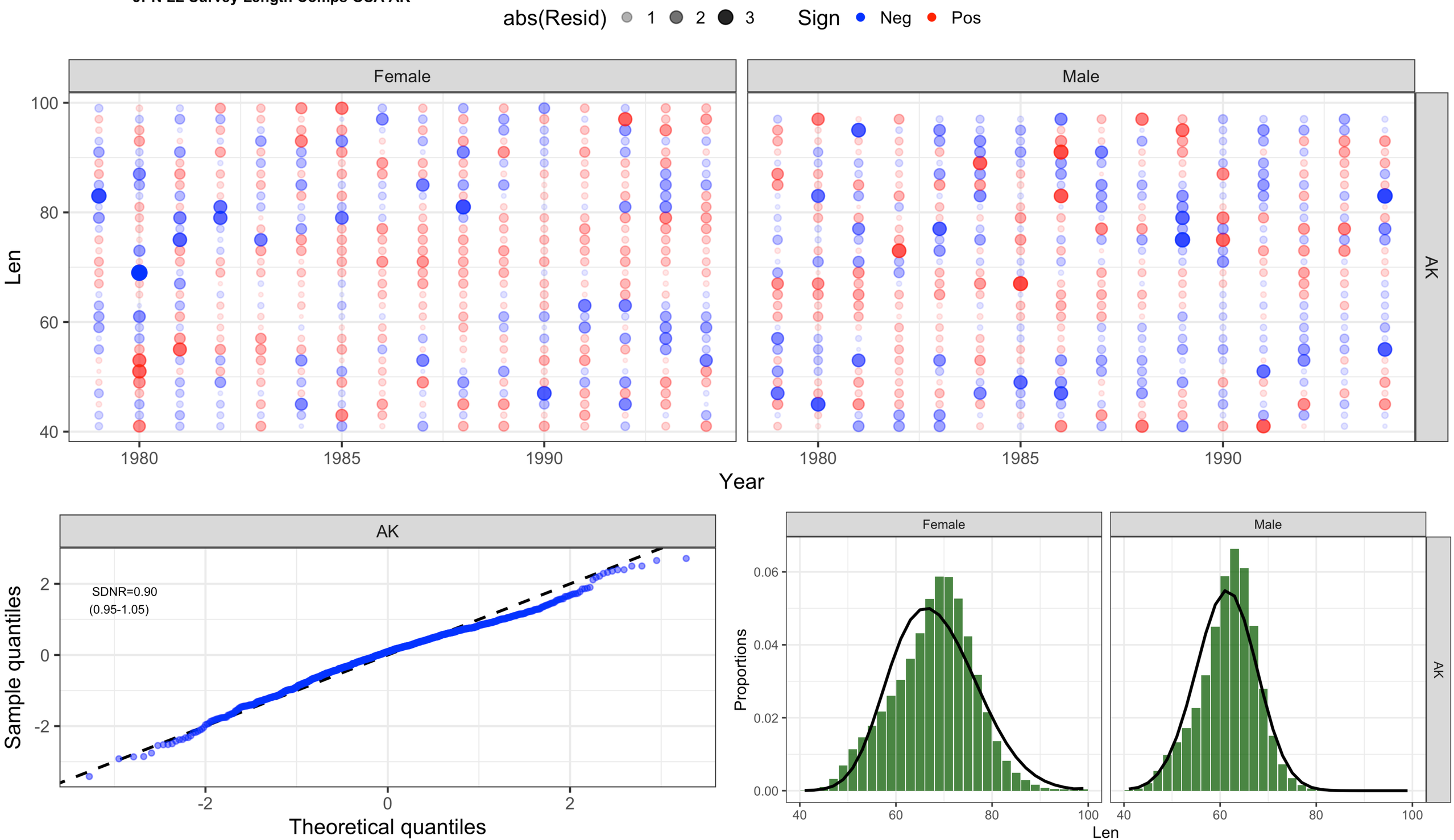
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

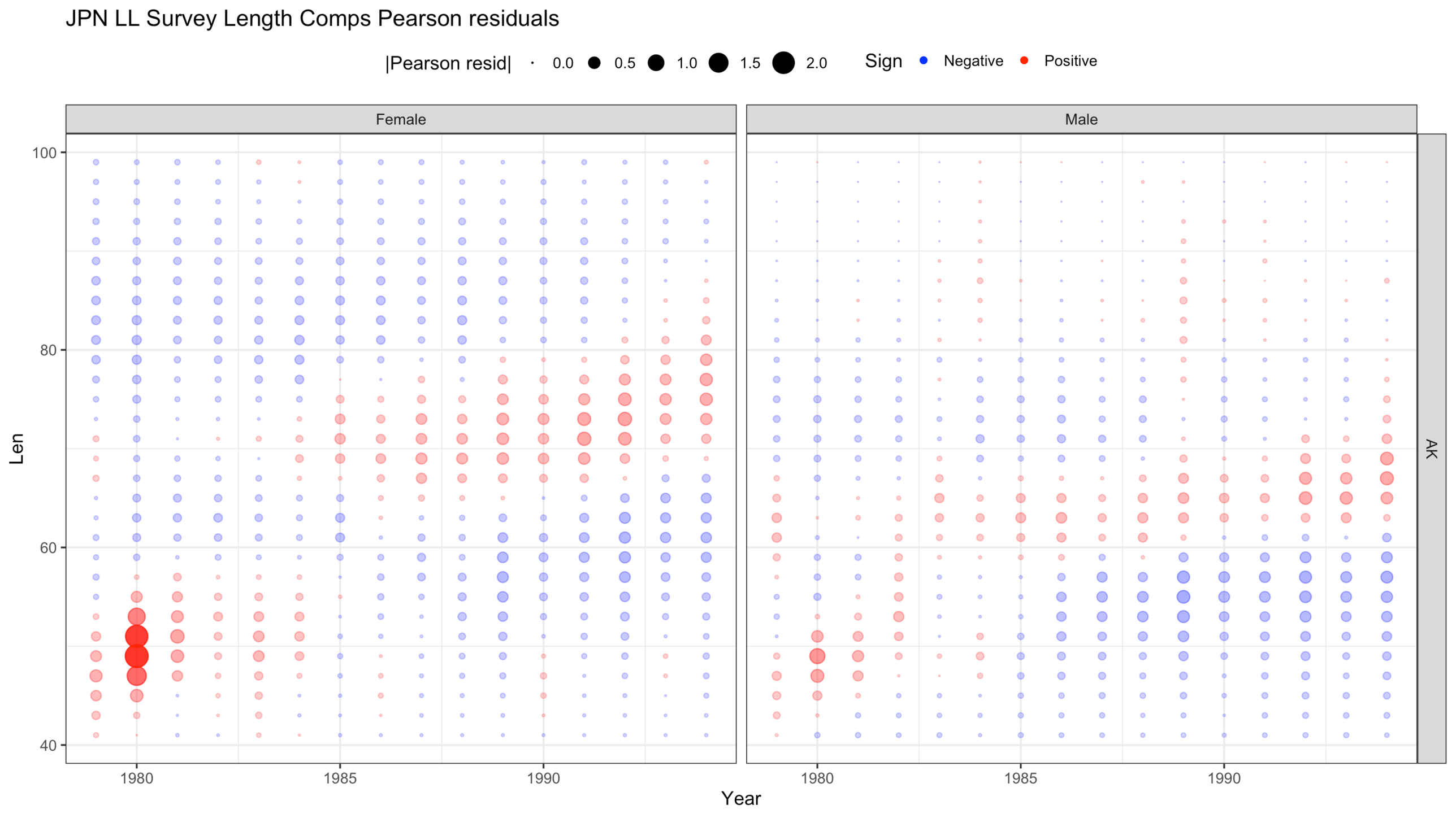
JPN LL Survey Length Comps Male AK

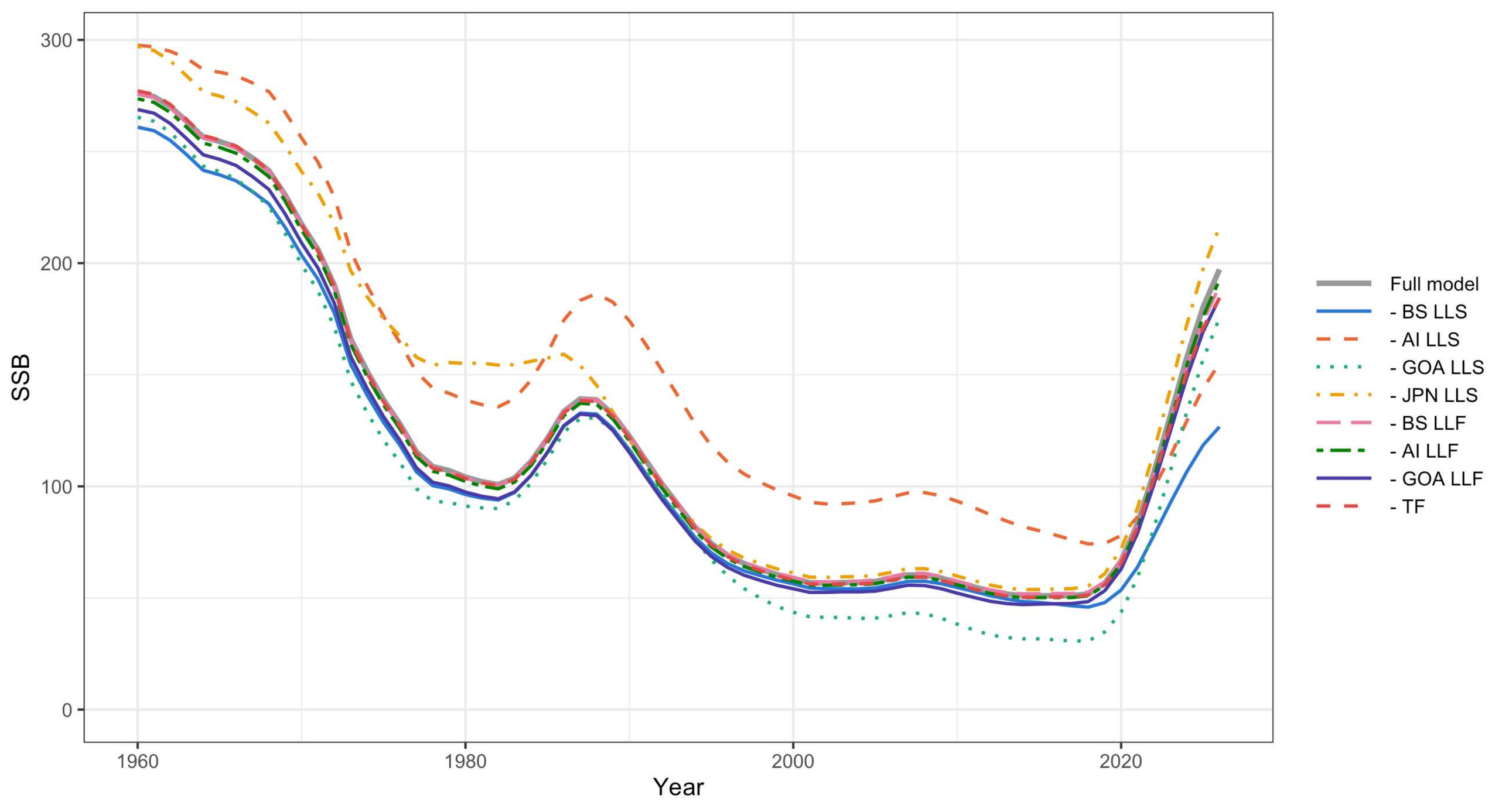
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

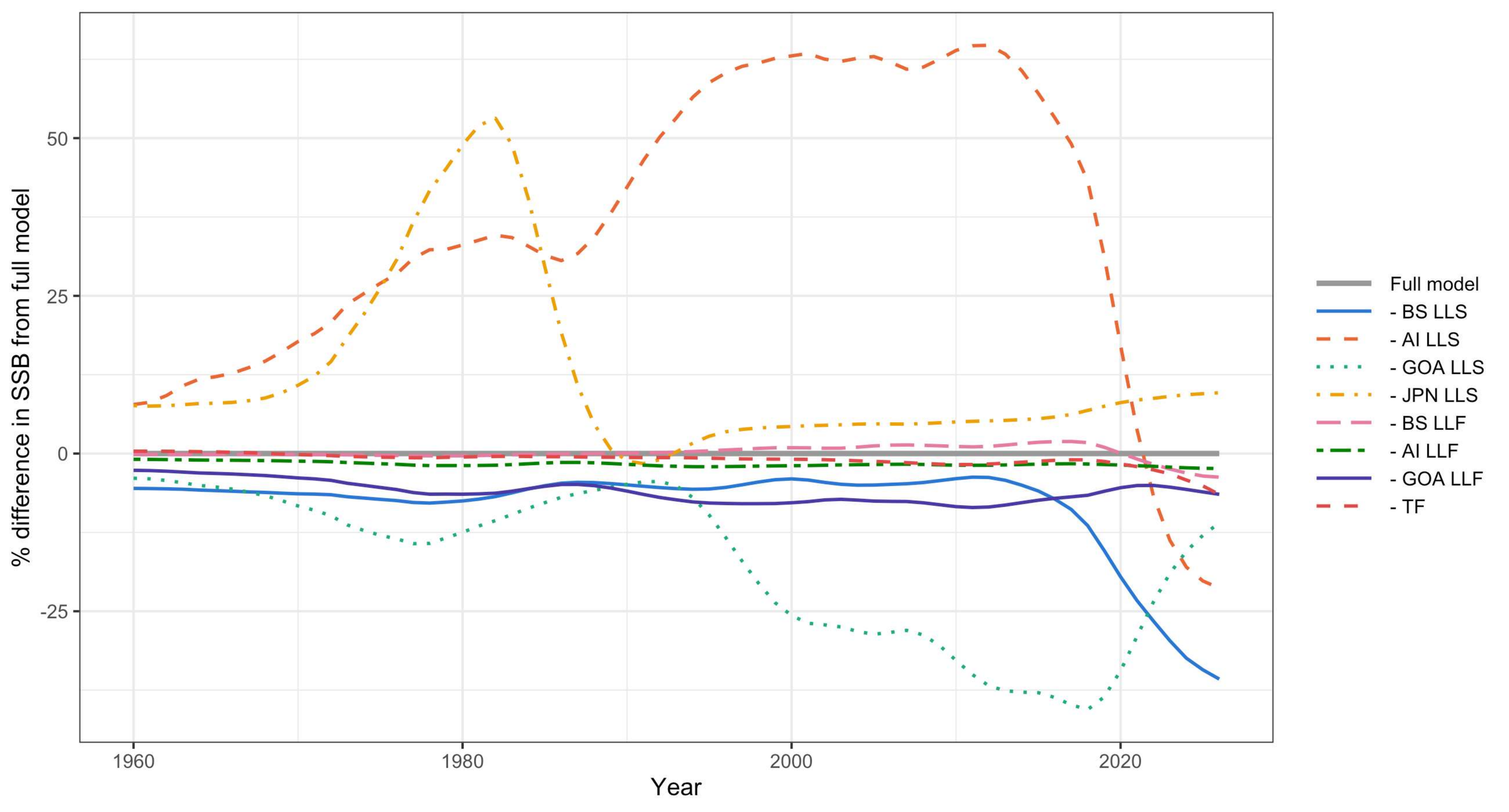
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

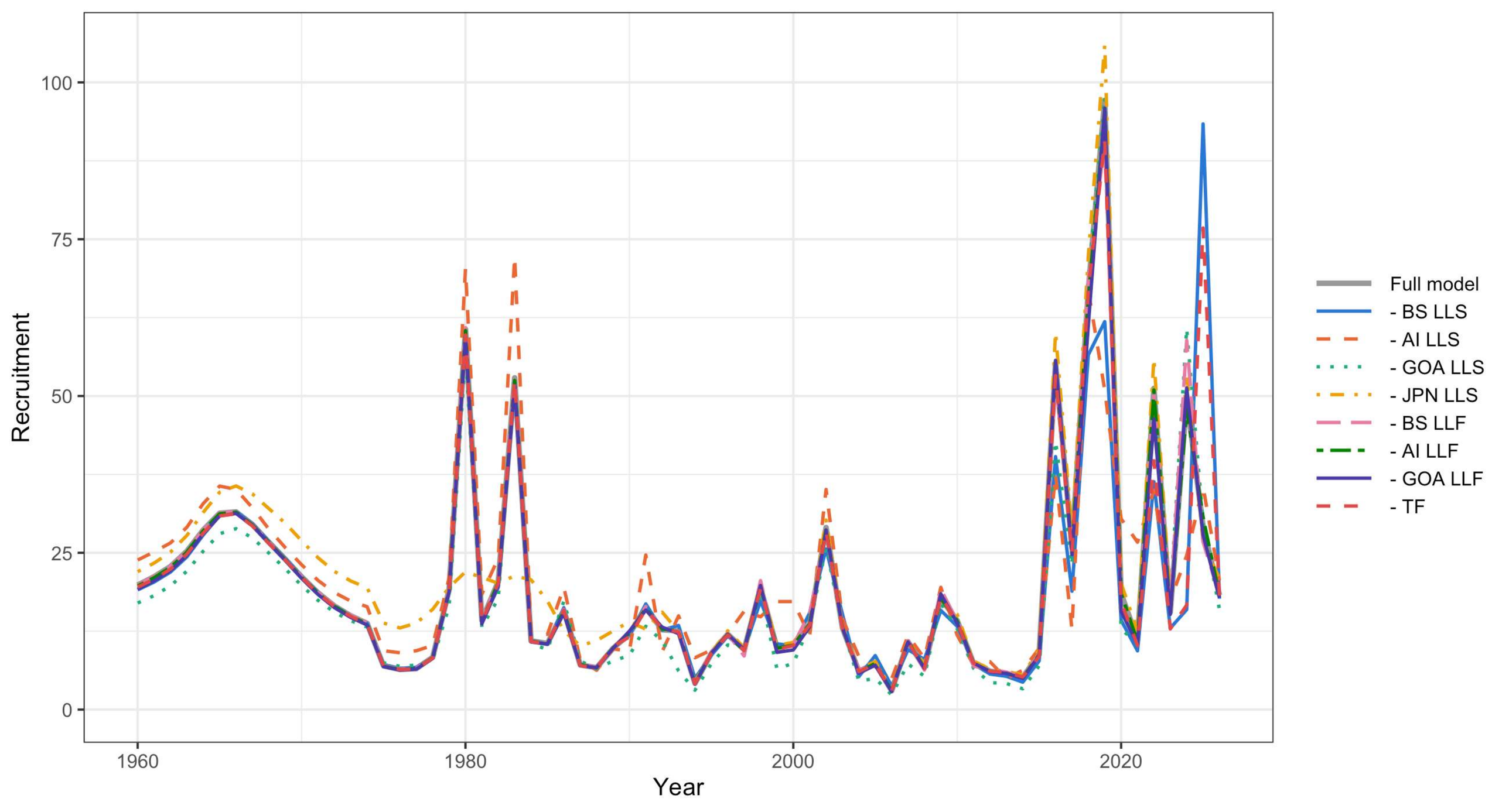
JPN LL Survey Length Comps OSA AK

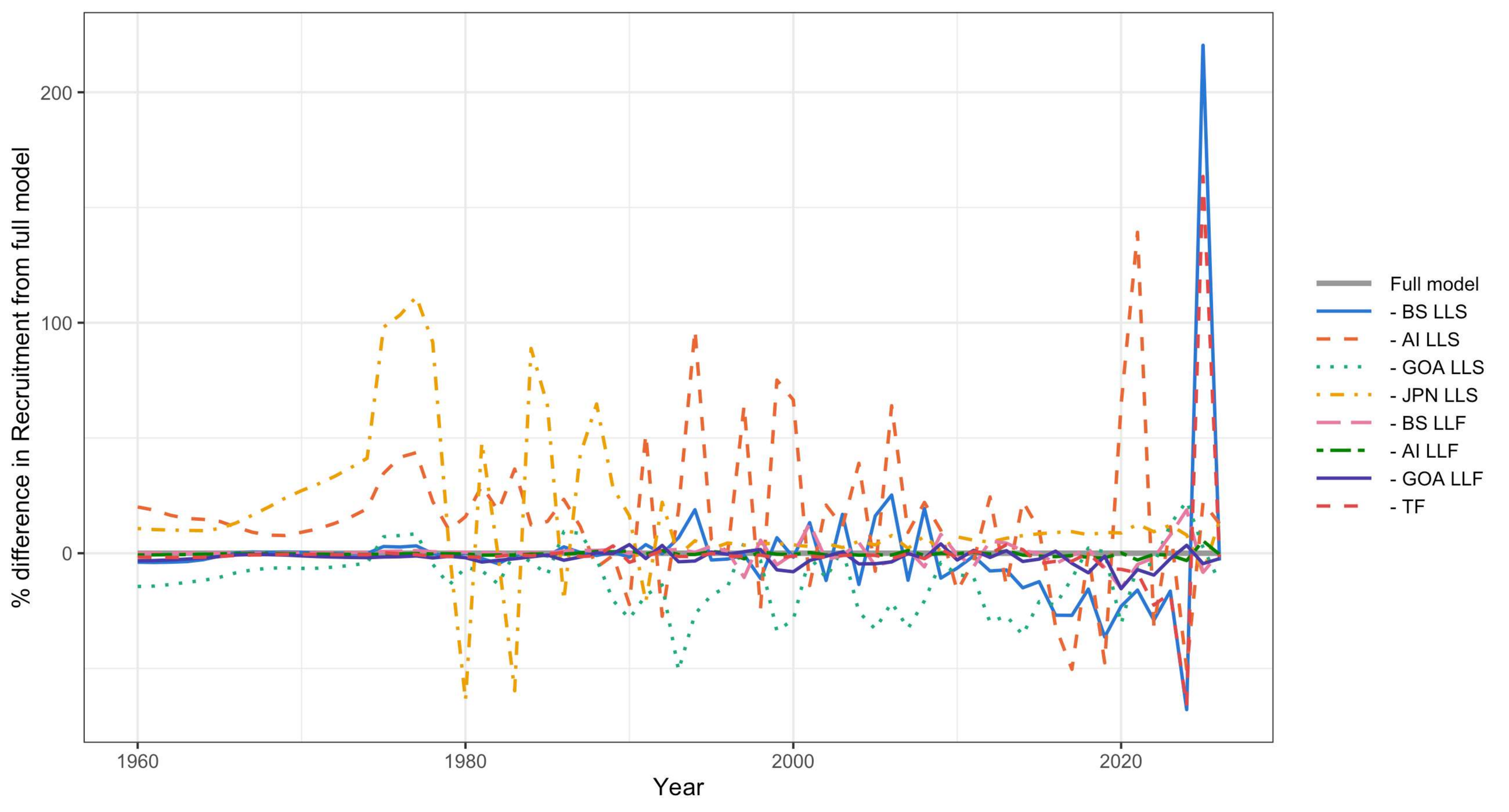



JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year


300
200
100
0
SSB
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- BS LLF
- AI LLF
- GOA LLF
- TF

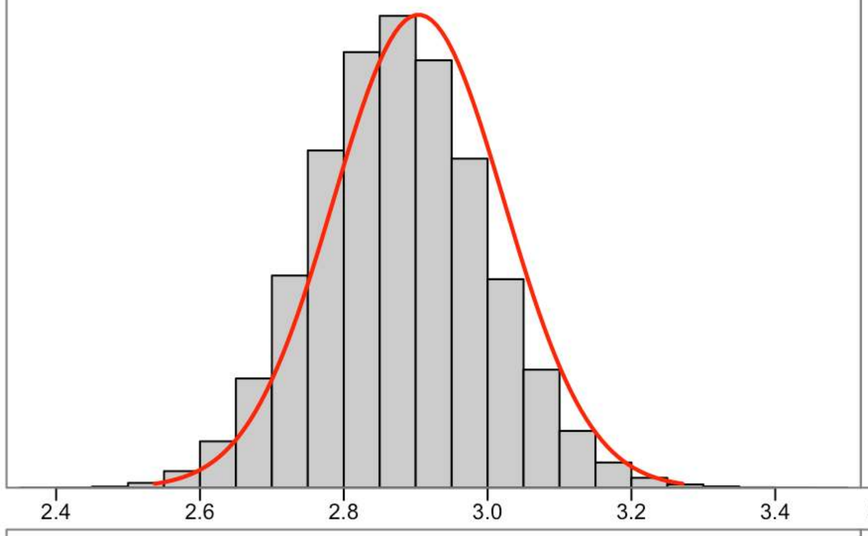

% difference in SSB from full model
50
25
0
-25
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- BS LLF
- AI LLF
- GOA LLF
- TF


Recruitment
100
75
50
25
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- BS LLF
- AI LLF
- GOA LLF
- TF


% difference in Recruitment from full model
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- BS LLF
- AI LLF
- GOA LLF
- TF

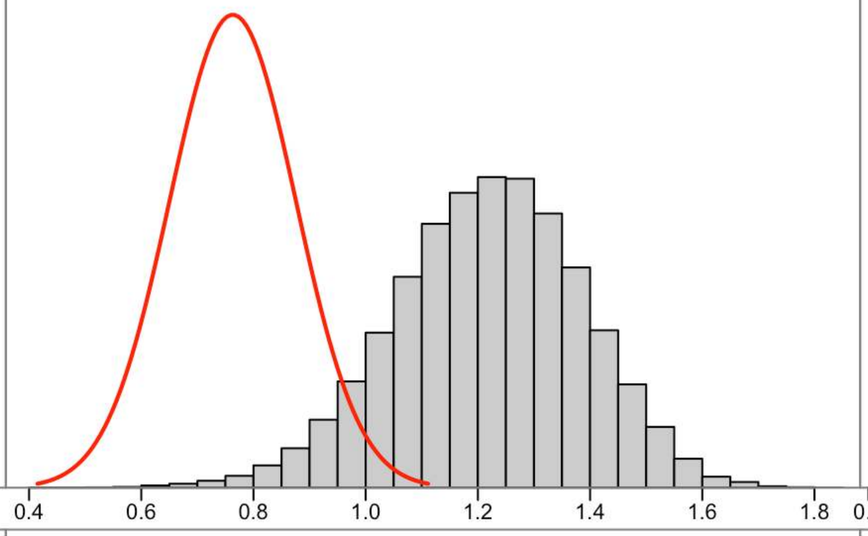
ln_global_R0

ESS=26590
Rhat=1



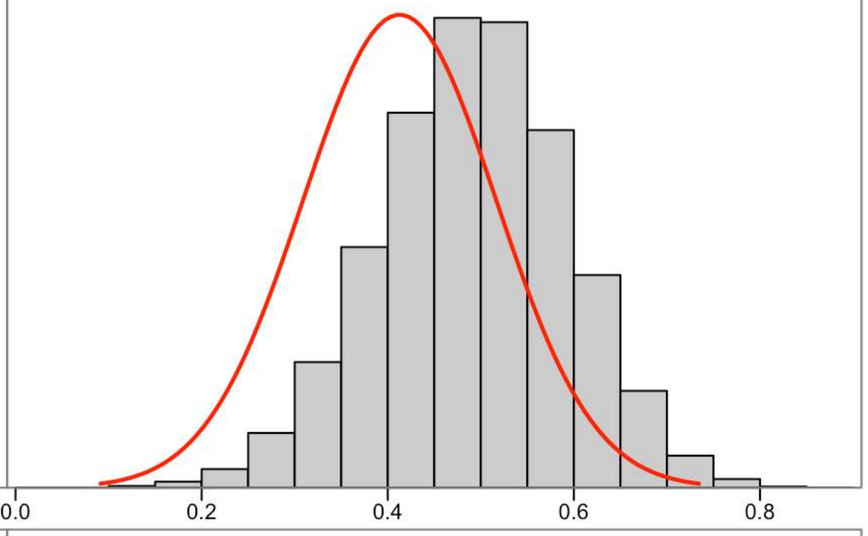
ln_srv_q[1]

ESS=16
Rhat=1.16



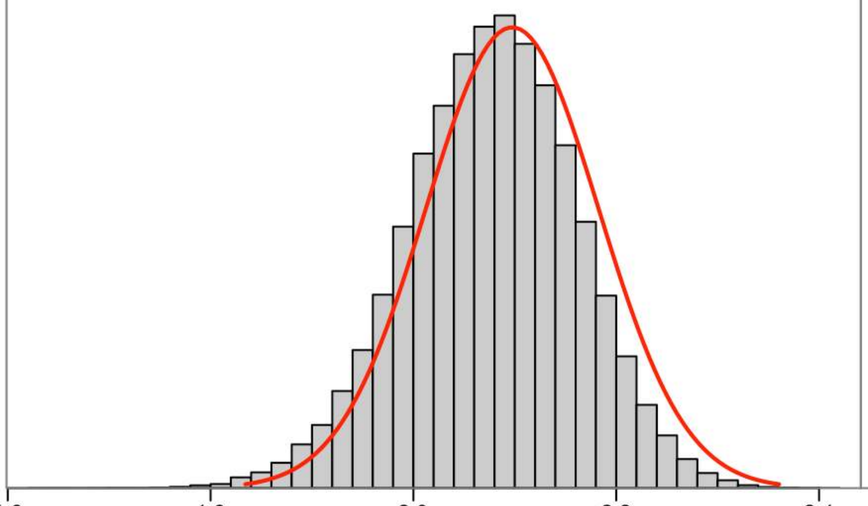
ln_srv_q[2]

ESS=638
Rhat=1.01



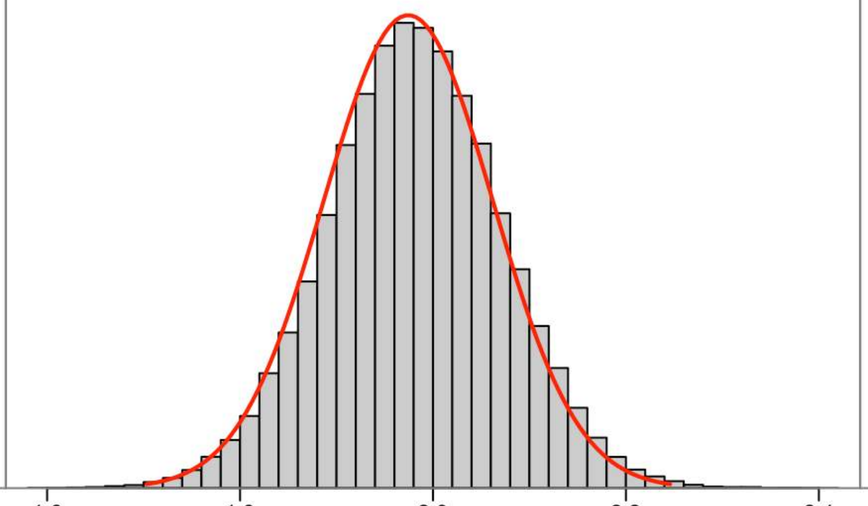
ln_srv_q[3]

ESS=18859
Rhat=1



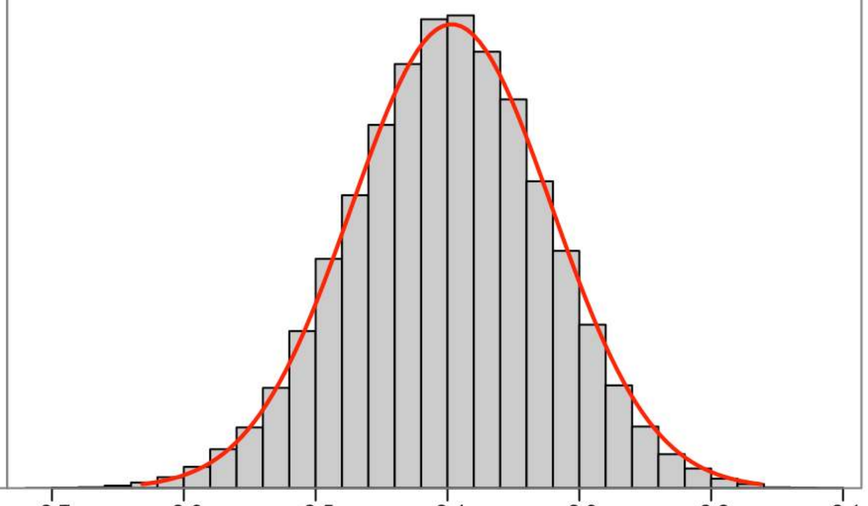
ln_srv_q[4]

ESS=2517
Rhat=1



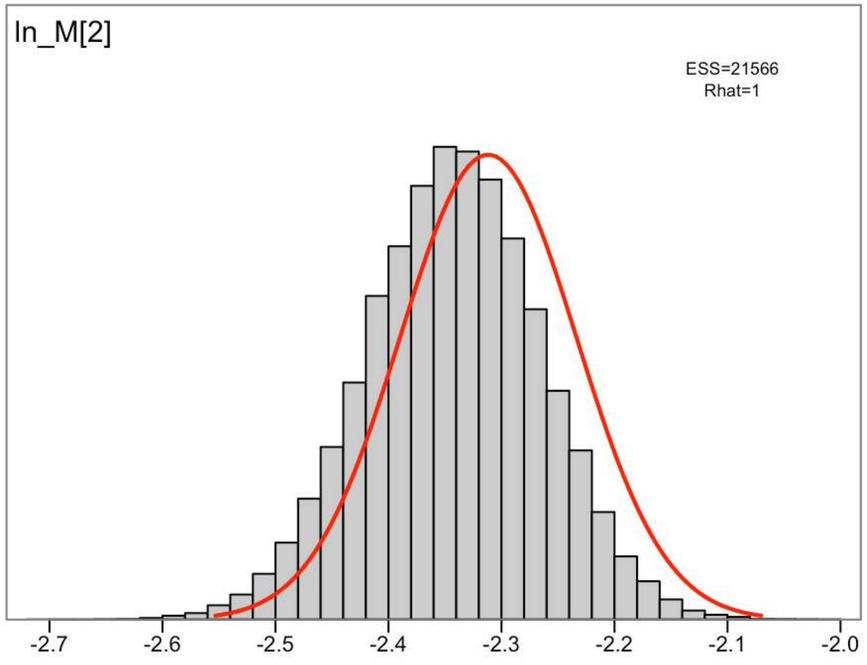
ln_M[1]

ESS=28571
Rhat=1

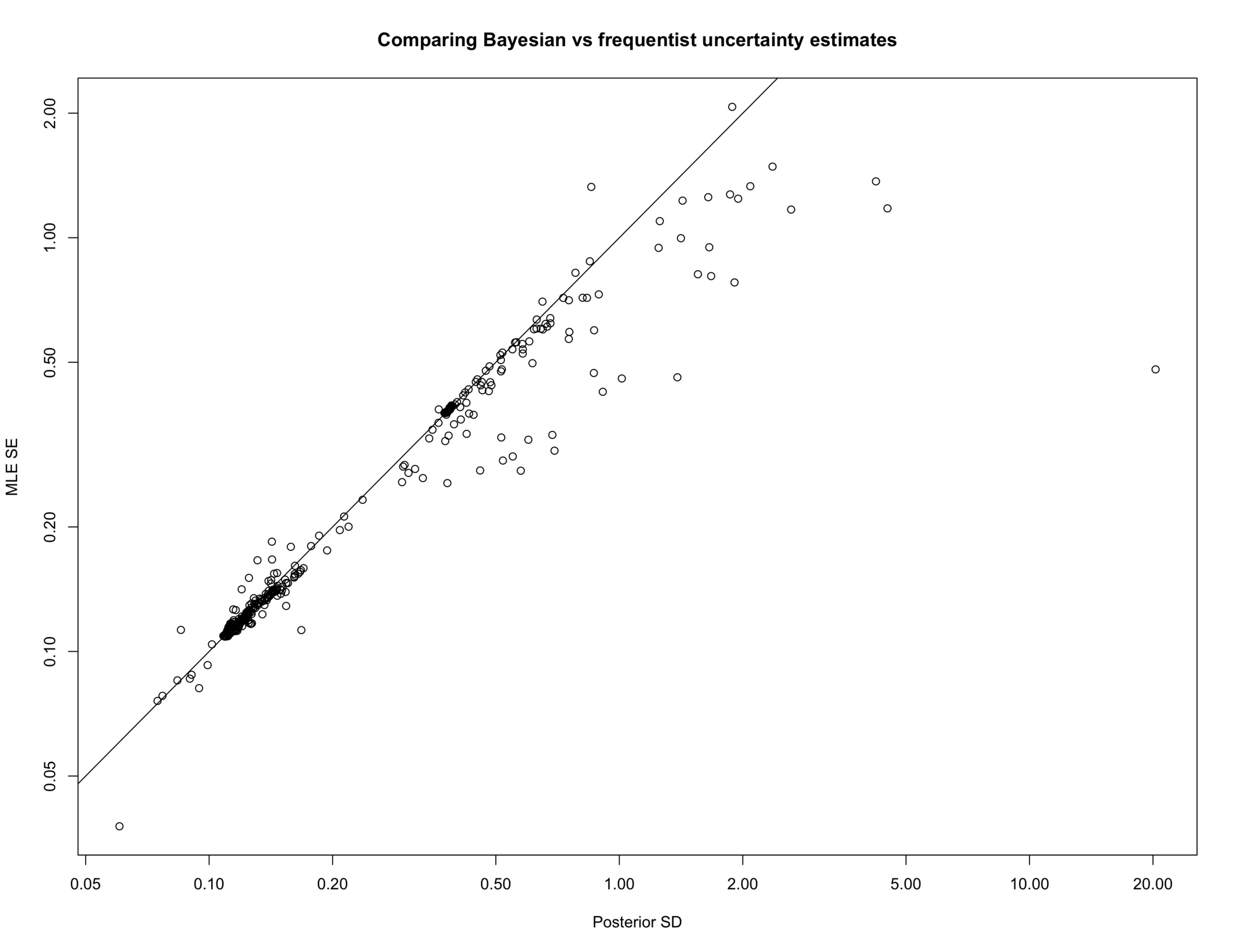


ln_M[2]

ESS=21566
Rhat=1



ln_global_R0	ln_srv_q[1]	ln_srv_q[2]	ln_srv_q[3]	ln_srv_q[4]	ln_M[1]	ln_M[2]	
ESS=26590 Rhat=1.00	-0.33	-0.54	-0.6	-0.54	0.72	0.68	ln_global_R0
	ESS=16 Rhat=1.16	0.49	0.38	0.33	-0.3	-0.34	ln_srv_q[1]
		ESS=638 Rhat=1.01	0.69	0.58	-0.52	-0.52	ln_srv_q[2]
			ESS=18859 Rhat=1.00	0.74	-0.59	-0.57	ln_srv_q[3]
				ESS=2517 Rhat=1.00	-0.49	-0.51	ln_srv_q[4]
					ESS=28571 Rhat=1.00	0.7	ln_M[1]
						ESS=21566 Rhat=1.00	ln_M[2]


Comparing Bayesian vs frequentist uncertainty estimates
MLE SE
Posterior SD
0.05
0.10
0.20
0.50
1.00
2.00
5.00
10.00
20.00

accept_stat
1.00
0.75
0.50
0.25
0.00
divergent
1.00
0.75
0.50
0.25
0.00
energy
2e+05
1e+05
0e+00
value
log_stepsize
2.5
0.0
-2.5
-5.0
-7.5
n_leapfrog
1000
750
500
250
0
0
5000
10000
15000
20000
iteration
chain
1
2
3
4

107 divergent transitions after warmup

Standardized value (posterior SDs)

2.5

0.0

-2.5

Divergent

Non-divergent

fish_fixed_sel_pars[10]

fish_fixed_sel_pars[8]

ln_F_devs[87]

fish_fixed_sel_pars[16]

ln_InitDevs[12]

ln_RecDevs[26]

ln_RecDevs[28]

ln_F_devs[110]

ln_F_devs[117]

ln_F_devs[89]

srv_fixed_sel_pars[7]

ESS=7
Rhat=1.61

0 20 40 60 80

srv_fixed_sel_pars[8]

ESS=7
Rhat=1.61

-10 -5 0 5

ln_srv_q[1]

ESS=16
Rhat=1.16

0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8

srv_fixed_sel_pars[3]

ESS=40
Rhat=1.06

-4 -2 0 2 4

srv_fixed_sel_pars[6]

ESS=88
Rhat=1.03

-10 -5 0 5 10 15

srv_fixed_sel_pars[23]

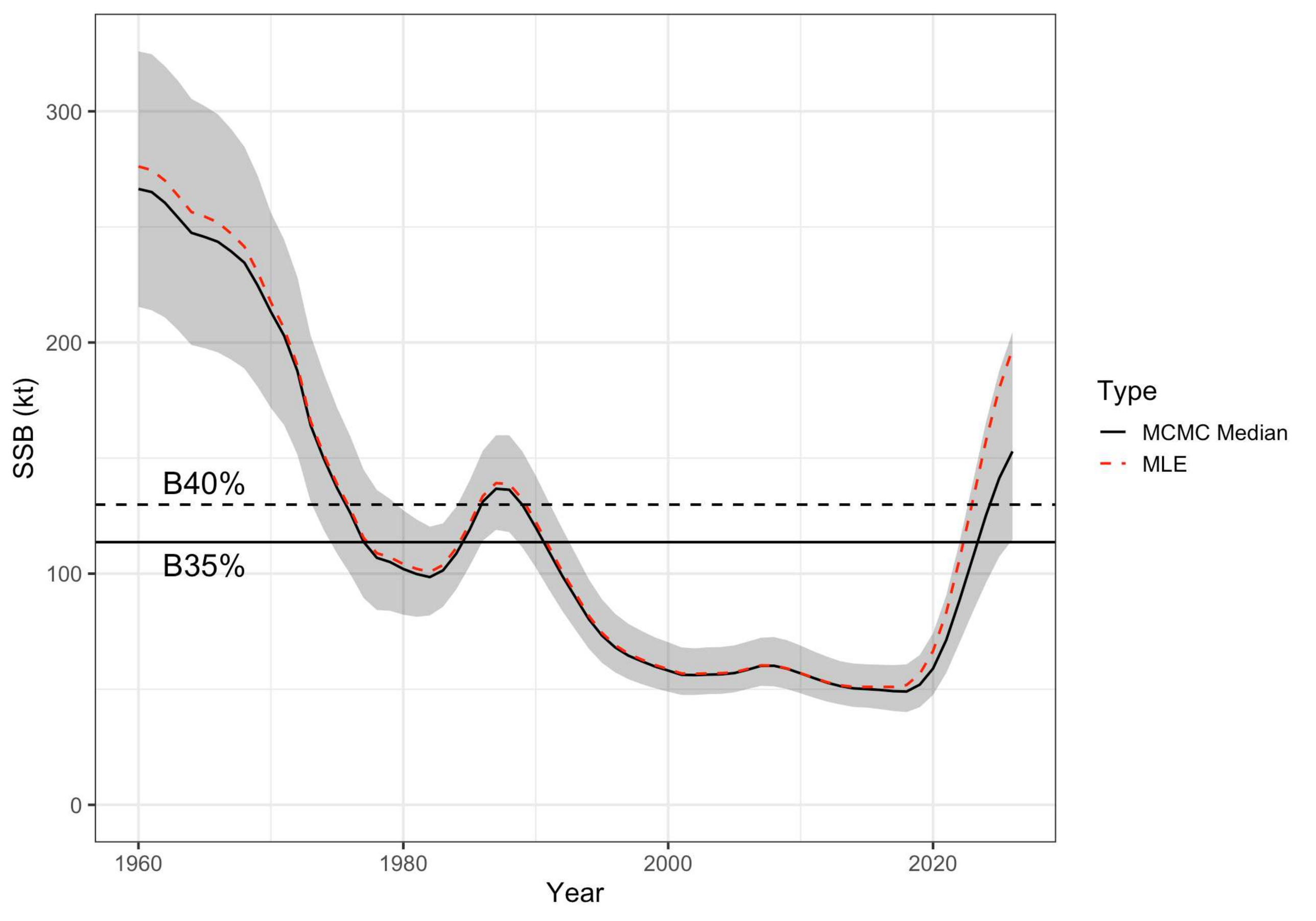
ESS=97
Rhat=1.03

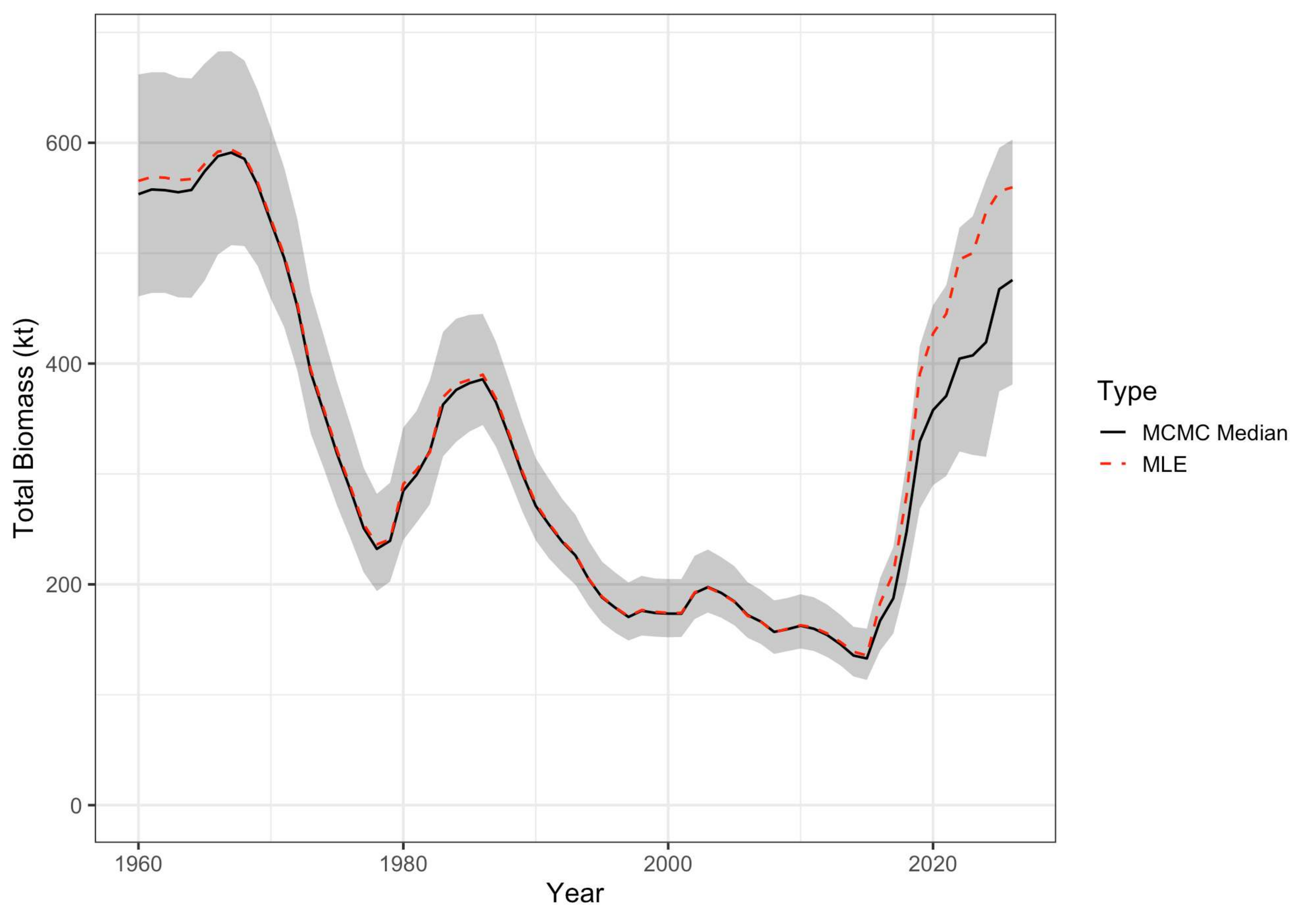
1.0 1.5 2.0 2.5

srv_fixed_sel_pars[8] srv_fixed_sel_pars[3] srv_fixed_sel_pars[23]

srv_fixed_sel_pars[7]		ln_srv_q[1]		srv_fixed_sel_pars[6]	
ESS=7 Rhat=1.61	0.9	0.54	-0.32	-0.23	-0.23
	ESS=7 Rhat=1.61	0.52	-0.32	-0.22	-0.23
		ESS=16 Rhat=1.16	-0.54	-0.34	-0.39
			ESS=40 Rhat=1.06	0.23	0.23
				ESS=88 Rhat=1.03	
					ESS=97 Rhat=1.03

srv_fixed_sel_pars[7] srv_fixed_sel_pars[8] ln_srv_q[1] srv_fixed_sel_pars[3] srv_fixed_sel_pars[6] srv_fixed_sel_pars[23]


SSB (kt)
300
200
100
0
B40%
B35%
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


Total Biomass (kt)
600
400
200
0
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

Age-2 Recruitment (millions of individuals)

100

50

0

1960

1980

2000

2020

Year

Type

MCMC Median

MLE

Type: Age, Catch, Index, jnLL, Length, Penalty, Prior

26.2_FAA_3LLF_LLS_BS_Dome_DN

Likelihood

500
250
0
-250
-500

Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

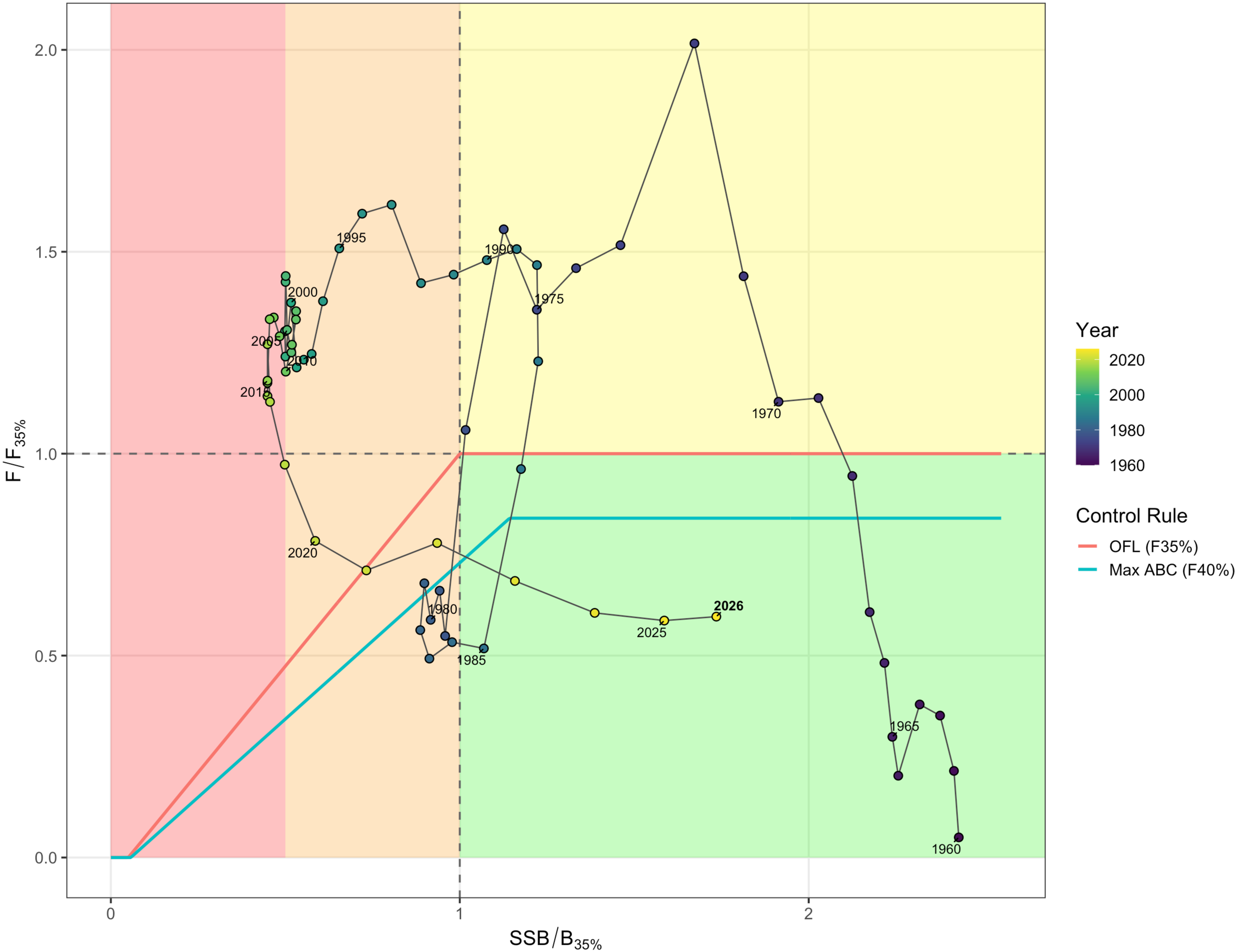
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year


2.0
1.5
1.0
0.5
0.0
F/F35%
0
1
2
SSB/B35%
1995
1990
2000
1975
2005
2010
2015
1970
2020
1980
2026
2025
1985
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Natural mortality profile

Scaled nLL

1000
750
500
250
0

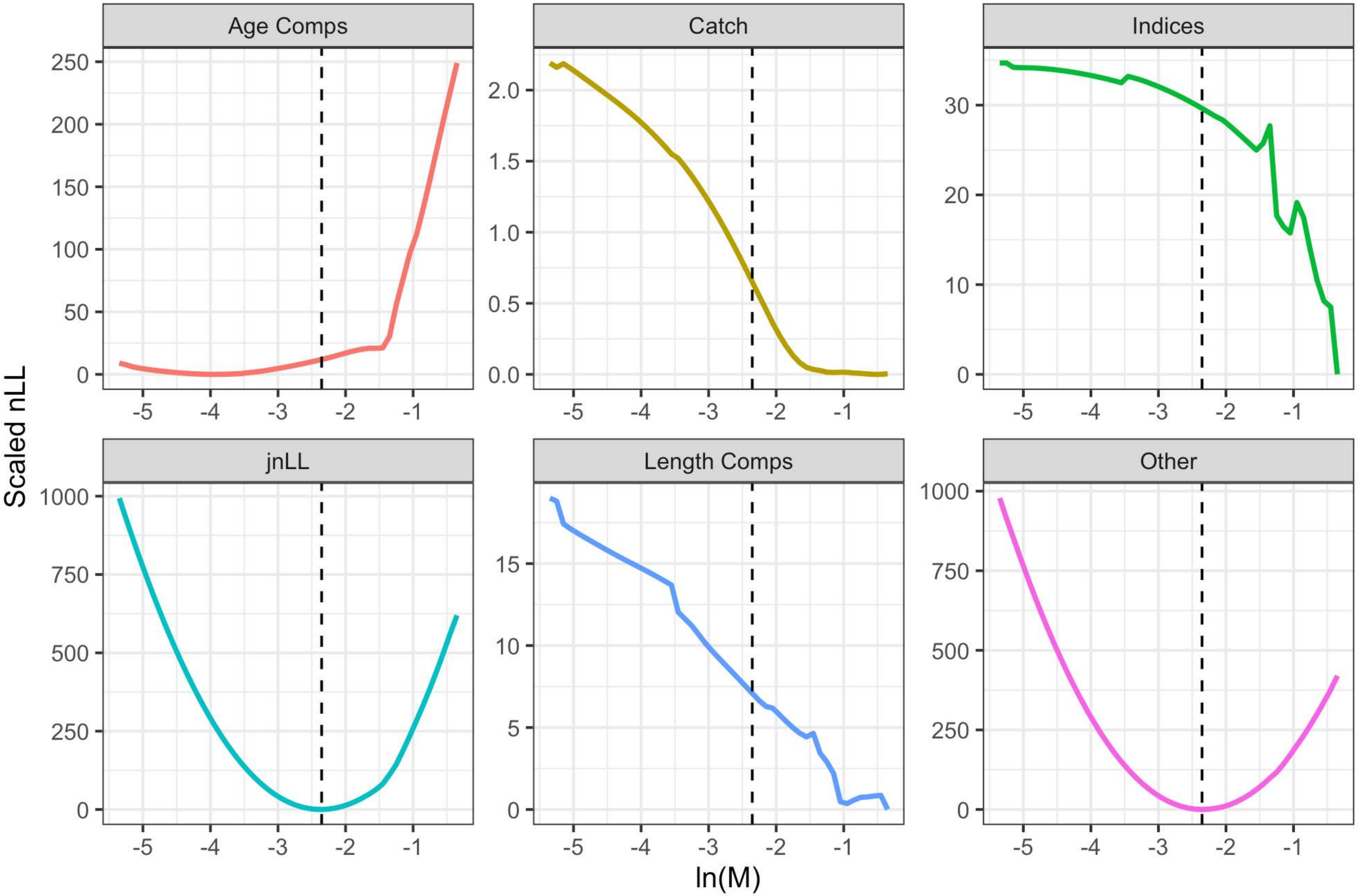
-5
-4
-3
-2
-1

ln(M)

Type

Age Comps
Catch
Indices
jnLL
Length Comps
Other

Natural mortality profile



Recruitment (R0) profile

Scaled nLL

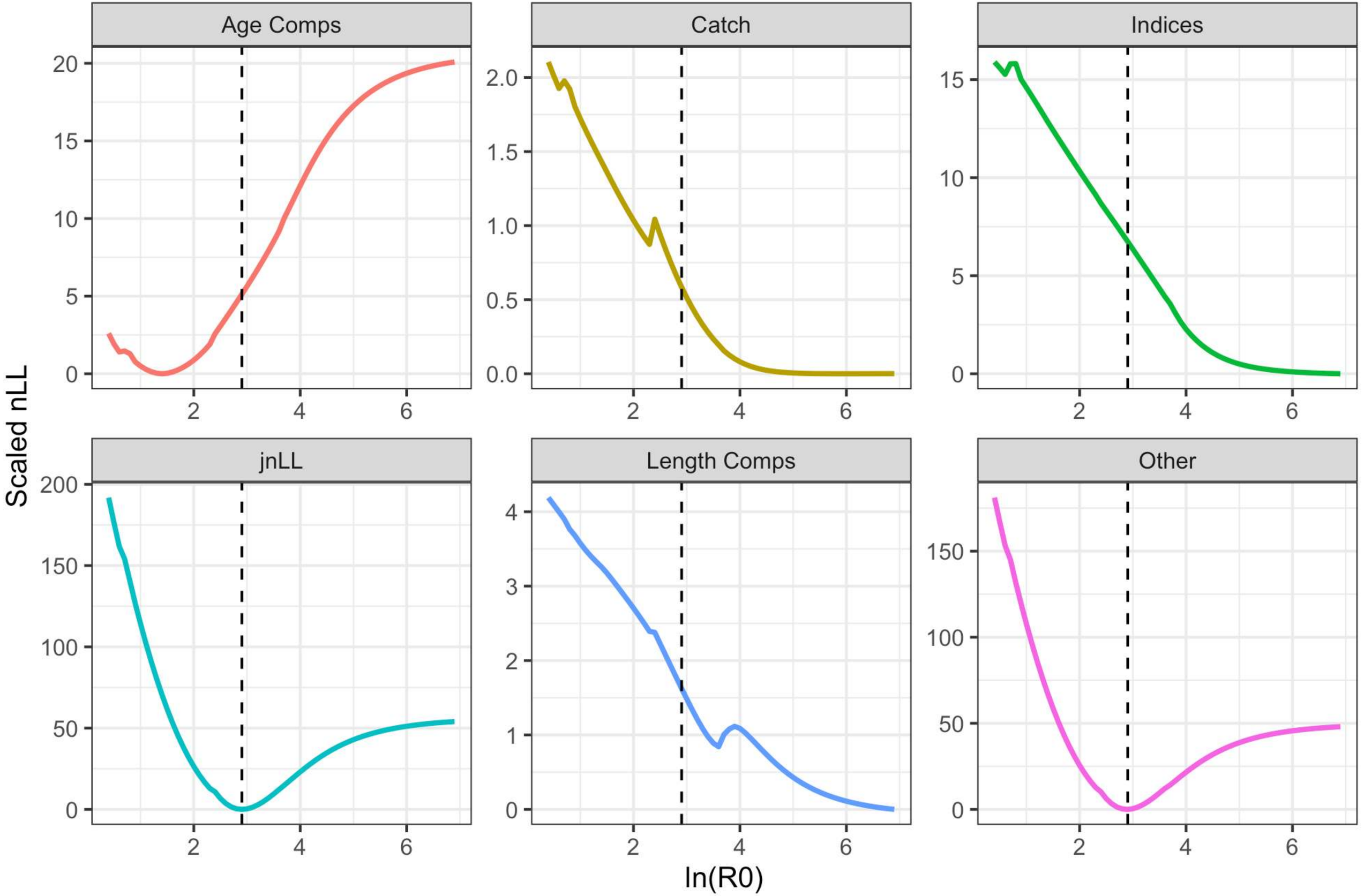
200
150
100
50
0

2
4
6

ln(R0)

Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

Recruitment (R0) profile



Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.2698

SSB

Mohns Rho: 0.1202

AK

Relative Difference from Terminal Year

0.5
0.0
-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

90
60
30
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

90

60

30

0

1

2

3

4

5

6

7

Years since cohort was last estimated

Cohort

2016

2017

2018

2019

2020

2021

2022

2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.2698

SSB

Mohns Rho: 0.1445

AK

Relative Difference from Terminal Year

0.5
0.0
-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

90
60
30
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

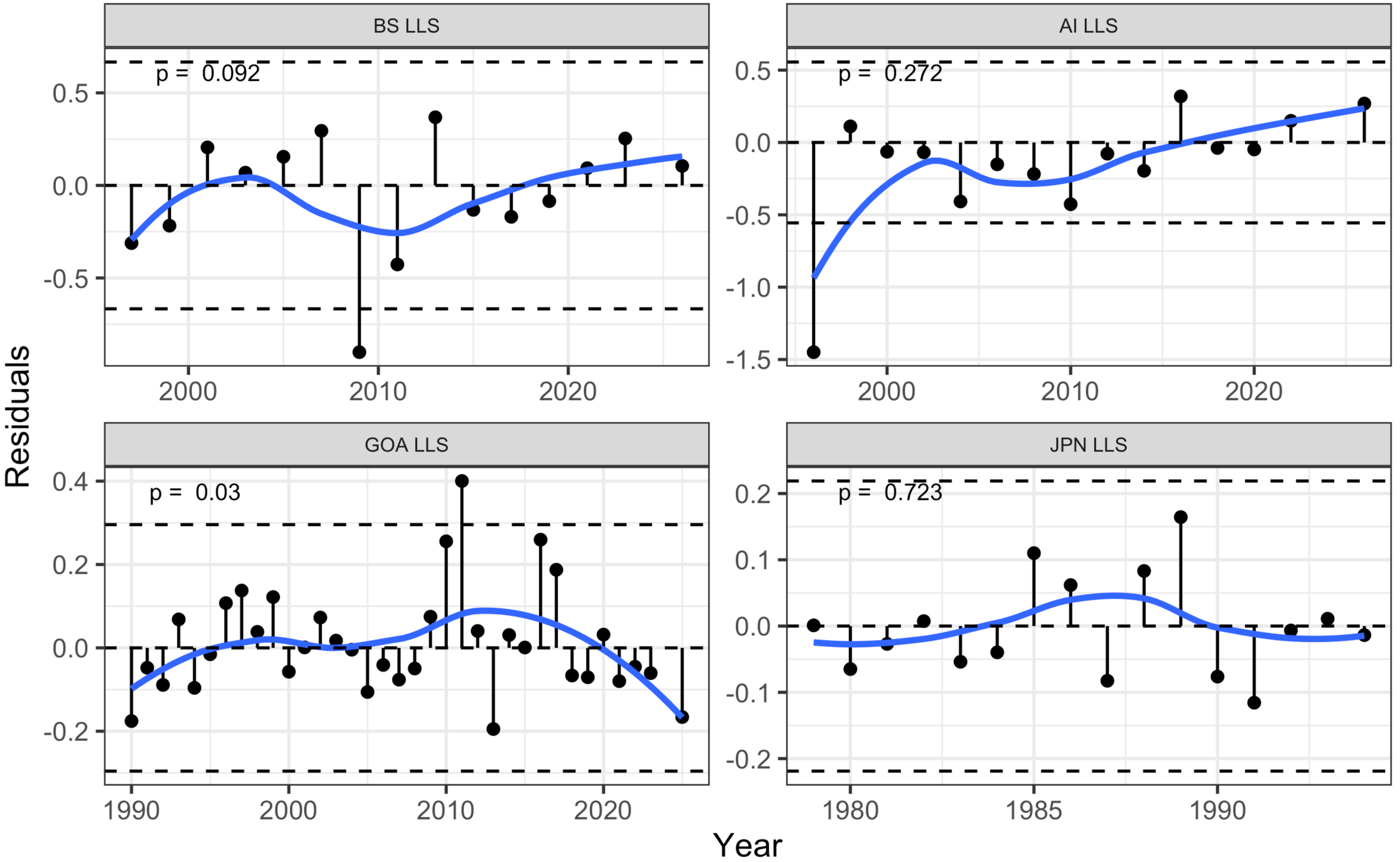
100
75
50
25

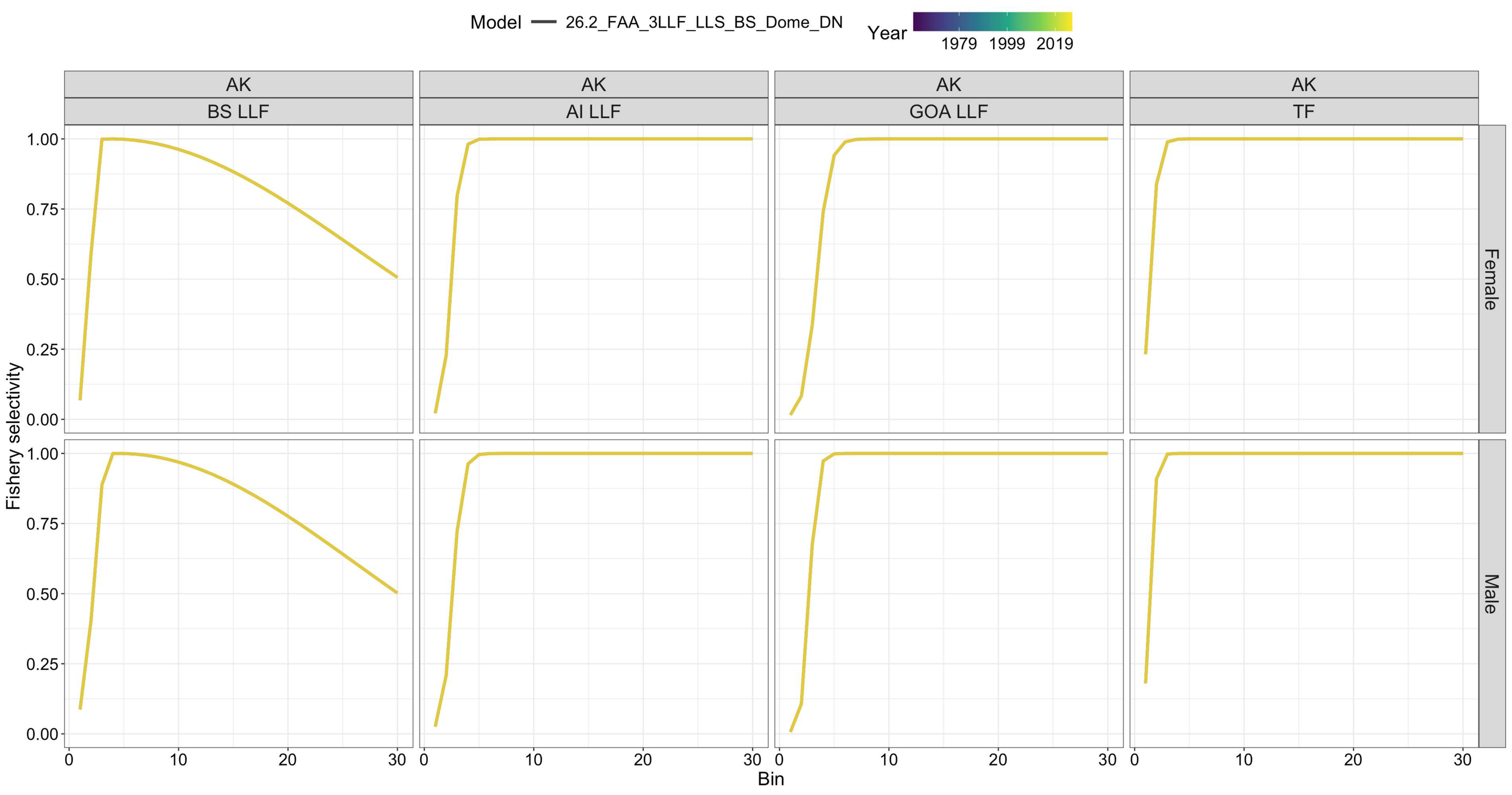
0
1
2
3
4
5
6
7

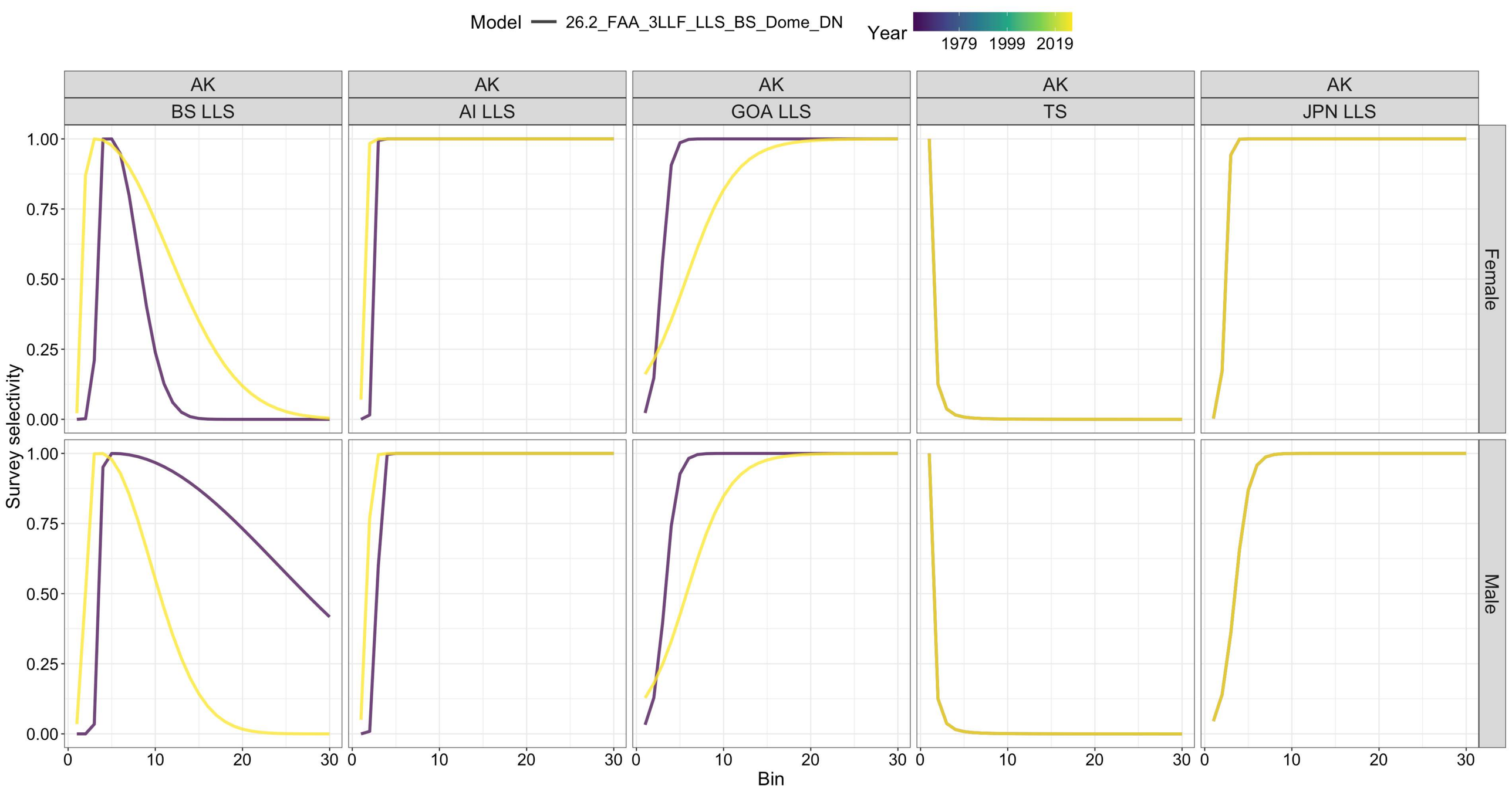
Years since cohort was last estimated

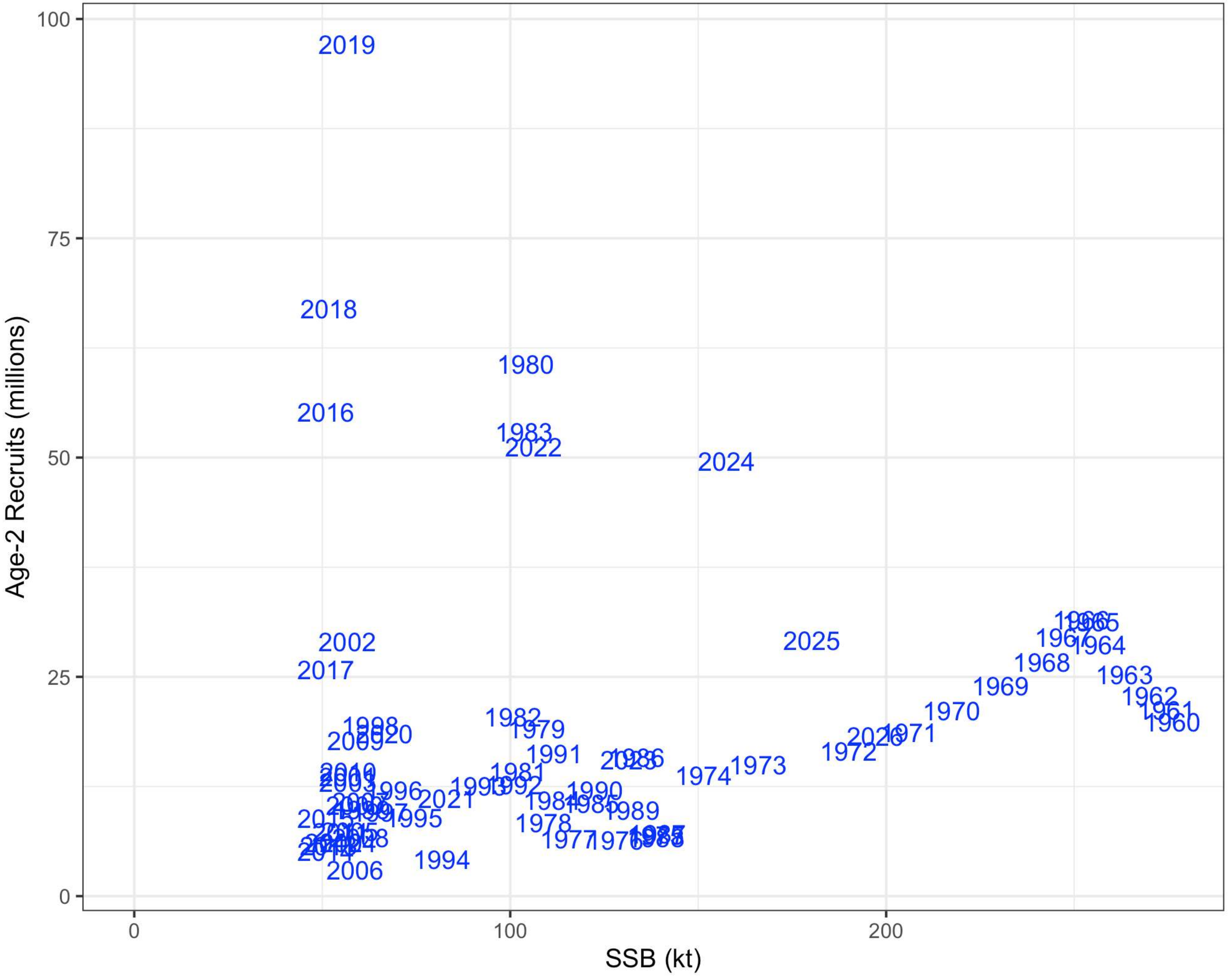
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test




Model
26.2_FAA_3LLF_LLS_BS_Dome_DN
Year
1979
1999
2019
AK
BS LLF
AK
AI LLF
AK
GOA LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_3LLF_LLS_BS_Dome_DN
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
100
75
50
25
0
0
100
200
SSB (kt)
2019
2018
1980
2016
1983
2022
2024
2025
2002
2017
2006
1994
2021
1995
1990
1989
1974
1973
1972
1971
1970
1969
1968
1963
1964
1962
1960
1979
1982
1991
1981
1978
1977

Population and Catch Trends

Recruitment (millions of fish)

SSB or Catch (kt)

100
75
50
25
0

300
200
100
0

1960
1980
2000
2020

Year

Catch
Recruitment
SSB

Model 26.2_FAA_3LLF_LLS_BS_Dome_DN

AK

Dynamic SSB0

Recruitment

SSB

Total Biom

BS LLF F

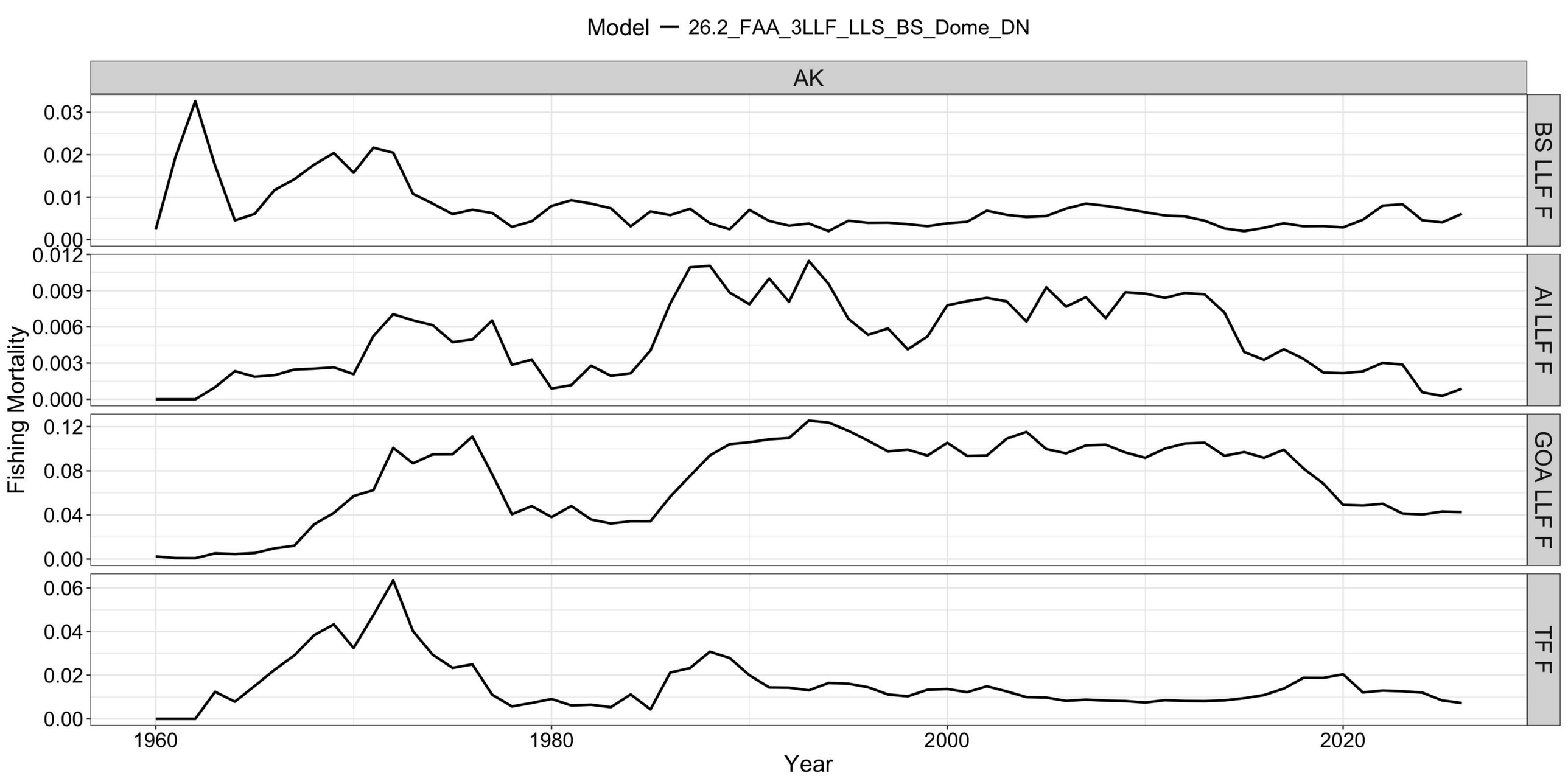
AI LLF F

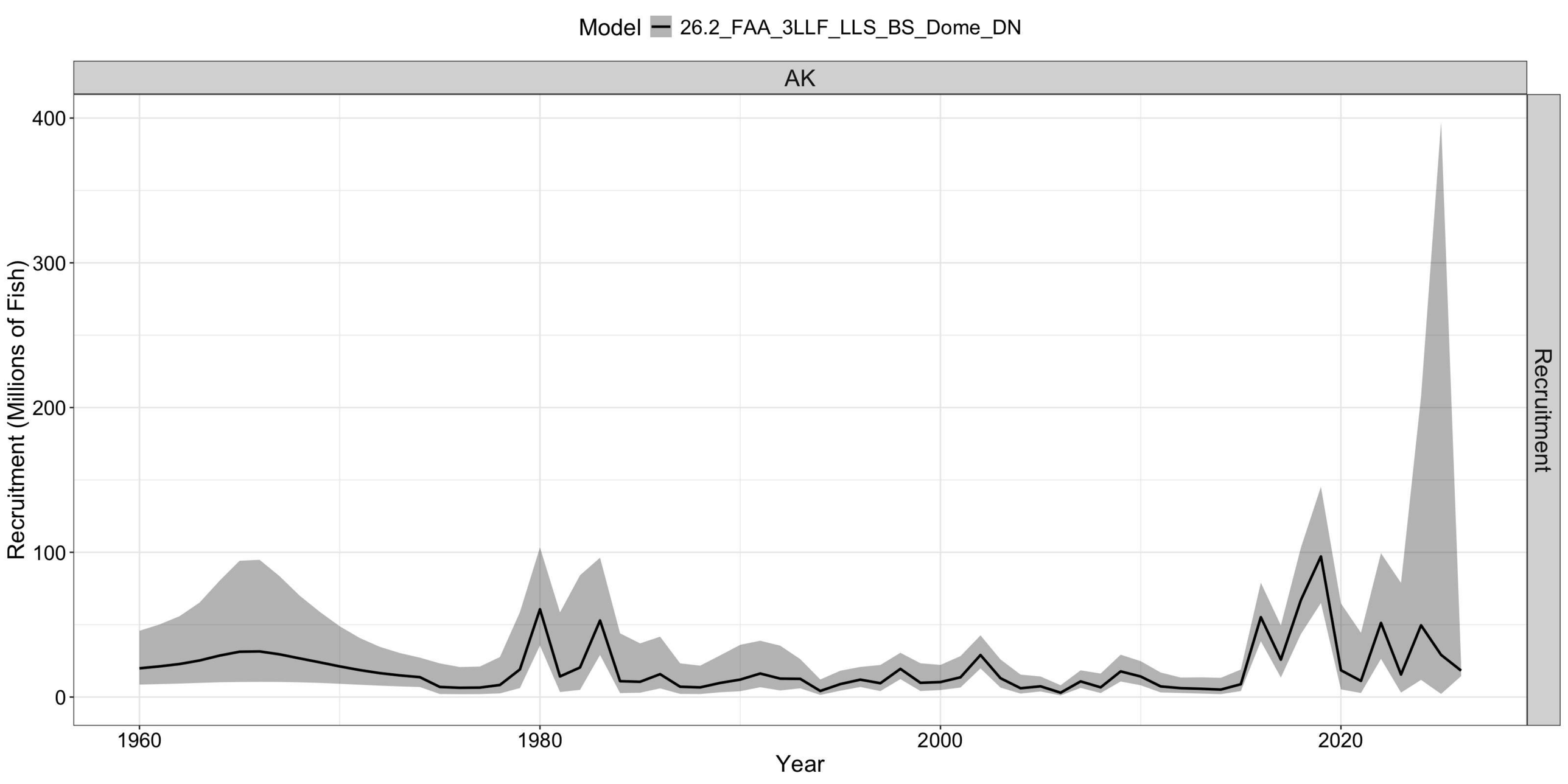
GOA LLF F

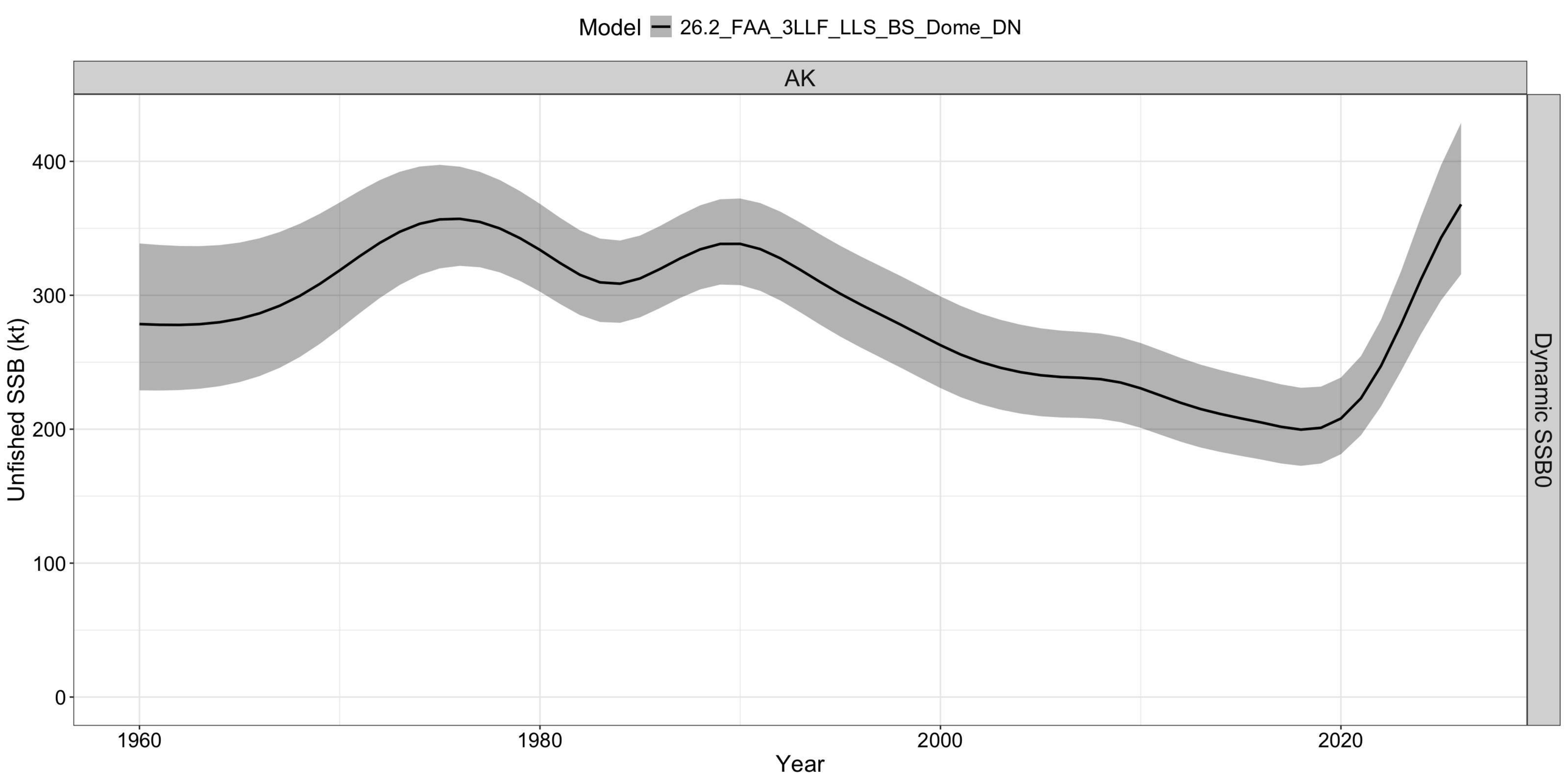
TF F

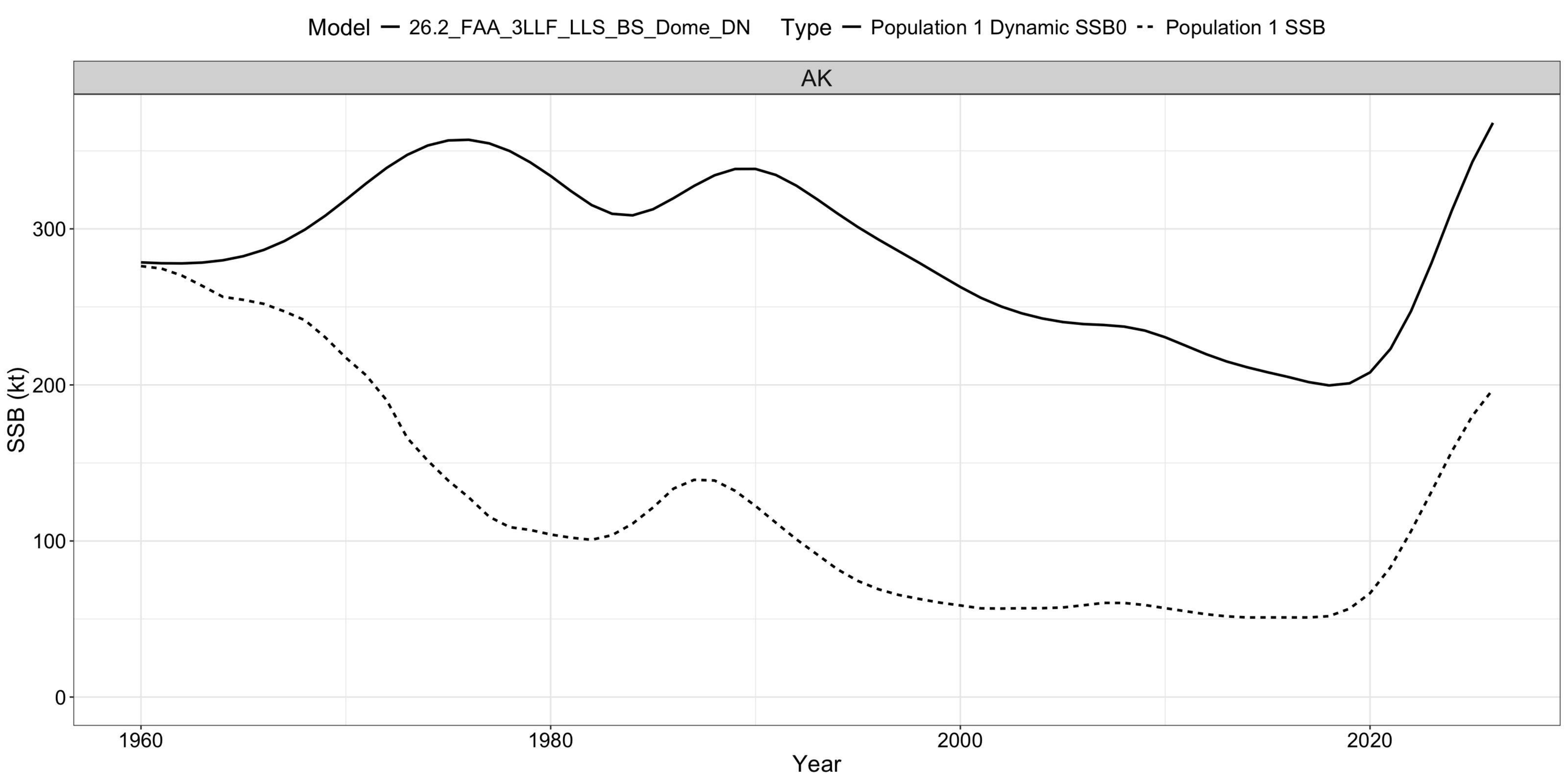
Value

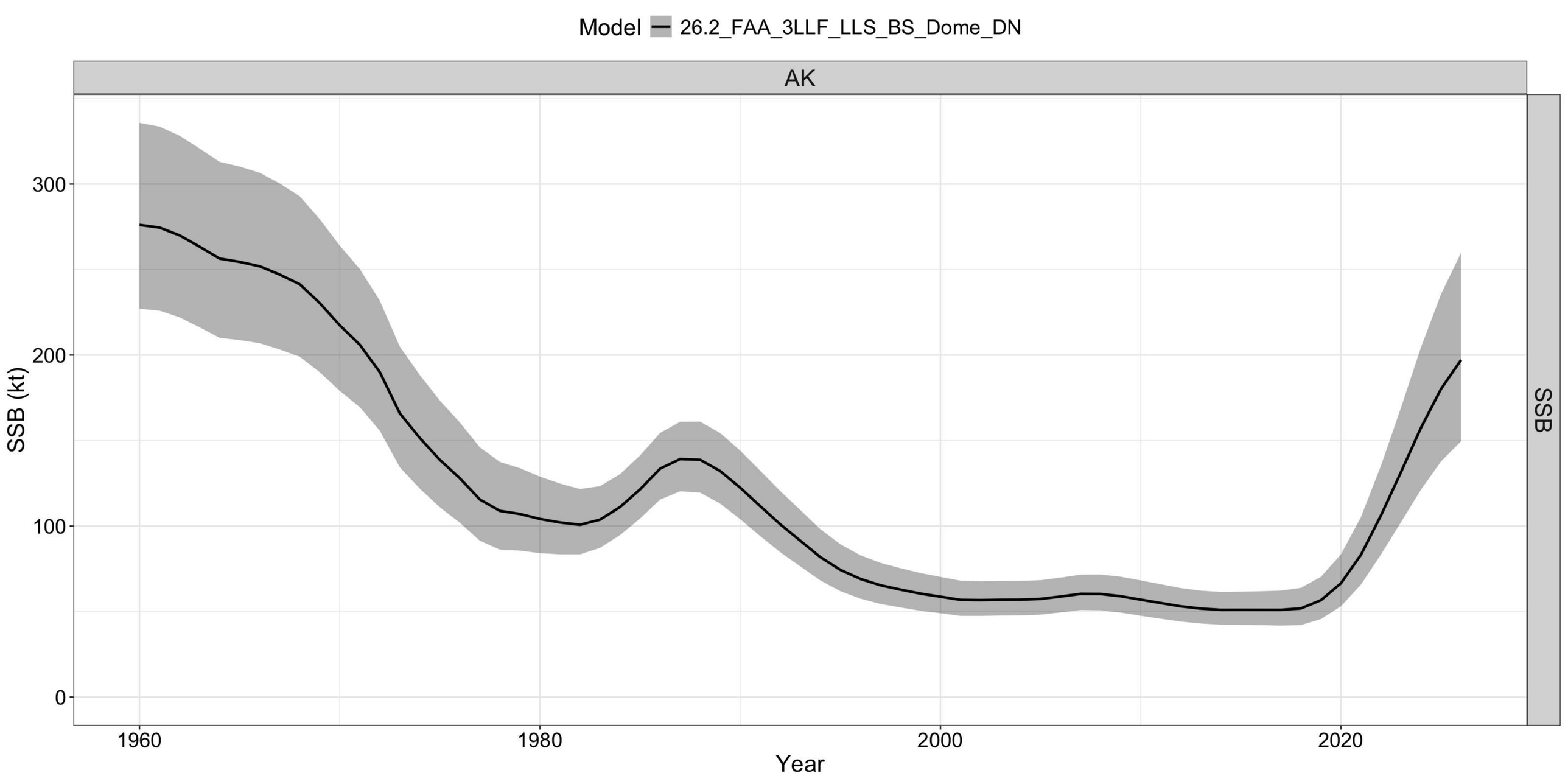
Year

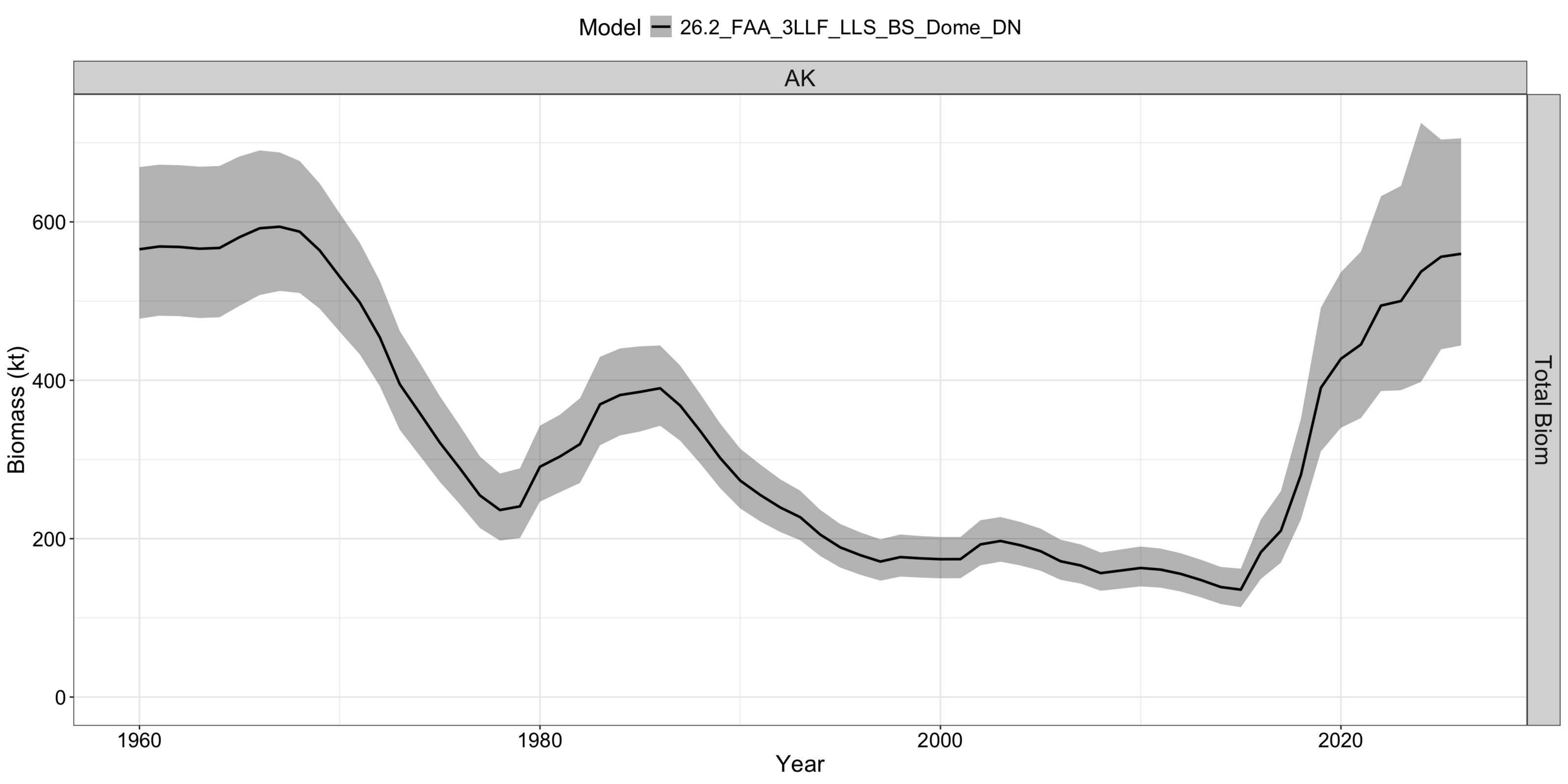

Model — 26.2_FAA_3LLF_LLS_BS_Dome_DN
AK
BS LLF F
AI LLF F
GOA LLF F
TF F
Fishing Mortality
0.03
0.02
0.01
0.00
0.012
0.009
0.006
0.003
0.000
0.12
0.08
0.04
0.00
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year


Model
26.2_FAA_3LLF_LLS_BS_Dome_DN
AK
Recruitment
Recruitment (Millions of Fish)
400
300
200
100
0
1960
1980
2000
2020
Year

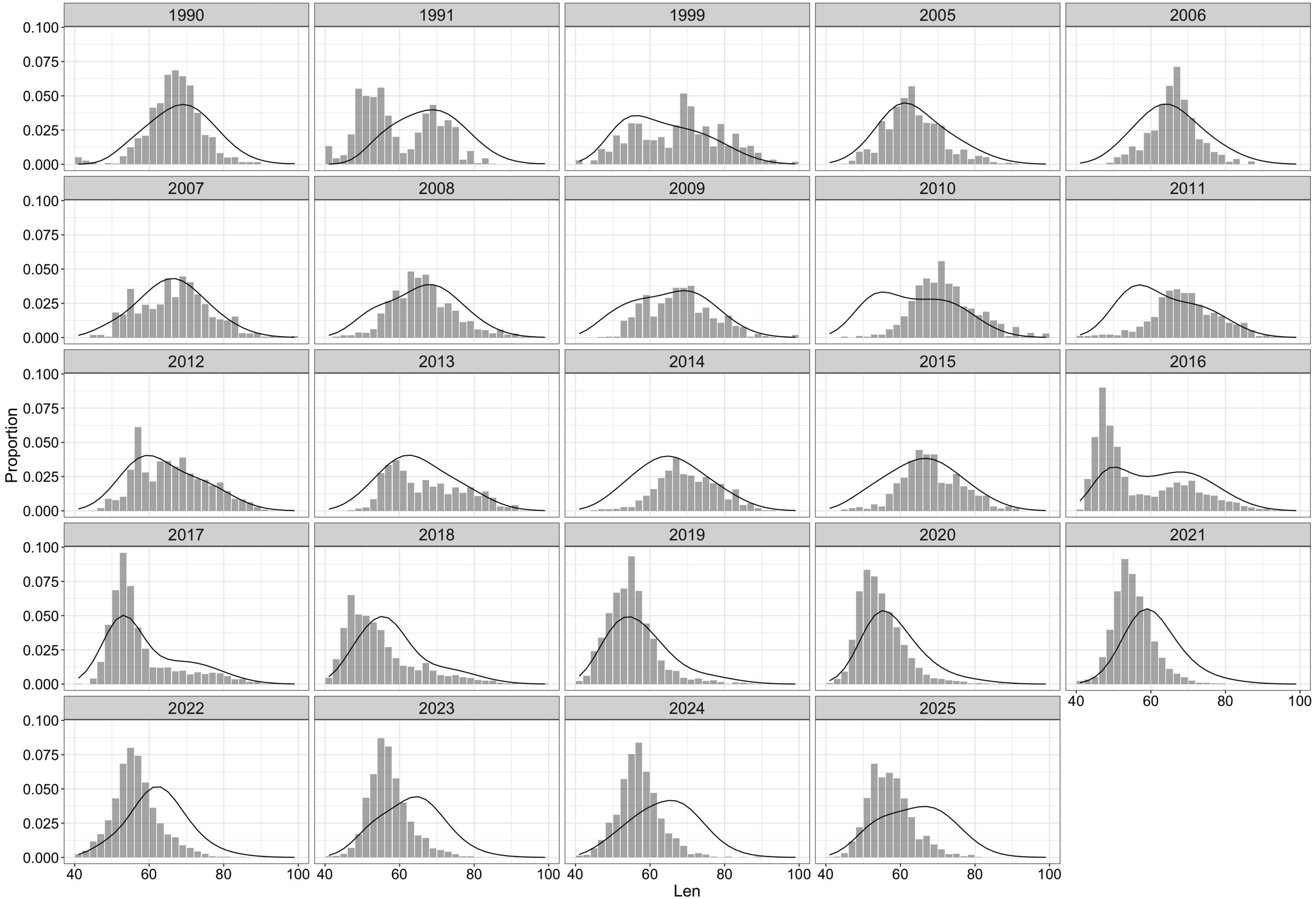

Model
26.2_FAA_3LLF_LLS_BS_Dome_DN
AK
Dynamic SSB0
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_3LLF_LLS_BS_Dome_DN
Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year

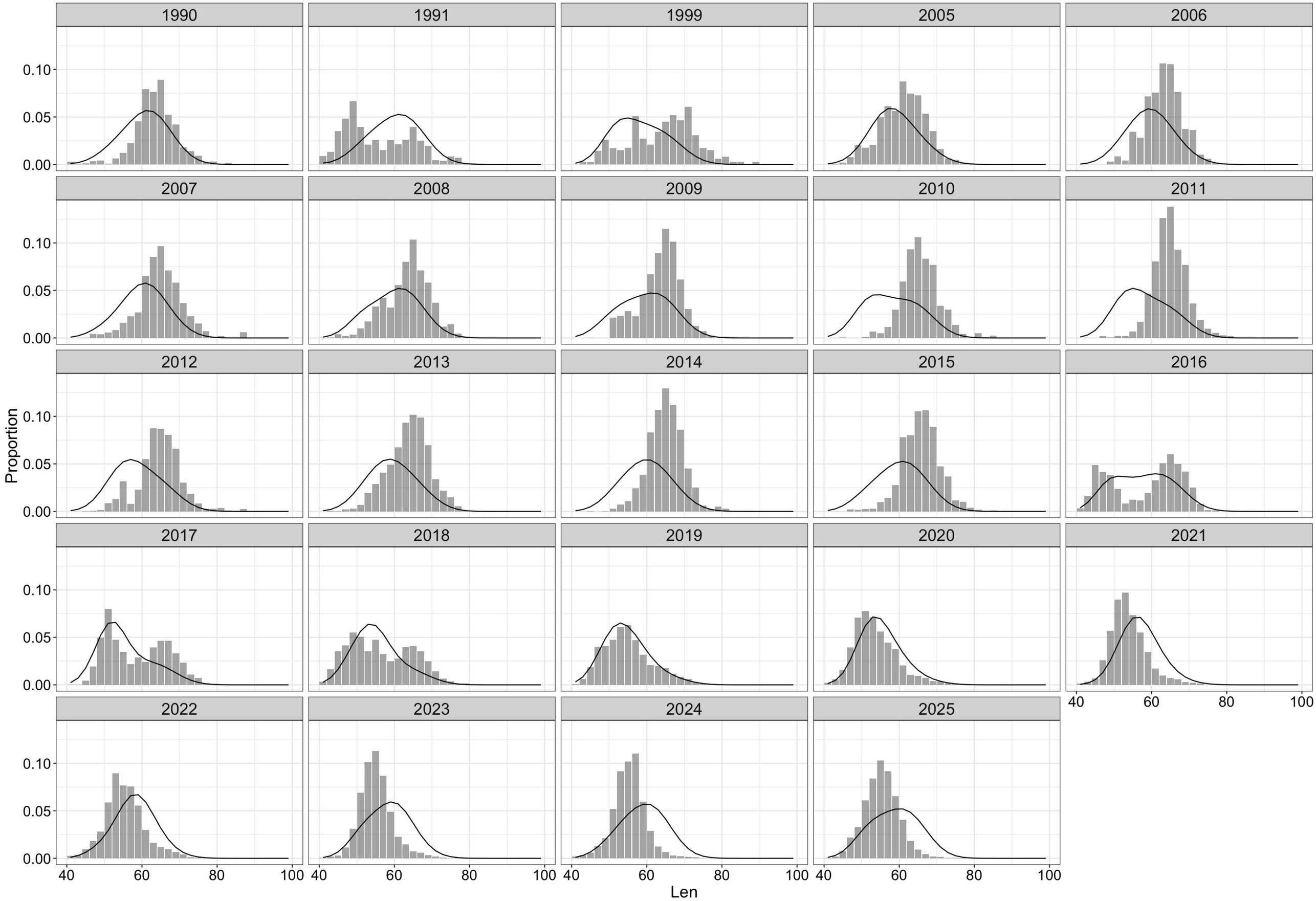

Model
26.2_FAA_3LLF_LLS_BS_Dome_DN
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model
26.2_FAA_3LLF_LLS_BS_Dome_DN
AK
Total Biom
Biomass (kt)
0
200
400
600
1960
1980
2000
2020
Year

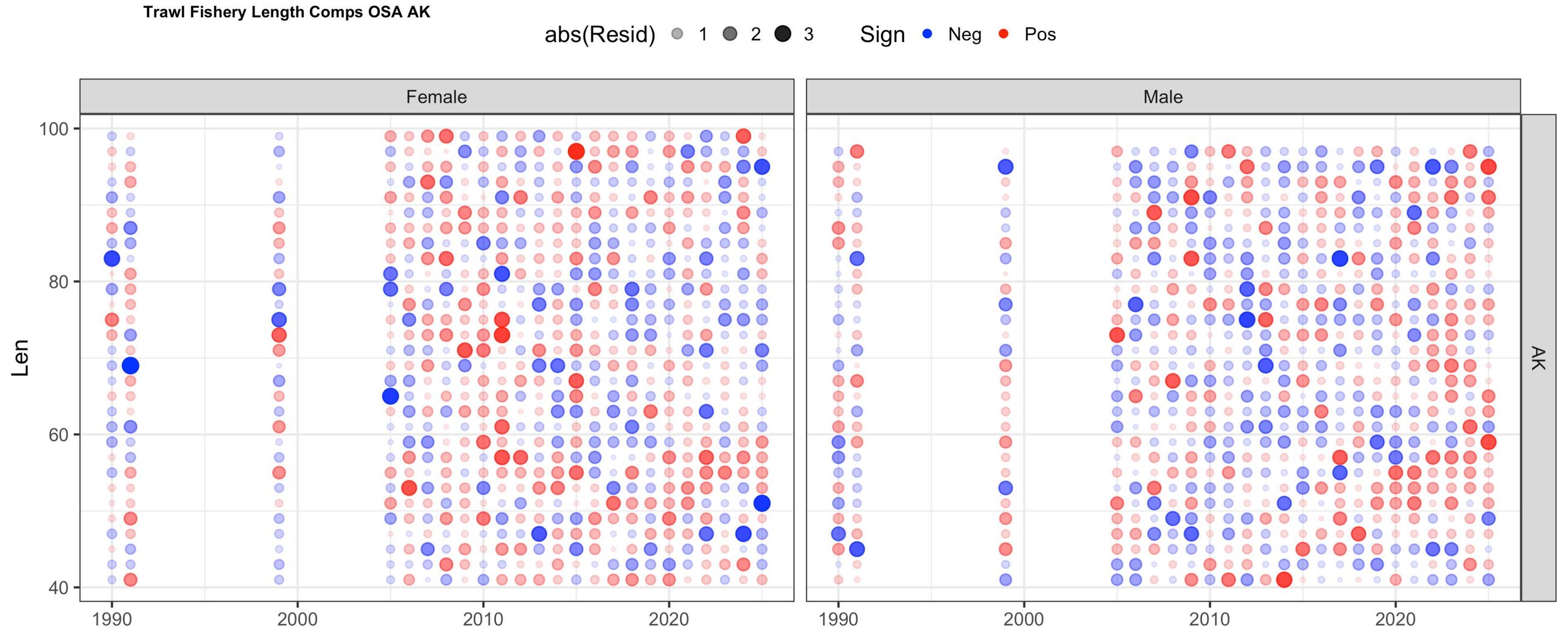
Trawl Fishery Length Comps Female AK

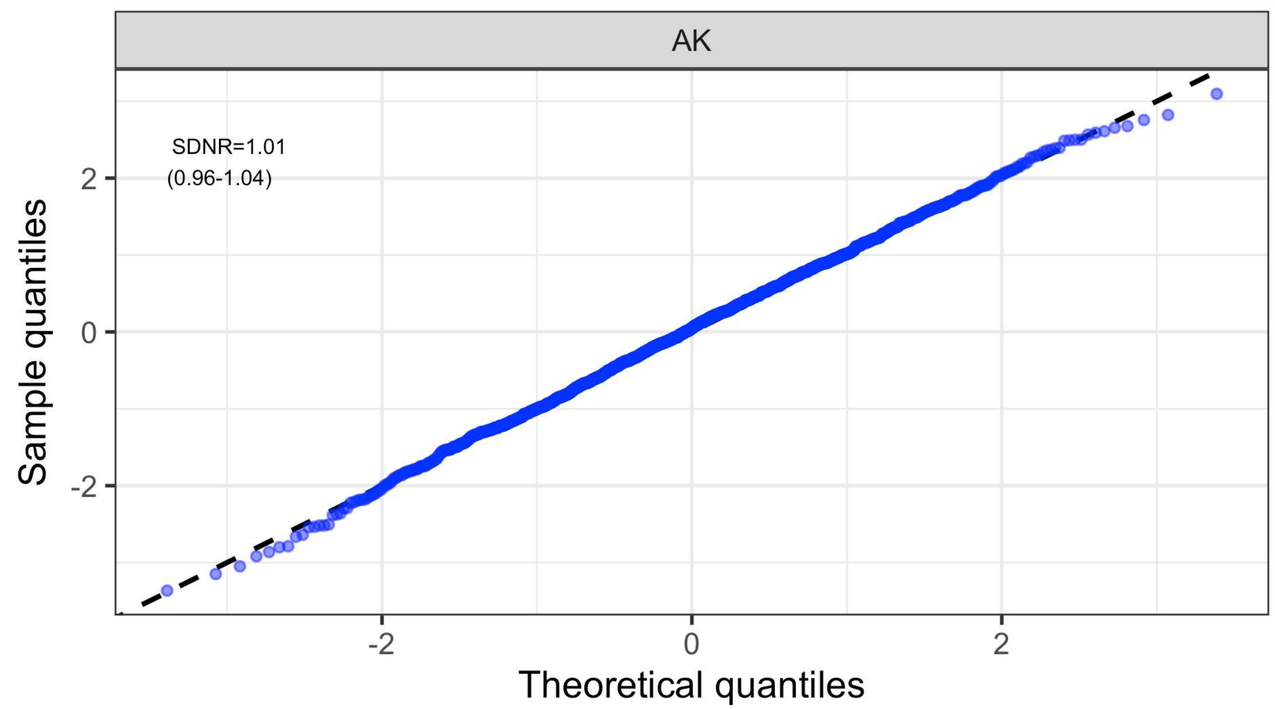


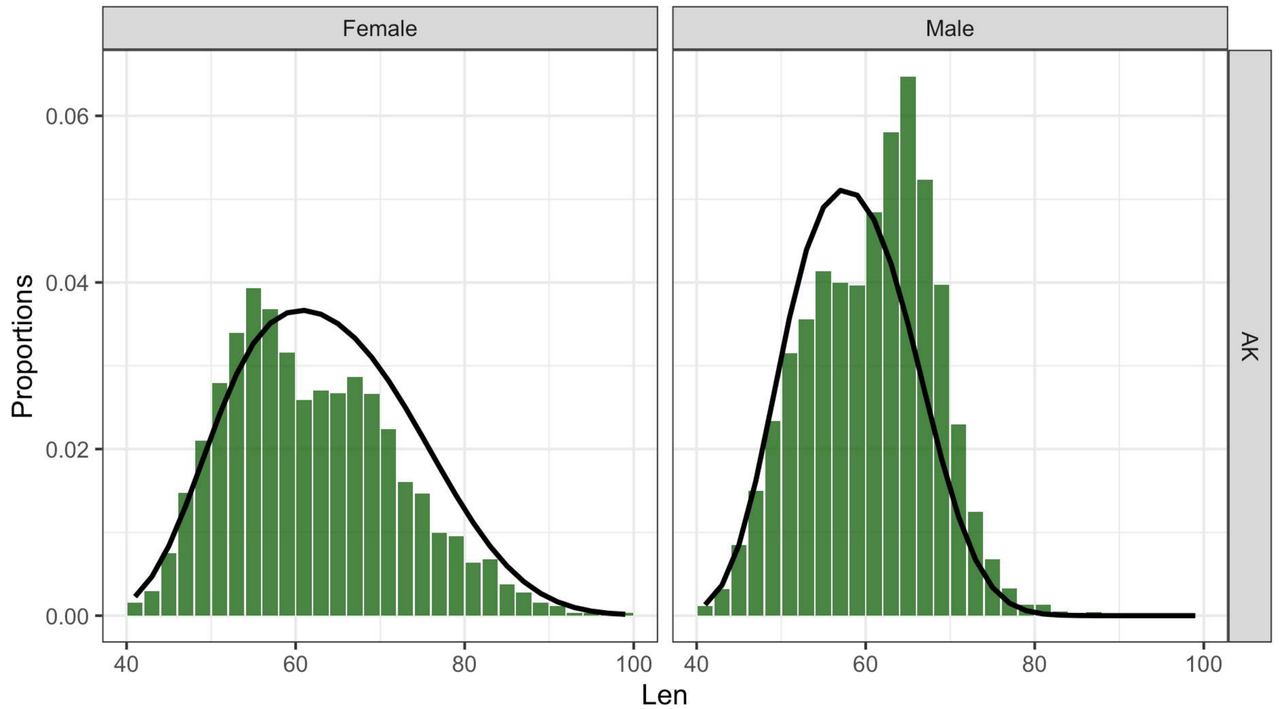
Trawl Fishery Length Comps Male AK

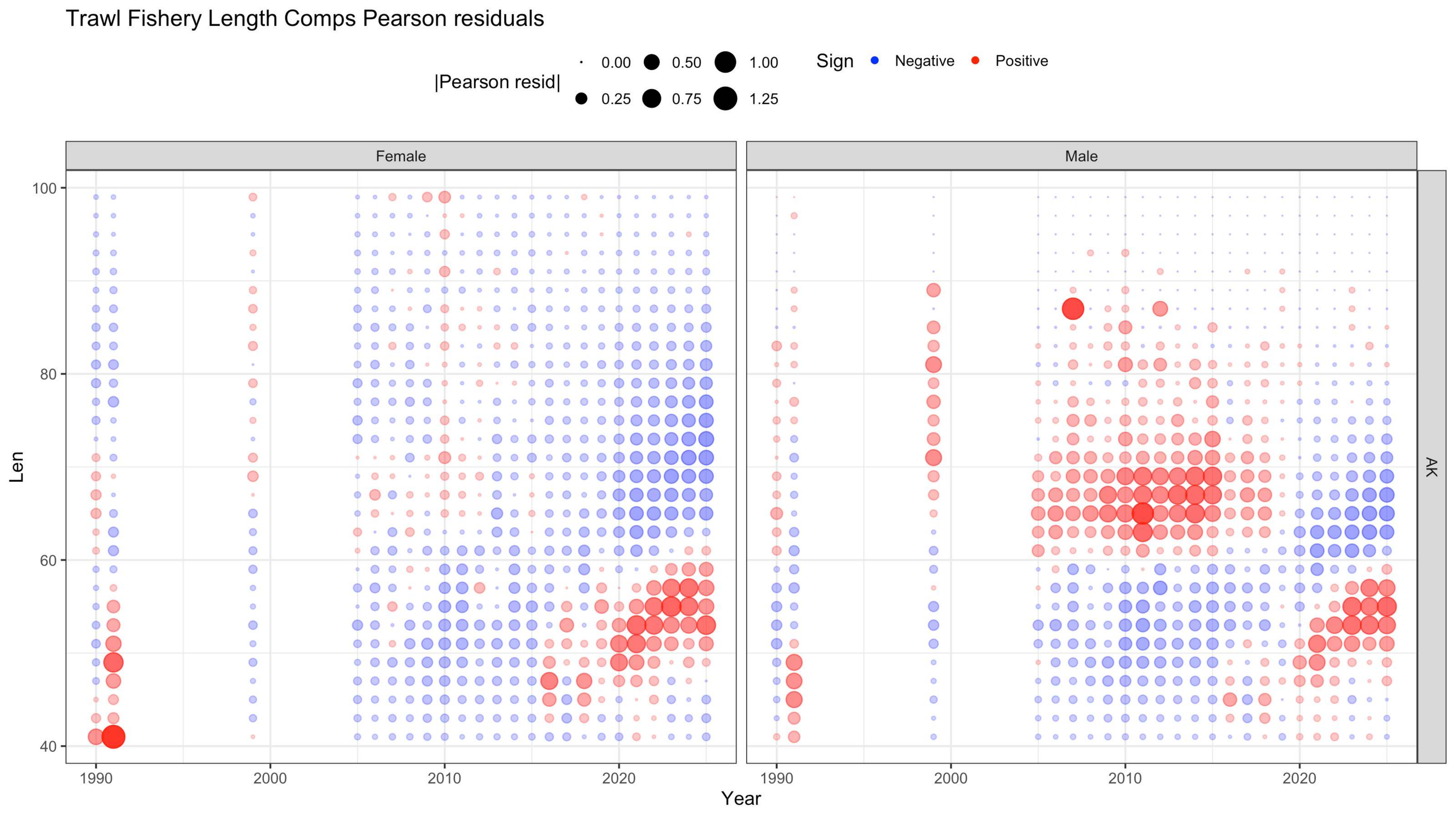


Trawl Fishery Length Comps OSA AK

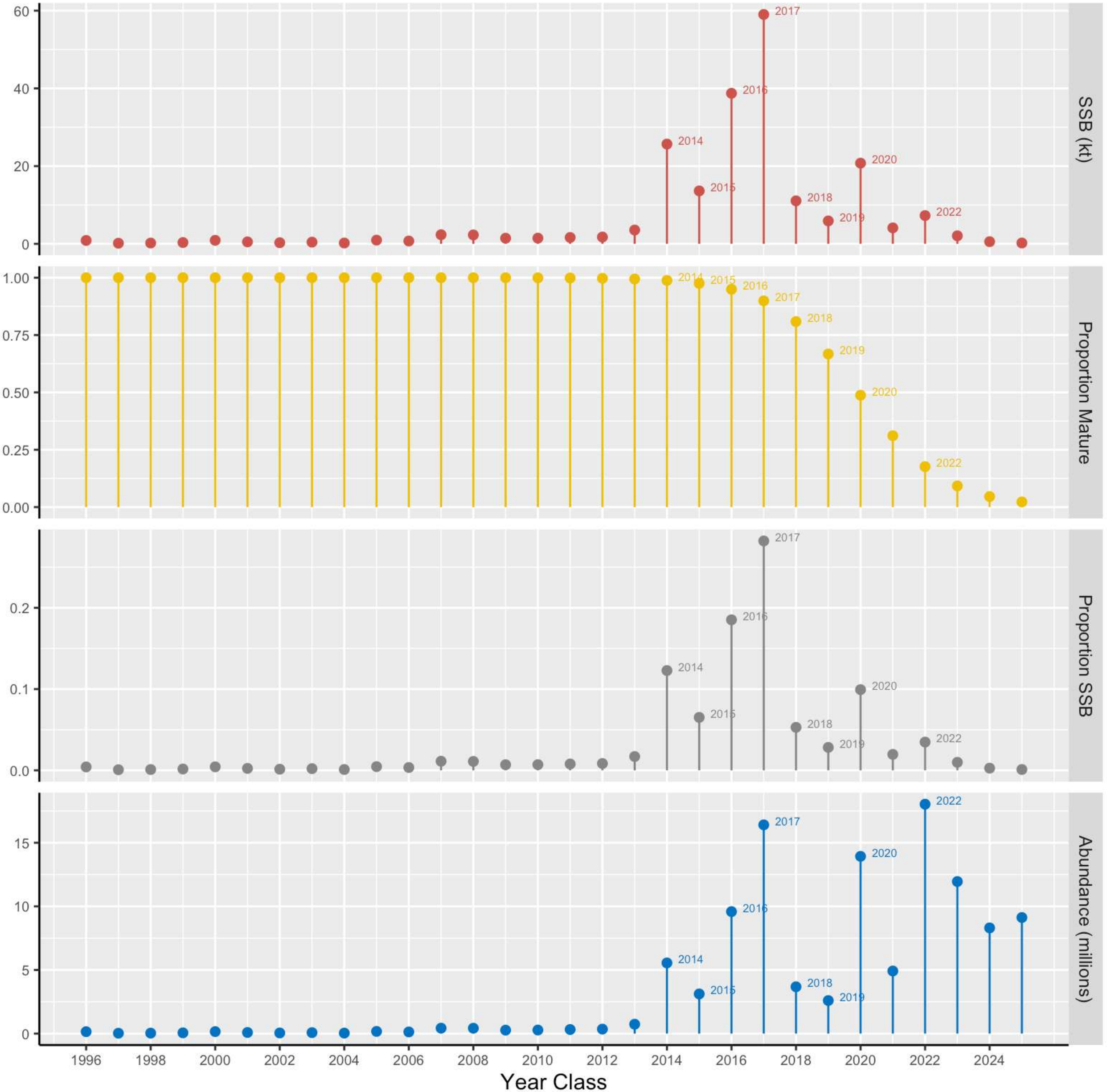






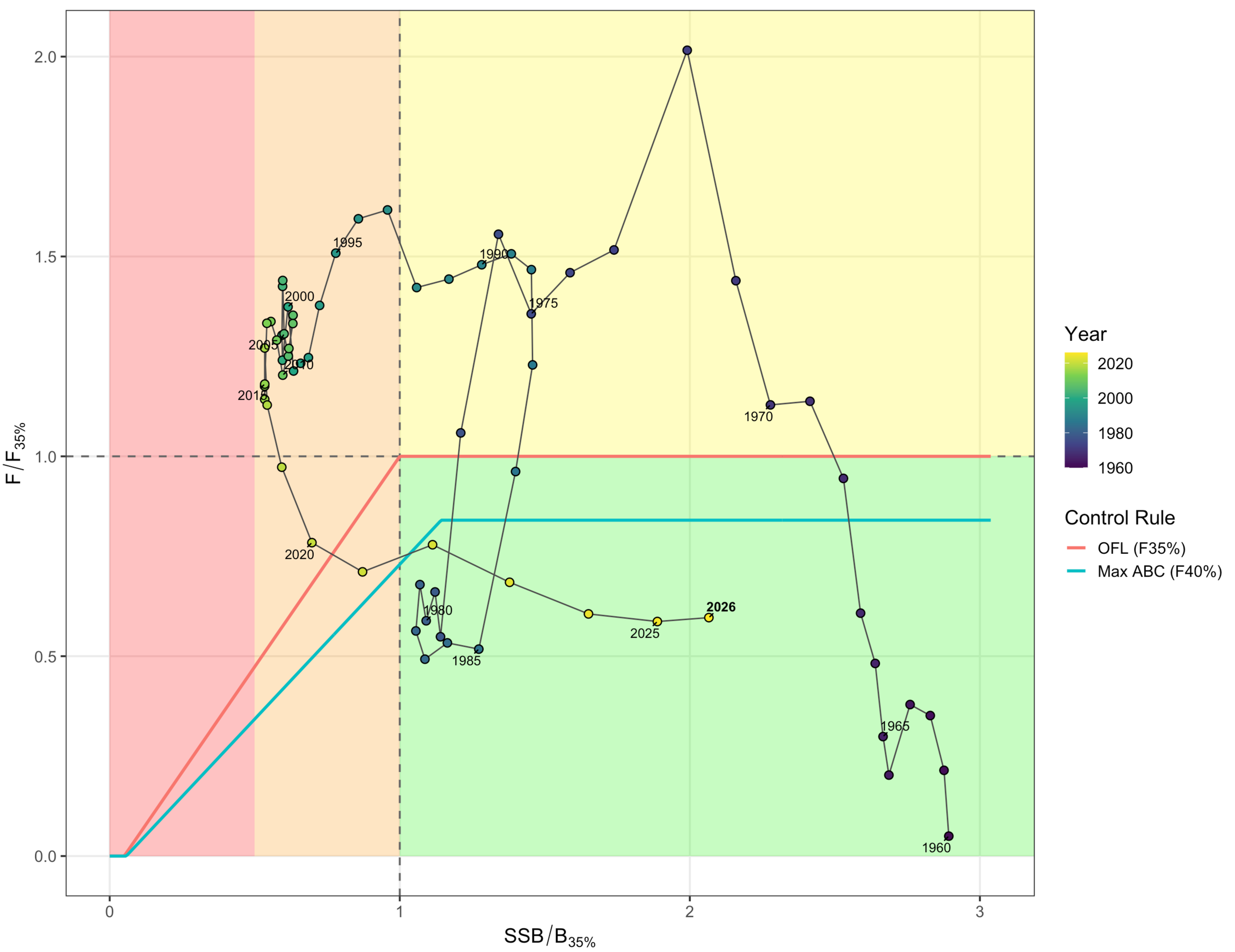

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.00
0.25
0.50
0.75
1.00
1.25
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery BS CPUE	NA	NA	NA	NA
Fixed Gear Fishery AI CPUE	NA	NA	NA	NA
Fixed Gear Fishery GOA CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.31	34.07	10.73
AI LL Survey Ages	NA	2.06	35.57	73.21
GOA LL Survey Ages	NA	0.41	67.17	27.45
JPN LL Survey Ages	NA	0.18	83.33	14.73
JPN LL Survey Lengths	NA	0.34	54.62	18.68
BS Fixed Gear Fishery Ages	NA	0.30	26.85	8.06
AI Fixed Gear Fishery Ages	NA	0.04	29.65	1.22
GOA Fixed Gear Fishery Ages	NA	0.11	63.44	7.21
Trawl Fishery Lengths	NA	0.10	47.62	4.77

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_ov
_3LLF_LLS_BS_Dome_DN	1	197.1824	367.9011	0.05674459	36.21291	129.868	0.07993	1.51833	0.53597	0.


2.0
1.5
1.0
0.5
0.0
F/F35%
0
1
2
3
SSB/B35%
1995
2000
2005
2010
2015
2020
1990
1975
1980
1985
2025
2026
1970
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.091	0.091
M Male (estimated)	0.114	0.114	0.099	0.099
Average Recruitment (millions)	25.9	25.9	19.6	19.6
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	557786	557903
Projected Female Spawning Biomass (t)	219,714	241,217	209117	220794
B100% (t)	302,672	302,672	324670	324670
B40% (t)	121,069	121,069	129868	129868
B35% (t)	105,935	105,935	113634	113634
FOFL	0.102	0.102	0.095	0.095
max FABC	0.087	0.087	0.08	0.08
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	42796	43381
OFLw (t)	58,532	57,797	42665	43250
max ABC (t)	50,283	49,651	36212	36710
ABC (t)	50,283	49,651	36212	36710
ABCw (t)	50,111	49,482	36081	36579
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

0.20

1960

1980

2000

2020

2040

Year

S1: FABC (F40)
S1a: Author's F (FABC Ratio) Year 1, FABC After
S2: Author's F (FABC Ratio) Years 1+2
S3: 5-year Average F
S4: Half max FABC (1/2 * F40)
S5: No Fishing
S6: Overfished (FOFL)
S6a: Author's F (FABC Ratio) Year 1, FOFL After
S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality
0.00
0.05
0.10
2010
2020
2030
2040
Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
250
500
750

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

250

500

750

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
20
40
60

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

1960
1980
2000
2020
2040

Year

Fleet
BS LLF
AI LLF
GOA LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

2010
2020
2030
2040

Year

Fleet
BS LLF
AI LLF
GOA LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
25
50
75
100

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

100

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

1960

1980

2000

2020

2040

Year

S1: FABC (F40)
S1a: Author's F (FABC Ratio) Year 1, FABC After
S2: Author's F (FABC Ratio) Years 1+2
S3: 5-year Average F
S4: Half max FABC (1/2 * F40)
S5: No Fishing
S6: Overfished (FOFL)
S6a: Author's F (FABC Ratio) Year 1, FOFL After
S7: Approaching Overfishing (FABC -> FOFL)
SSB (kt)
0
100
200
300
400
2010
2020
2030
2040
Year

Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	10824	7216	18041	3969	7577	2165	4330	36081
2027	REMA_LLS_No_wAI	10103	8298	17680	3608	7577	2165	4330	36081
2027	Carryover	9742	9381	16958	3608	7216	2165	3969	36081
2027	REMA_LLS	9381	10824	15876	3247	6855	1804	3970	36081
2027	GOA_Fix	10824	8659	16597	3247	7216	1804	4331	36081
2028	Carryover_No_wAI	10974	7315	18290	4024	7682	2195	4389	36579
2028	REMA_LLS_No_wAI	10242	8413	17924	3658	7682	2195	4389	36579
2028	Carryover	9876	9510	17192	3658	7316	2195	4024	36579
2028	REMA_LLS	9511	10973	16095	3292	6950	1829	4024	36579
2028	GOA_Fix	10974	8779	16826	3292	7316	1829	4389	36579

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-17	-37	-20	-15	-21	-18	-23	-23
2027	REMA_LLS_No_wAI	-23	-27	-22	-23	-21	-18	-23	-23
2027	Carryover	-25	-18	-25	-23	-25	-18	-29	-23
2027	REMA_LLS	-28	-5	-30	-31	-29	-32	-29	-23
2027	GOA_Fix	-17	-24	-26	-31	-25	-32	-23	-23
2028	Carryover_No_wAI	-16	-36	-19	-14	-20	-17	-21	-22
2028	REMA_LLS_No_wAI	-21	-26	-21	-22	-20	-17	-21	-22
2028	Carryover	-24	-17	-24	-22	-24	-17	-28	-22
2028	REMA_LLS	-27	-4	-29	-30	-28	-31	-28	-22
2028	GOA_Fix	-16	-23	-25	-30	-24	-31	-21	-22

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	10824	2028	BS	NA	10974
2026	AI	NA	11421	7440	632	2027	AI	NA	7216	2028	AI	NA	7315
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	18041	2028	GOA	NA	18290
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	3969	2028	WGOA	NA	4024
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	7577	2028	CGOA	NA	7682
2026	WY	NA	2652	2652	2271	2027	WY	NA	2165	2028	WY	NA	2195
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	4330	2028	SEO	NA	4389
2026	Total	57797	47008	38986	22161	2027	Total	42665	36081	2028	Total	43250	36579

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	238899	NA	18041	NA	NA
GOA	2028	238169	NA	18290	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	130816	NA	10824	NA	NA
BS	2028	130416	NA	10974	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	144017	NA	7216	NA	NA
AI	2028	143577	NA	7315	NA	NA